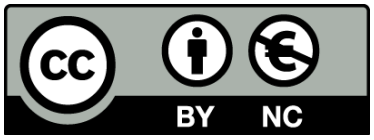


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A computational study of the Low-Density Lipoprotein Receptor-Related Protein 1 and its ligands

Gian Marco Tuveri



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Tesi doctoral

A computational study of the Low-Density Lipoprotein Receptor-Related Protein 1 and its ligands

Autor
Gian Marco Tuveri

Directors
Prof. Dr. Giuseppe Battaglia
Prof. Dr. Giancarlo Franzese



UNIVERSITAT DE BARCELONA

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Memoria presentada para optar al grado de doctor por
la Universitat de Barcelona

Programa de Doctorado en Física

Autor
Gian Marco Tuveri

Directores
Prof. Dr. Giuseppe Battaglia
Prof. Dr. Giancarlo Franzese

Tutor
Prof. Dr. Assumpta Parreño García



UNIVERSITAT DE BARCELONA

Declaration

I hereby declare that the thesis entitled *A computational study of the Low-Density Lipoprotein Receptor-Related Protein 1 and its ligands*, submitted by me for the award of the degree of Philosophiae Doctor in Physics to the Universitat de Barcelona (University of Barcelona), is a record of bonafide work carried out by me under the supervision of Prof. Dr. Giuseppe Battaglia at the Institute for Bioengineering of Catalonia and Prof. Dr. Giancarlo Franzese at the Department de Fisica de la Materia Condensada, Facultat de Fisica, Universitat de Barcelona.

I further declare that the work in this thesis has not been and will not be submitted, in a whole or in part, for the award of any other degree or diploma at this institute or any other institute or university.

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Abstract

This thesis investigates the structure, dynamics, and molecular recognition properties of the low-density lipoprotein receptor–related protein 1 (LRP1), a key receptor at the blood–brain barrier (BBB). Through a combination of molecular modeling, molecular dynamics simulations, docking analyses, and coarse-grained nanoparticle studies, we provide an integrated description of how modular motifs, hydration, glycans, and nanocarrier electrostatics define receptor function and therapeutic potential.

We first generated structural models of LRP1 by integrating domain-by-domain neural network predictions with homology modeling from LRP2. These models revealed the modular repetition of calcium-binding, β-propeller, and EGF-like motifs, and highlighted the role of glycosylation in shaping receptor architecture. Molecular dynamics showed that the flexible R domains undergo large-scale coiling, emphasizing that LRP1 is not a static scaffold but a dynamic receptor whose organization depends on solvent and steric effects.

Hydration analysis revealed that conserved amino acids display heterogeneous water interactions depending on their structural context. By introducing a corrected radial distribution function method and solving the Ornstein–Zernike closure iteratively, we constructed free-energy maps of residue–water interactions. This approach was extended to estimate desolvation enthalpies for ligand binding: for example, the binding of angiopep2 to CB19 and CB25 was found to require the removal of hydration shells

corresponding to $\sim 20 k_B T$, resulting in a predicted binding timescale of ~ 0.6 ms. These results demonstrate that hydration is not passive but an active regulator of binding specificity and kinetics at the BBB.

Docking analyses further showed that LRP1 presents extended binding regions rather than a single site, consistent with its ability to recognize multiple ligands. Glycans were found to stabilize the dimeric canopy through protein–glycan and glycan–glycan hydrogen bonds, while their steric repulsion modulates ligand access. Finally, coarse-grained simulations of PEG–PLA micelles reproduced their self-assembly into PLA cores and PEG coronas, while zeta-potential calculations demonstrated how spontaneous hydrolysis at the PEG–PLA junction can generate charged fragments that tune micelle electrostatics over time.

Together, these findings emphasize that LRP1 function emerges from the interplay of structure, dynamics, solvation, and glycosylation, and that nanoparticle design must consider not only affinity but also hydration and charge dynamics. This integrated perspective provides both a molecular basis for LRP1-mediated transport and guidelines for the rational engineering of nanocarriers for brain delivery.

Resum en català

Tìtol: Estudi computacional de la proteïna *Low-Density Lipoprotein Receptor-Related Protein 1* (LRP1) i els seus lligands

Aquesta tesi explora l'estructura, la dinàmica i la solvatació de la proteïna *Low-Density Lipoprotein Receptor-Related Protein 1* (LRP1), un receptor multifuncional implicat en el trànsit i la transitosi de lligands a través de la barrera hematoencefàlica. Mitjançant tècniques de modelització molecular, simulacions de dinàmica molecular, estudis de *docking* i models *coarse-grained* de nanopartícules terapèutiques, s'ofereix una visió multiescala de com l'estructura, la hidratació i l'electrostàtica determinen la funció del receptor i la seva interacció amb lligands i nanotransportadors.

El Capítol 1 introdueix la rellevància biològica i biomèdica de la LRP1, destacant el seu paper central en l'endocitosi i la transitosi a través de la barrera hematoencefàlica, la seva modularitat estructural i la seva importància en la nanomedicina dirigida al cervell. Aquest capítol també estableix els objectius del treball: reconstruir l'estructura del receptor, estudiar-ne la flexibilitat i la hidratació, quantificar-ne les interaccions amb lligands i caracteritzar nanopartícules terapèutiques dirigides a la LRP1.

El Capítol 2 descriu les metodologies computacionals desenvolupades i utilitzades al llarg de la tesi. Detalla els procediments de modelització molecular, els protocols de *docking* les simulacions de dinàmica molecular atòmica i coarse-grained, i les noves

eines analítiques per calcular la hidratació, les superfícies d'energia lliure i els potencials electrostàtics.

El Capítol 3 estableix el context i la modelització estructural de la LRP1. Les prediccions basades en xarxes neuronals (AlphaFold2, RoseTTAFold) es van combinar amb models d'homologia derivats de la LRP2 per reconstruir les formes monomèrica i dimèrica del receptor. Els models resultants van revelar la seva organització modular, amb dominis d'unió al calci (CB), estructures de β-*propeller* (B) i dominis tipus EGF, i van evidenciar la seva forta similitud estructural amb el LRP2.

El Capítol 4 analitza la flexibilitat del receptor mitjançant simulacions de dinàmica molecular dels clústers R. Les simulacions mostren una clara tendència a l'enrotllament i grans fluctuacions conformacionals no impulsades per nous enllaços d'hidrogen, sinó per efectes mediats pel solvent. Els resultats destaquen la importància de la hidratació i de la geometria terciària en la definició de la flexibilitat proteica.

A més, el Capítol 4 quantifica l'estabilitat energètica de l'organització terciària de la LRP1. Mitjançant simulacions d'umbrella sampling del domini β-*propeller* 8 (B8), es va calcular una energia d'uns 180 kT per separar les cadenes α i β després del tall per la furina, demostrant que les interaccions no covalents mantenen la integritat del receptor. En el mateix capítol, es confirma que els ions de calci romanen fortament coordinats dins de les gàbies àcides dels dominis CB, amb energies de dissociació d'aproximadament 16 kT, la qual cosa explica l'estabilitat mecànica d'aquests dominis.

El Capítol 5 defineix una superfície d'energia lliure d'unió per al complex LRP1-angio-pep2. A partir de centenars de posicions de docking, es van calcular les energies lliures per residu, combinant contribucions entàlpiques i entròpiques. L'anàlisi va revelar múltiples regions d'unió superposades en lloc d'un únic lloc definit, coherent amb la capacitat de la LRP1 per reconèixer diversos lligands. A més, mitjançant un model

polimèric, es va mostrar que les N-glicosilacions generen camps repulsius locals que modulen l'accés del solvent i del lligand, especialment a la interfície de la volta dimèrica.

El Capítol 6 se centra en la hidratació i la termodinàmica de la solvatació. Una nova correcció de les funcions de distribució radial, combinada amb l'equació d'Ornstein-Zernike, va permetre calcular els potencials d'interacció aminoàcid–aigua. Els càlculs van demostrar que residus idèntics poden presentar perfils d'hidratació molt diferents segons el seu entorn estructural. A partir d'aquests resultats, es van estimar les entalpies de desolvatació per a la interacció LRP1–angiopep2, obtenint una barrera energètica d'uns 20 kT i un temps d'unió d'aproximadament 0,6 ms, en concordança amb les dades experimentals.

El Capítol 7 amplia l'estudi a les nanopartícules terapèutiques dirigides a la LRP1. Les simulacions *coarse-grained* de micel·les de PEG–PLA van reproduir la seva morfologia experimental, amb un nucli hidrofòbic de PLA i una corona hidratada de PEG. Els càlculs del potencial zeta van mostrar una dependència lineal entre el potencial superficial i la càrrega de la micel·la, d'acord amb les observacions experimentals, i van suggerir que la hidròlisi del polímer modula la càrrega amb el temps.

En conjunt, aquesta tesi presenta un marc molecular complet que relaciona estructura, solvatació i electrostàtica amb la funció biològica de la LRP1. A més, introdueix noves eines computacionals, com la correcció de funcions RDF i el mapeig de superfícies d'energia lliure, aplicables a l'estudi d'altres sistemes macromoleculars implicats en el transport receptor-mediat i en el disseny de nanomedicines dirigides al cervell.

Paraules clau: LRP1, Barrera hematoencefàlica, Dinàmica molecular, Modelització molecular, Glicosilació, Hidratació proteica, Energia lliure d'unió, Angiopep2, Desolvatació, Potencial zeta, Nanopartícules terapèutiques, Micel·les PEG–PLA, Simulacions coarse-grained, Transitosi

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Chapter 1

Introduction

The low-density lipoprotein receptor (LDLR) family comprises transmembrane proteins that play crucial roles in various biological processes, including lipid uptake, endocytosis, protease clearance, and signal transduction. Key members of this family include LDLR, low-density lipoprotein receptor-related protein 1 (LRP1), LRP2 (also known as megalin), LRP5/6, and LRP8 (also known as apolipoprotein E receptor 2, ApoER2). They are characterized by their unique structure, which includes extracellular ligand-binding repeats and cytoplasmic motifs that help facilitate signal relay and adaptor recruitment [1, 2]. Among these proteins, LRP1 is remarkably versatile, exhibiting diverse functionalities across various tissues and systems.

LRP1 is synthesized as a 600-kDa precursor and then cleaved by the enzyme furin into an extracellular α-chain (approximately 515 kDa) and a membrane-bound β-chain (approximately 85 kDa). This protein can bind over 40 ligands, including apoE, proteases, extracellular matrix proteins, and growth factors [3, 4]. Its dual role as an endocytic receptor and signaling modulator makes LRP1 essential for processes related to vascular biology, immune regulation, cancer, and neurodegeneration.

1 LRP1 structural composition: a sequence of motifs

The low-density lipoprotein receptor-related protein 1 (LRP1) is a large endocytic receptor composed of repeating modular motifs. As illustrated in Figure 1.1, its extracellular region is organized into three characteristic structural elements: the calcium-binding (CB) motifs, shown in pink; the β-propeller (B) motifs, shown in blue; and the epidermal growth factor-like (EGF) motifs, shown in orange [2]. The ordered repetition of these motifs defines the modular architecture of the receptor and is a hallmark of the LDL receptor family.

Each motif carries specific structural and functional features. The CB motifs, also called class A repeats or cysteine-rich motifs, are short cysteine-rich domains of about 40 residues. They contain six conserved cysteines forming three disulfide bridges, and a cluster of acidic residues that coordinate calcium ions [2]. The first CB was identified in the LDLR [5]. Later structural studies revealed that the Ser-Asp-Glu triplet and other acidic residues form a "calcium cage" that stabilizes local folding and creates high-affinity binding sites [6, 7]. The β-propeller motifs are built from six Tyr-Trp-Thr-Asp (YWTD) repeats, forming toroidal blades of antiparallel β-sheets, typically around 250 residues in size [8]. They act as scaffolds for domain organization and are crucial for pH-dependent conformational changes during endocytosis [9]. The EGF-like motifs, consisting of 40–45 residues and three disulfide bridges, are often positioned adjacent to β-propellers and act as stabilizing linkers [8]. Unlike CBs, they lack the acidic triplet and do not coordinate calcium; however, they support structural flexibility and ligand release under endosomal conditions.

The modular design of LRP1 is not unique. The LDLR itself contains seven CB repeats, a single β-propeller, and three EGF-like domains [9]. This compact arrangement enables it to function as a high-affinity receptor for LDL and VLDL (very low-density

lipoprotein) particles. By contrast, LRP2 (megalin) exhibits a domain organization strikingly similar to that of LRP1, with four clusters of CB repeats separated by β-propellers and interspersed with EGF-like motifs [10]. The main differences lie at the termini: LRP2 contains additional CB motifs at the N-terminus, whereas LRP1 carries extra EGF-like motifs near its C-terminus [10]. These variations reflect functional specialization, but the strong conservation of modular structure across the family highlights their shared evolutionary origin and common mechanistic principles [9].

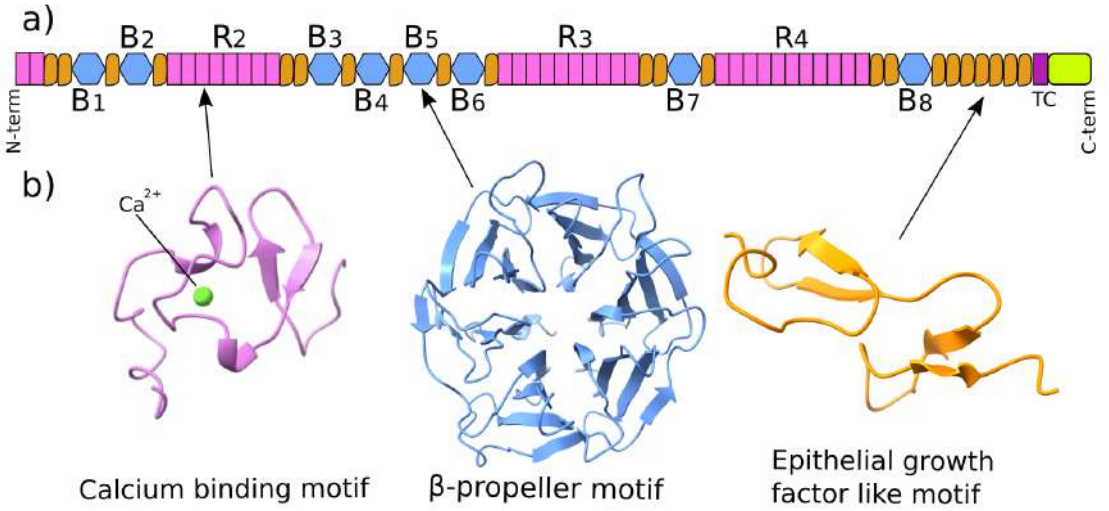


Figure 1.1: Schematic representation of the three main LRP1 structural motifs. Calcium-binding (CB) motifs are highlighted in pink, β-propeller (B) motifs in blue, and EGF-like motifs in orange. Their ordered repetition along the sequence defines the modular architecture of LRP1 and other LDL receptor family members [2].

2 LRP1 known ligands and shared interactions

The large dimension of LRP1 is accompanied by its ability to bind to many ligands. The first discovered binding ligand was apolipoprotein E, apoE [11]. This small protein, composed of approximately 300 amino acids, is involved in the transport of lipid particles in the bloodstream and facilitates the interaction between the lipid particles and their

receptors. Another member of the lipoprotein family, apoB, which is also involved in lipid transport, is an LRP1 ligand. The two apolipoproteins have been observed to bind not only to LRP1 but also to LDLR and VLDLR [12]. An unbalanced removal of these lipids from the bloodstream, caused by receptor malfunction, can lead to cardiovascular diseases, among which familial hypercholesterolemia is the most studied. The introduction of mutations in the LDL receptor CB units proved that the protein-protein binding with the apoproteins apoE and apoB-100 happens within a combination of different CBs and is probably due to the acid-basic interactions [13]. The study highlights the importance of CB units 3-5 and 7 in LDLR, underscoring the superior binding capacity of unit 5 and linking it to the presence of a Glu residue after the globally conserved motif Ser-Asp-Glu. Another recent study highlights the role of punctual mutations in the CB motif SxSDE in regulating apical membrane targeting in renal epithelial cells [14]. The study investigated the binding kinetics of various ligands with the calcium-binding clusters from LRP1 [14]. The CB cluster fragments were separately expressed, and binding affinity experiments were conducted using the fragments. The study reiterates the fundamental role of calcium and its "activation role" for CB units; moreover, it enhances the similar kinetics of the second (8 units) and fourth (11 units) CB clusters with diverse ligands, identifying them as major binding sites. The study also proposed a model for the RAP-LRP1 binding.

Receptor Associated Protein, or RAP (UniProt P30533), is a crucial chaperone that binds to LRP1 and the other members of the family in the presence of calcium ions [15]. RAP has been located within the endoplasmic reticulum and the Golgi apparatus. Its presence in the early secretory pathway is due to its role in the folding of LDL family proteins. Due to its binding specificity to the LDL family receptors, RAP has been used as an antagonist in binding affinity studies since the early 90s [16]. The natural inhibition capacity led researchers to develop an LRP1 inhibitor by mutating the RAP protein, stabilizing its structure, and making the bond resistant to pH changes and high temperatures [17]. A complete list of identified ligands is displayed in Table 1.1.

Receptor	Ligands	References
LRP1-2	Albumin	Cui et al. 1996
LRP1	alpha2-macroglobulin	Kristensen et al., 1990
LRP1	Angiopep2	Demeule et al. 2008
LRP1-2	Angiotensin II	Gonzalez-Villalobos et al. 2005
LRP1-2	Angiotensin 1-7	Gonzalez-Villalobos et al. 2006
LRP1-2	ApoB	Stefansson et al. 1995
LRP1-2-8	ApoE	Willnow, 1999, He X. et al. 2007
LRP1-2	ApoH	Moestrup and Verroust, 2001
LRP1	Apoj (Clusterin)	Kounnas et al. 1995
LRP1-2	ApoM	Faber et al. 2006
LRP1-2	Aprotinin	Moestrup and Verroust, 2001
LRP1-2	Bone morphogenetic protein 4	Spoelgen et al. 2005
LRP1-2	Ca^{2+}	Christensen and Nielsen, 2007
LRP1-2	Cathepsin b	Nielsen et al. 2007
LRP1-2	Coagulation Factor VIII	Ananyeva et al. 2008
LRP1-2	Connective tissue growthfactor	Gerritsen et al. 2010
LRP1-2	Cytochrome C	Kawata et al. 2012
LRP1-2	Cystatin C	Lee et al. 2012
LRP1-2	Epidermal growth factor	Orlando et al. 1998
LRP1-2	Folate binding protein	Birn et al. 2005
LRP1	Frizzled-1	Zilberberg et al. 2004
LRP1-2	Hemoglobin	Gburek et al. 2002
LRP1-2	Insulin	Orlando et al. 1998
LRP1-2	Lactoferrin	Willnow, 1999
LRP1-2	Leptin	Dietrich et al. 2008
LRP1-2	Lipoprotein lipase	Knauer et al. 1993
LRP1	Metallothionein	Klassen et al. 2004
LRP1-2	Plasminogen	Kanalas and Makker, 1993
LRP1-2	Plasm. activator inhib. type1 tissue	Kanalas and Hopfer, 1997
LRP1-2	Plasm. activator	Moestrup and Verroust, 2001

Table 1.1: List of ligands known to interact with LRP1. Some ligands are shared among other members of the LDLR family, namely LRP2 and LRP8.

One of the most used LRP1 ligands is Angiopep2. The peptide and its use in LRP1 targeting will be described further in this Chapter.

3 Ectodomain shedding

An interesting feature of LRP1 is the presence of a cleavage site in the ectodomain portion of the receptor, which is the part of the membrane receptor that extends into the extracellular space and is close to the cell membrane. The cleavage does not affect the structure of the receptor: the chains (515 and 85 kDa, respectively) are held together by non-covalent bonds, namely hydrogen bonds and hydrophobic effective interactions, the latter excluding water between adjacent β sheets. In 1988, Herz and colleagues observed that a specific LRP1 C-terminal-interacting antibody bound strongly to an 85 kDa protein [11], which is approximately 10 times smaller than the expected 600 kDa full-length LRP1 chain. Two years later, the same authors defined the cleavage site in correspondence to the amino acids Arg-His-Arg-Arg (RHRR) [18]. The four-amino-acid-long sequence is similar to the cleavage of other peptide precursors [19] and membrane proteins like the insulin receptor [20]. The RHRR motif is conserved in other species: M. musculus (Uniprot Q91ZX7), R. norvegicus (Uniprot G3V928), and G. gallus (Uniprot P98157). In the latter case of G. gallus, Histidine is mutated in Asparagine (RNRR). In 1996, Willnow, Herz, and colleagues [21] proved the existence of a furin-mediated cleavage site. The study was carried out on various cell lines, but mainly on furin-deficient RPE.40 shows how the cleavage takes place in the Golgi apparatus and doesn't affect the transport of the protein complex to the membrane surface. Thus, the cleavage doesn't interfere with the intracellular transport of LRP1. However, researchers observed a slight reduction in the endocytosis of LRP1 ligands, which could be directly attributed to the presence of a covalently bonded LRP1. The endocytosis returns to expected values when furin is introduced into furin-deficient cells. Another study [22] has found that the lack of a cleavage site affects the persistence of LRP1 in the endoplasmic reticulum, suggesting the presence of a possible compensatory mechanism that activates the protein's functionality.

In the same period, the soluble LRP1 species was identified in the human plasma [23] and characterised by Quinn and colleagues [24] in vitro.

Counterintuitively, in the study, the sLRP1 ('s' stands for soluble or shedded in literature) was found to be made of the entire α chain plus a small amount of the β chain, quantified in 55 kDa. Considering that the whole β chain weight is 85 kDa, the cut must happen very close to the membrane. In Quinn's work, researchers hypothesize that a metallo proteinase on the extracellular membrane is responsible for shedding in human cells, as a metalloproteinase inhibitor (hydroxamic acid) reduces the presence of sLRP1 in solution. Researchers also verified that sLRP1 can bind the tissue plasminogen activator (tPA) and plasminogen activator inhibitor (PAI) complex. This suggests that sLRP1 acts as an antagonist of its anchored counterpart, thereby reducing overall clearance activity. The in vivo sLRP1 has been reported to increase with aging, with no significant variations in patients with and without Alzheimer's disease of the same age [25]. In vitro, Amyloid β (Aβ) appears to stimulate LRP1 shedding, with the metalloproteinases ADAM10 and ADAM17 supporting this process. Indeed, the increase of sLRP1 in the presence of AB was not observed in ADAM knock-out cells. Interestingly, in this work, researchers investigated the presence of LRP1 α and β chains, without considering sLRP1, which is composed of both α and β chains.

4 LRP1 in Immunology

LRP1 has significant immunomodulatory roles in both the innate and adaptive immune systems. Its expression varies by cell type; it is found in high levels in macrophages, dendritic cells, and monocytes but is either low or inducible in lymphocytes [26].

In innate immune cells, LRP1 plays a crucial role in regulating inflammation, phagocytosis, and the clearance of dead cells. It aids in the elimination of apoptotic cells by binding to calreticulin and complement C1q, while also reducing the release of pro-inflammatory

cytokines by suppressing the MAPK, JNK, and NF-κB pathways. In studies, knocking down the Lrp1 gene in microglia has been shown to increase sensitivity to LPS and boost the secretion of TNF-α and IL-6 [27]. In macrophages, a deficiency of LRP1 results in decreased expression of anti-inflammatory genes and heightened pro-inflammatory responses.

In dendritic cells (DCs), LRP1 aids in the uptake of antigens and the cross-presentation of MHC-I-restricted antigens. Deletion of Lrp1 in CD11c+ cells results in impaired T cell priming, rendering the host more susceptible to tumor growth [28]. Additionally, LRP1 deficiency in DCs has been linked to heightened Th2-type inflammation and eosinophilia in allergic asthma models, indicating its role in controlling excessive adaptive immune responses [29].

In T cells, LRP1 expression is low when T cells are resting but increases with activation. It facilitates adhesion and movement through interactions with thrombospondin-1 and CD47 [30]. Notably, LRP1 rapidly sheds from the cell surface upon activation, which inhibits IL-2 signaling and cell proliferation [31]. T cells lacking LRP1 show a diminished ability for alloreactivity, and experiments have demonstrated that their absence can protect against severe graft-versus-host disease [26].

5 LRP1 in Cancer

The LDL receptor-related proteins (LRPs) are a diverse group of endocytic receptors that influence cell signaling, matrix remodeling, and metabolism. Among these, LRP1 has emerged as a crucial regulator in cancer, with roles that can vary depending on the specific context.

LRP1 is often found in tumor-associated stromal cells, especially at the invasive edges of several types of cancers, such as colorectal, pancreatic, and glioblastoma [32].

In colon cancer, lower levels of LRP1 correlate with disease progression and poor survival outcomes. Research suggests that reduced LRP1 expression, partly regulated by miR-103/107, decreases cell growth and enhances apoptosis, creating a feedback loop dependent on p53 and linked to aggressive tumor behavior [33].

In pancreatic ductal adenocarcinoma (PDAC), the role of LRP1 is complex. While high levels of LRP1 are related to poor prognosis and increased invasiveness, facilitated by interactions with extracellular HSP90 and the activation of ERK and AKT signaling pathways [34], LRP1 may also function as a growth suppressor when its expression promotes apoptotic signaling via MIF-p53-dependent pathways. Additionally, micro-RNAs, such as miR-429, regulate LRP1 levels, influencing proliferation and migration in PDAC cells [35].

In glioblastoma, LRP1 is highly expressed not only in tumor cells but also in the surrounding tumor microenvironment. Its levels rise under low oxygen conditions and promote the movement of glioblastoma cells through AKT activation [36]. Furthermore, LRP1 facilitates the transport of therapeutic agents across the blood-brain barrier, thereby contributing to effective drug delivery strategies for brain tumors [37], a theme treated in the next section. It also impacts CXCR3-dependent invasion and interacts with tPA and uPAR complexes, highlighting its role in modulating the extracellular matrix and facilitating tumor spread [38].

6 LRP1 at the Blood-Brain Barrier

The blood-brain barrier (BBB) maintains the homeostatic environment of the central nervous system (CNS) by tightly regulating the passage of molecules between the circulation and the brain. Among the key proteins facilitating selective transport is LRP1, which is expressed on the luminal and abluminal membranes of brain endothelial cells, as well as in associated cells such as pericytes and astrocytes [3, 39].

LRP1 facilitates receptor-mediated transcytosis (RMT), enabling the selective transport of ligands across endothelial cells. This function has made LRP1 a target of interest for CNS drug delivery. Studies show that LRP1 can interact with around 30 ligands, including apolipoproteins, proteases, growth factors, and receptor-associated proteins [3].

LRP1 also plays a significant role in the central nervous system beyond the BBB. It regulates the clearance of amyloid-β and modulates tau protein spread in Alzheimer's disease. LRP1 mediates tau internalization through interactions involving the microtubule -binding domain and participates in prion-like propagation of pathogenic aggregates across neural circuits [40]. Its interaction with the N-methyl-D-aspartate (NMDA) receptor and postsynaptic proteins further implicates it in synaptic function and neuroprotection [41, 18].

Genetic and experimental studies have confirmed the critical role of LRP1 in Aβ clearance. Endothelial-specific deletion of the LRP1 gene in mice impairs Aβ1–42 efflux from brain to blood, increases soluble brain Aβ, and worsens cognitive decline in Alzheimer's models [42, 43]. Moreover, pericytes contribute to Aβ clearance via LRP1 in an apoE isoform-specific manner: apoE3 supports, and apoE4 impairs, this process [39]. These findings reinforce the neurovascular hypothesis of Alzheimer's disease and position LRP1 as a promising therapeutic target for enhancing CNS clearance mechanisms.

6.1 The nanotherapeutics approach

The BBB is a significant challenge in treating brain disorders. It limits the entry of therapeutic agents due to tight junctions, active efflux transporters, and restricted transcytosis. However, nanotherapeutics have emerged as a powerful strategy to overcome this barrier by facilitating targeted and efficient delivery of drugs across the BBB [44].

Nanoparticles (NPs) are engineered at the nanoscale to carry therapeutic payloads, like drugs, genes, or imaging agents, across physiological barriers. Their design can be customized for optimal size, shape, surface chemistry, and ligand functionalization to enhance brain targeting [45]. For instance, functionalization with ligands such as Angiopep2 enables receptor-mediated transcytosis via LRP1, which is one of the most commonly targeted receptors for CNS delivery [44].

In cancer therapy, particularly for conditions such as glioblastoma and brain metastases, nanocarriers can be modified to cross the BBB and target tumor cells. Angiopep2-modified nanoparticles, for example, can increase LRP1 expression on both brain microvascular endothelial cells and tumor cells. This creates a self-reinforcing loop that improves both BBB penetration and tumor targeting. These nanocarriers have demonstrated effectiveness in delivering chemotherapeutics, such as doxorubicin, and have significantly enhanced survival rates in animal models of brain metastasis [46, 47].

For neurodegenerative disorders such as Alzheimer's disease, LRP1 also plays a crucial role in clearing amyloid-β and facilitating the transport of tau protein. LRP1-targeted nanotherapeutics may improve the removal of pathogenic proteins or deliver neuroprotective agents directly to neurons or glial cells. This is particularly important in tauopathies, where LRP1 has been identified as a key regulator of tau uptake and spread [40].

A key parameter in nanocarrier design is avidity, defined as the cumulative binding strength generated by the multivalent interaction of multiple ligands with their receptors. Unlike affinity, which reflects the strength of a single ligand–receptor pair, avidity captures the cooperative effect of many simultaneous interactions. Recent insights into intracellular trafficking at the BBB, such as the role of syndapin-2–mediated tubular carriers, demonstrate that avidity critically determines cargo fate. Excessively high avidity promotes retention and lysosomal degradation, while too low avidity fails to

support transcytosis. Only an intermediate binding strength, the so-called “Goldilocks zone”, ensures efficient transport across the BBB while limiting degradation, thus optimizing sustained brain delivery [45].

Nanotherapeutic strategies extend beyond oncology and neurodegeneration; they also show promise in treating epilepsy, stroke, and CNS infections. For example, responsive nanogels have been developed to release antiepileptic drugs in response to brain electrical activity, providing temporal and spatial control of therapy [48].

In high-grade gliomas, the overexpression of LDLR family members, such as LRP1 and LRP8, supports receptor-mediated targeting. These receptors are involved not only in transport but also in signaling pathways that regulate tumor proliferation and migration. Nanoparticles functionalized to engage these receptors can deliver therapies while also modulating pathological signaling [47].

Overall, nanotherapeutic approaches offer several advantages for treating brain diseases, including enhanced BBB penetration, reduced systemic toxicity, increased targeting precision, and the potential for multimodal therapies. Future developments are likely to focus on optimizing receptor-ligand interactions, enhancing endosomal escape, and integrating diagnostic and therapeutic capabilities into a single platform.

6.2 The Angiopep2 ligand

Angiopep2 (sequence: TFFYGGSRGKRNNFKTEEY) is a synthetic nineteen amino acid peptide designed to enhance drug delivery across the BBB, which poses a significant challenge in treating CNS disorders due to its selective permeability that limits the entry of most therapeutic agents. Angiopep2 effectively targets LRP1 and crosses the BBB through receptor-mediated transcytosis. By interacting with LRP1, Angiopep2 facilitates the transport of therapeutic compounds into brain tissue, making it a valuable tool for CNS drug delivery. The structure of Angiopep2 is shown in Figure 1.2.

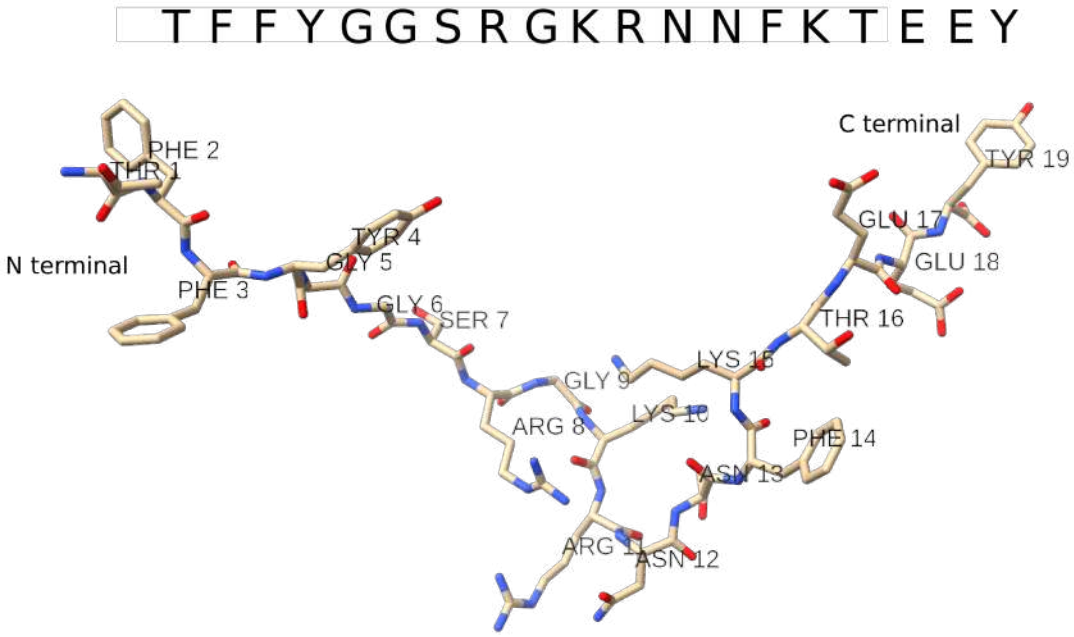


Figure 1.2: Sequence and stick cartoon structure of the Angiopep2 peptide.

Angiopep2 binds with high affinity to LRP1, particularly engaging specific binding domains, including the CB5-6 and CB17 motifs. Molecular docking studies indicate that the binding utilizes most of the peptide's residues; importantly, conjugating drugs to Angiopep2 does not significantly hinder its binding capacity. This characteristic allows Angiopep2 to effectively transport a variety of therapeutic agents, including small molecules, proteins, genes, and nanoparticles [49].

One of the early applications of Angiopep2 was in the creation of ANG1005, a conjugate of paclitaxel and Angiopep2. This construct demonstrated significantly better penetration across the BBB and higher drug accumulation in brain tissues compared to unconjugated paclitaxel. The enhanced delivery is attributed to LRP1-mediated transcytosis, which remains unaffected by the P-glycoprotein (P-gp) efflux system that typically limits CNS

drug exposure. Both in vivo and in vitro studies confirmed that Angiopep2 displays superior transport properties compared to other ligands, such as transferrin or lactoferrin [50, 51].

The versatility of Angiopep2 has also been demonstrated in various disease models. For instance, in epilepsy, Angiopep2 was conjugated to electro-responsive hydrogel nanoparticles for the delivery of phenytoin sodium. These nanoparticles can release the drug in response to pathological EEG signals, thus targeting seizures with precise timing and location. The Angiopep2 modification enabled efficient BBB penetration and delivery to areas of seizure activity, significantly improving seizure control compared to traditional therapies [48].

In the context of Parkinson's disease, Angiopep2 was used to enhance dendrigraft poly-L-lysine (DGL)-based gene vectors. These nanoparticles carried the gene for glial cell line-derived neurotrophic factor (GDNF). They showed better brain targeting, increased gene expression in neuronal tissue, and neuroprotective effects in a chronic rotenone-induced model. This approach offers a non-viral alternative for sustained gene therapy, utilizing the receptor-mediated pathway of Angiopep2 for effective CNS delivery [52].

Angiopep2 has also shown promise in treating CNS infections. In a model of cryptococcal meningoencephalitis, amphotericin B encapsulated in Angiopep2-conjugated polymeric micelles achieved significantly higher drug concentrations in the brain, reduced fungal burden, and improved survival rates. These findings underscore the potential of Angiopep2 to enhance the efficacy of hydrophobic antifungals in brain infections where the BBB remains mostly intact [53].

Furthermore, Angiopep2 is being actively researched for the treatment of glioblastoma. A recent study investigated the impact of Angiopep2 density on polymeric nanoparticles' ability to penetrate the BBB, revealing a complex and model-dependent relationship.

While higher densities improved binding to endothelial cells in static 2D cultures, they hindered transport across Transwell BBB models. In contrast, dynamic BBB-on-a-chip systems that replicate physiological flow conditions showed enhanced transport with optimized Angiopep2 density. In vivo data supported these observations, indicating that moderate surface density strikes a balance between effective receptor engagement and successful transcytosis. The resulting nanoparticles loaded with doxorubicin showed increased accumulation in brain tumors and improved antitumor efficacy [54].

The molecular interaction between Angiopep2 and LRP1 has been characterized and involves strong, site-specific binding energies, particularly to the CR5-6 domain. This interaction mimics how LRP1 engages with receptor-associated proteins. Notably, the binding site of Angiopep2 facilitates chemical conjugation without compromising its interaction with LRP1, making it suitable for use in nanoparticle functionalization and drug conjugates [49].

In conclusion, Angiopep2 is a highly effective BBB shuttle that utilizes the receptor-mediated transport properties of LRP1. Its dual functionality, crossing the BBB and targeting brain tumors or diseased cells, establishes it as a versatile platform for CNS drug delivery.

7 Objectives of this Thesis

This thesis aims to bridge that gap through a multiscale computational framework that spans different levels of organization. The main objectives are as follows:

First, to reconstruct the three-dimensional structure of LRP1 using state-of-the-art prediction methods, generating models of both the monomeric and dimeric states. These models provide the foundation for studying conformational dynamics, with special

emphasis on the flexibility of calcium-binding motifs, the stability of the furin cleavage site, and the contribution of glycosylation to interdomain and intermonomer interactions.

Second, to quantify energetic contributions that govern receptor stability and function. This includes assessing calcium binding within conserved acidic cages, evaluating the noncovalent interactions stabilizing α- and β-fragments after furin cleavage, and analyzing how glycans shape the free energy landscape of the receptor.

Third, to investigate ligand recognition at multiple scales. Global and local docking analyses of angiopep2 were carried out to reveal extended binding regions, shaped by both motif architecture and solvent accessibility, rather than isolated pockets. The concept of multivalent avidity is examined as a key principle for effective transcytosis.

Fourth, to characterize hydration as a determinant of both protein flexibility and ligand binding. A novel correction of radial distribution functions and iterative solutions of the Ornstein-Zernike closure were developed to measure residue-water interaction potentials. These data were used to build free-energy maps, extend the analysis beyond conserved residues, and estimate desolvation enthalpies that define the energetic barriers for docking poses. In this way, hydration links directly to the energetic stability of the receptor and to the kinetics of ligand recognition.

Finally, to extend the study to therapeutic nanoparticles that exploit LRP1 for delivery. Coarse-grained simulations of PEG-PLA micelles were used to reproduce their self-assembly and morphology. Their electrostatic properties were then analyzed through zeta potential calculations, testing how chemical modifications and spontaneous hydrolysis alter surface charge and colloidal stability, in agreement with experimental trends.

Taken together, these objectives provide an integrated picture of how structural motifs, solvent interactions, post-translational modifications, ligand binding, and nanoparticle

electrostatics converge to define the function of LRP1 and its potential exploitation in brain drug delivery.

Chapter 2

Methodologies

1 Structure prediction

Protein structure prediction seeks to determine the three-dimensional arrangement of atoms in a protein based on its amino acid sequence. This area of research is grounded in the principle that a protein's native structure is encoded within its primary sequence under physiological conditions, a concept first proposed by Anfinsen in 1973 [55]. Although experimental techniques such as X-ray crystallography, cryo-electron microscopy, and nuclear magnetic resonance spectroscopy provide high-resolution structural data, these methods can be expensive, technically challenging, and may not be feasible for every protein target. As a result, computational approaches have become essential in modern structural biology [56].

1.1 Template-Based Modeling and MODELLER

Template-based modeling leverages the evolutionary conservation of protein structure to infer the conformation of an unknown protein, referred to as the target. This is achieved

by aligning the target with one or more proteins of known structure, called templates, and transferring spatial information from the templates to the target. Homology modeling, the most established form of template-based modeling, achieves high accuracy when the sequence identity between the target and template exceeds approximately 30% [57, 58].

MODELLER is among the most widely utilized tools for this purpose. It uses a restraint-based approach, representing the target-template relationship through a set of spatial constraints derived from sequence alignments and statistical observations from known structures [57]. These constraints encompass distances, angles, dihedral preferences, hydrogen bonding patterns, and stereochemical rules.

The MODELLER workflow begins with the identification of appropriate templates using sequence search tools like BLAST, HHsearch, or PSI-BLAST. Multiple templates can be selected to capture different regions of similarity. The alignments between the target and templates define the spatial restraints, which MODELLER optimizes using a combination of conjugate gradient minimization and simulated annealing molecular dynamics [59]. The software can model loops that are absent from the template, refine sidechain conformations, and incorporate experimental restraints such as NMR NOE distances or cross-linking data.

MODELLER offers a Python interface that provides substantial flexibility, allowing for automation, custom scoring, and hybrid modeling strategies [60]. The method is computationally efficient, generating multiple alternative models that can be ranked according to both internal and external quality assessment criteria. While MODELLER achieves high accuracy with sufficiently similar quality templates, its performance declines significantly when sequence identity falls below approximately 25%. Additionally, this method is not designed to predict novel protein folds that are not represented in existing template databases.

1.2 Deep Learning–Based Prediction

In recent years, deep learning has revolutionized protein structure prediction, transitioning from a modular, stepwise approach to an end-to-end differentiable framework where a neural network directly predicts atomic coordinates from the input sequence and relevant features. This shift has been driven by advancements in model architecture and the availability of large-scale structural and sequence datasets [56, 61].

AlphaFold2, built by DeepMind, revolutionized the field during the CASP14 competition [62], achieving median GDT-TS scores above 90 for many targets and nearing experimental resolution [63, 61]. Its architecture includes several innovations, such as the Evoformer module, which processes both a multiple sequence alignment representation that captures evolutionary relationships and a pairwise residue representation that encodes inter-residue geometries. Information is exchanged between these streams through attention mechanisms, allowing the integration of co-evolutionary signals with geometric context. The Invariant Point Attention system enables the network to reason directly in three-dimensional space, independent of coordinate frames. Additionally, a recycling strategy where the model feeds its predictions back into the input for refinement leads to progressively improved accuracy. Unlike earlier methods that relied on predicting intermediate features for a separate modeling phase, AlphaFold2 optimizes the entire pipeline against experimentally determined structures in a single, end-to-end training process. It excels in single-chain domains and many multi-domain proteins, particularly when deep multiple sequence alignments are available. However, it requires significant computational resources, is less reliable for intrinsically disordered regions, and has diminished accuracy for sequences with limited evolutionary information.

Building on AlphaFold2, AlphaFold3 [64, 61] expands its capabilities to multi-chain complexes, protein–nucleic acid assemblies, and protein–ligand interactions. It enhances handling of multiple sequence alignments, making it more effective in situations with

few homologous sequences. Its training encompasses a variety of structural contexts, including bound and unbound conformational pairs, which improves modeling of flexible interfaces. Refinements in attention mechanisms also enhance its ability to capture long-range inter-chain dependencies, making AlphaFold3 more robust for complex systems and allowing it to address a broader range of biological questions.

RoseTTAFold, developed by the Baker Laboratory, takes a different yet conceptually related approach [65, 56]. It employs a three-track network where one-dimensional sequence features, two-dimensional geometric constraints (such as predicted distances and orientations), and three-dimensional coordinates are updated in parallel. This design enables simultaneous refinement across sequence, geometry, and structure spaces, facilitating continuous information exchange among the tracks. RoseTTAFold is more computationally efficient than AlphaFold2, generating structures more quickly and requiring less hardware. It performs well for both single-chain and multi-chain systems and is particularly adept at modeling large assemblies. Additionally, it has been adapted for protein design applications where iterative optimization of sequence and structure is necessary. While its accuracy for single-chain targets is generally slightly lower than that of AlphaFold2, it remains competitive and excels in scenarios where computational efficiency and flexibility are critical.

1.3 Comparative Perspective

MODELLER, AlphaFold2, AlphaFold3, and RoseTTAFold each represent unique approaches in the protein structure prediction landscape. MODELLER excels in interpretability, speed, and seamless integration with experimental restraints, making it the tool of choice when high-quality templates are available. AlphaFold2 established a new standard for accuracy in single-chain and domain-level predictions, especially when extensive evolutionary data is accessible. AlphaFold3 broadens the framework

to complex, multi-component systems and enhances robustness in contexts with low homology. RoseTTAFold provides a more computationally light yet adaptable option, well-suited for large assemblies and design scenarios. The future of protein structure prediction may likely incorporate hybrid pipelines that leverage both the interpretability and constraint-handling strengths of template-based modeling, as well as the generalized capabilities of deep learning, further enhanced by physics-based refinement to yield models that are both accurate and mechanistically informative [56, 61, 63, 65].

2 Molecular dynamics

Molecular dynamics (MD) simulation is a computational method used to model and predict the temporal evolution of molecular systems at both atomistic and coarse-grained resolutions. By applying the principles of classical mechanics, MD integrates Newton's equations of motion to obtain trajectories of particles that evolve according to the forces derived from a predefined potential energy function. This enables the study of structural, dynamical, and thermodynamical properties of biomolecules with a temporal resolution down to the femtosecond scale, providing a bridge between theoretical models and experimental observations [66, 67].

At its conceptual foundation, molecular dynamics represents a simplification of quantum mechanics: instead of solving the time-dependent Schrödinger equation for the electronic wavefunction, which is computationally intractable for large systems, MD replaces the quantum potential energy surface with a classical, empirical potential that approximates interatomic forces. This approximation allows efficient sampling of atomic motions while retaining a physically meaningful description of molecular energetics.

In contrast to static structural techniques, such as X-ray crystallography, cryo-electron microscopy, or nuclear magnetic resonance, MD provides a dynamic, time-resolved view

of molecular processes, allowing for the direct observation of conformational transitions, binding events, and fluctuations that underpin function.

2.1 Fundamental principles

At its core, molecular dynamics (MD) simulation applies the laws of classical mechanics to a system of N interacting particles, usually atoms or coarse-grained sites, to predict their time evolution under a chosen potential energy function. Each particle i of mass m_i is characterized by its position vector $\vec{r}_i(t)$ and velocity $\vec{v}_i(t)$. Newton's second law governs the motion of each particle:

$$m_i \frac{d^2\vec{r}_i}{dt^2} = \vec{F}_i(t) \tag{2.1}$$

$$\vec{F}_i(t) = -\nabla_{\vec{r}_i} U(\vec{r}_1, \vec{r}_2, ..., \vec{r}_N) \tag{2.2}$$

where U is the total potential energy of the system, and $\vec{F}_i$ is the net force acting on particle i at time t [66, 67].

A molecular mechanics force field defines the potential energy U. For an all-atom biomolecular simulation, the general form of the force field can be written as:

$$U_{total} = U_{bonded} + U_{non-bonded} \tag{2.3}$$

$$U_{bonded} = \sum_{bonds} k_b(r - r_0)^2 + \sum_{angles} k_\theta(\theta - \theta_0)^2 + \sum_{dihedrals} k_\phi[1 + \cos(n\phi - \delta)] \tag{2.4}$$

$$U_{non-bonded} = \sum_{i<j} \left[4\varepsilon_{ij} \left(\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^{6} \right) + \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} \right] \tag{2.5}$$

Here:

- k_b, k_θ, and k_ϕ are force constants for bond stretching, angle bending, and dihedral torsions, respectively;
- r_0 and θ_0 are equilibrium bond lengths and angles;
- n and δ are the multiplicity and phase shift for the dihedral angles;
- ε_{ij} and σ_{ij} are Lennard–Jones parameters for van der Waals interactions;
- q_i and q_j are atomic partial charges; ε_0 is the vacuum permittivity.

Once forces are computed, the equations of motion are integrated numerically in discrete time steps Δt. Commonly used algorithms include velocity-Verlet:

$$\vec{r}_i(t+\Delta t) = \vec{r}_i(t) + \vec{v}_i(t)\Delta t + \frac{\vec{F}_i(t)}{2m_i}(\Delta t)^2 \tag{2.6}$$

$$\vec{v}_i(t+\Delta t) = \vec{v}_i(t) + \frac{\vec{F}_i(t) + \vec{F}_i(t+\Delta t)}{2m_i}\Delta t \tag{2.7}$$

The velocity-Verlet scheme is symplectic and time-reversible, which helps maintain long-term energy stability in microcanonical (NVE) simulations. In practical biomolecular MD, the system is rarely simulated under NVE conditions alone. Instead, statistical mechanical ensembles are enforced through additional algorithms:

- NVT ensemble (canonical ensemble) uses thermostat algorithms to control the temperature T;
- NPT ensemble (isothermal-isobaric ensemble) add barostats to also control the pressure P.

The initial conditions, coordinates, and velocities are equally critical. Coordinates are typically derived from experimentally determined structures (Protein Data Bank) or homology models. Velocities are assigned from the Maxwell–Boltzmann distribution:

$$P(v_{i\alpha}) \propto \exp\left(-\frac{m_i v_{i\alpha}^2}{2k_B T}\right)$$

where $v_{i\alpha}$ is the velocity component in the α direction, k_B is Boltzmann's constant, and T is the target temperature.

2.2 Simulation ensembles and boundary conditions

Molecular dynamics simulations are performed within statistical mechanical ensembles that control the thermodynamic conditions under which the system evolves. The choice of ensemble determines how temperature, pressure, and volume fluctuate during the simulation and reflects the physical environment being modeled. In practice, biomolecular simulations rely primarily on the canonical (NVT) and isothermal–isobaric (NPT) ensembles, which require coupling schemes to maintain the desired temperature and pressure.

In the NVT ensemble, the number of particles, the volume, and the temperature are kept constant. Temperature control is achieved by coupling the system to an external heat bath, typically through Nosé–Hoover [68, 69], Langevin [70], or velocity-rescaling [71] thermostats. These algorithms regulate the kinetic energy of the particles to maintain thermal equilibrium while preserving the correct canonical distribution. The NVT ensemble is particularly useful during equilibration stages or when exploring conformational changes at fixed volume.

In the NPT ensemble, the number of particles, the pressure, and the temperature are conserved. Pressure is regulated by barostat algorithms such as Berendsen [72] or

Parrinello–Rahman [73], which rescale the simulation box to achieve the target pressure. This ensemble is commonly used for production simulations of solvated biomolecules because it reproduces conditions close to experimental measurements and allows the system to adopt its equilibrium density.

Most biomolecular simulations employ periodic boundary conditions, which eliminate surface effects by replicating the simulation cell in all spatial directions [74]. When a particle exits one side of the box, its periodic image enters from the opposite side, creating the illusion of an infinite bulk environment. This approach avoids artificial confinement and enables the use of long-range electrostatic methods such as particle mesh Ewald, which rely on translational symmetry. Periodic boundary conditions also ensure that solvent molecules form a continuous environment around the solute, maintaining uniform density throughout the simulation.

The efficient treatment of non-bonded interactions is central to the stability and accuracy of molecular dynamics. Short-range interactions, such as van der Waals forces and near-field electrostatics, are evaluated within a cutoff radius beyond which their contributions become negligible [75]. To avoid discontinuities at the cutoff, switching or smoothing functions are often applied. Neighbor lists are used to track atom pairs located within or near this radius, allowing the simulation engine to update interactions efficiently and reduce computational cost by avoiding unnecessary pair evaluations [76].

Long-range electrostatics require a more sophisticated approach, because Coulomb interactions decay slowly and cannot be truncated without significant artifacts. The particle mesh Ewald (PME) method [77] addresses this challenge by splitting the electrostatic potential into a short-range component computed in real space and a long-range component evaluated in reciprocal space using fast Fourier transforms. This decomposition enables accurate treatment of electrostatics under periodic boundary conditions while maintaining computational efficiency close to linear scaling. PME has

become the standard method in biomolecular simulations due to its balance between precision and performance.

Together, the choice of ensemble, the application of periodic boundary conditions, and the algorithms used to evaluate non-bonded interactions define the thermodynamic and spatial context of a molecular simulation. They ensure that the system samples physically meaningful configurations while avoiding finite-size artifacts and inaccuracies in electrostatic interactions. These considerations are essential for generating equilibrated trajectories that can be reliably analyzed in the subsequent sections of this thesis.

2.3 Classical Force Fields

As described in Section 2.1, molecular dynamics employs an additive potential energy function that replaces the quantum-mechanical description of electrons with a parametrized, empirical form. Building on this classical approximation, modern biomolecular force fields differ mainly in the physical philosophy and the strategy adopted for parameter optimization.

Two principal families dominate the field: the CHARMM [78] and AMBER [79] force fields. Both rely on non-polarizable additive potentials (with the form of Eq. 2.3, 2.4 and 2.5) in which atomic charges are fixed and polarization effects are implicitly included through parametrization. However, they diverge in the data sources and validation targets used to derive parameters.

- CHARMM follows an ensemble-based philosophy. Parameters are refined to reproduce condensed-phase observables—such as crystal structures, NMR data, and thermodynamic properties—ensuring that the balance between intramolecular and solvent-mediated interactions remains physically realistic.

- AMBER, conversely, adopts a quantum-derived philosophy. Parameters are fitted primarily to high-level quantum mechanical calculations on small molecules, using restrained electrostatic potentials and torsional scans, followed by limited empirical correction to match experimental data.

This distinction reflects two complementary objectives: CHARMM seeks to capture the average conformational behavior observed experimentally, while AMBER emphasizes microscopic accuracy grounded in quantum reference data.

The CHARMM family and CHARMM36m

The CHARMM (Chemistry at HARvard Macromolecular Mechanics) force field family was originally designed to reproduce condensed-phase properties of biomolecules, including hydration, folding, and association equilibria. Its parameter optimization integrates experimental and theoretical inputs, iteratively adjusting bonded and non-bonded terms to reproduce both structural and thermodynamic observables.

The version employed in this work, CHARMM36m [80], is a refinement that improves backbone dihedral potentials through CMAP corrections and updates side-chain parameters to reproduce solution-state conformational ensembles. It achieves a balanced representation of folded and intrinsically disordered proteins, making it particularly suitable for modular systems such as LRP1, where both compact and extended states coexist. The force field was specifically tuned for compatibility with the CHARMM-modified TIP3P water model, ensuring coherent treatment of solute-solvent interactions and accurate reproduction of experimental hydration thermodynamics.

The AMBER family and ff14SB

The AMBER (Assisted Model Building with Energy Refinement) force field family originated from the idea that accurate quantum-mechanical reference data could be

transferred to macromolecular simulations through well-defined small-molecule analogues. Bonded terms were derived from high-level torsional scans, while atomic partial charges were obtained from restrained electrostatic potential (RESP) fitting at the HF/6-31G* level. This approach provides internal physical coherence and high accuracy for local geometries and energetics.

The ff14SB parameter set [81] represents a substantial revision of the ff99SB lineage. It refines both backbone and side-chain torsions to improve agreement with experimental scalar couplings and crystallographic data, while maintaining numerical stability and transferability. In this work, ff14SB was used during the initial energy minimization and relaxation steps of the modeled domains, where its robust local energetics ensured smooth convergence prior to explicit solvation.

Water Models

The description of solvent is a defining element of any molecular dynamics simulation. In this work, two complementary representations were adopted: the explicit TIP3P water model and the implicit GBn2 continuum model.

The TIP3P model [82] represents water as three interaction sites, each carrying fixed partial charges. Here, Lennard–Jones parameters are also assigned to the hydrogen atoms, improving the reproduction of density, dielectric constant, and diffusion properties while maintaining computational efficiency. Its compatibility with CHARMM36m ensures a consistent description of hydration structure, ion coordination, and hydrogen-bond dynamics.

The Generalized Born GBn2 model [83] provides an implicit treatment of solvation by approximating the solvent as a continuous dielectric medium. It estimates electrostatic screening via pairwise descreening and atomic “neck” corrections that improve agreement with Poisson–Boltzmann reference energies. GBn2 accelerates equilibration and geome-

try optimization steps, offering a computationally efficient alternative when explicit solvent is unnecessary or inaccessible.

Together, these models describe solvent polarization and dielectric response at different levels of resolution: explicit water resolves molecular detail and hydrogen bonding, whereas implicit solvation provides averaged electrostatic effects for rapid relaxation.

Beyond atomistic frameworks, further simplification is achieved by coarse-graining, where groups of atoms are represented as interaction sites to extend spatial and temporal scales.

2.4 Coarse-Grain MD simulations

Coarse-grained (CG) molecular dynamics simplifies atomic detail by grouping atoms into larger interaction sites, known as beads. This approach smooths the potential energy surface and reduces the number of degrees of freedom, allowing for simulations over much longer timescales and larger system sizes compared to all-atom models. Although this method sacrifices atomic resolution, coarse-graining facilitates the investigation of mesoscale organization, significant conformational changes, and collective dynamics that are computationally challenging for fully atomistic descriptions [84].

Among the various CG models available, the Martini force field is one of the most widely utilized. Initially developed for lipid membranes [85], the Martini model has since been expanded to include proteins, nucleic acids, carbohydrates, polymers, nanoparticles, and hybrid systems [86]. Its design is based on a building-block approach, where small chemical fragments are mapped to standard bead types based on polarity, size, and hydrogen-bonding capability. Nonbonded interactions between beads are described using Lennard-Jones and Coulombic potentials, with parameters calibrated against experimental thermodynamic data, such as partition free energies and reference atomistic simulations.

Over time, Martini has developed into a general-purpose CG framework. The latest version, Martini 3 [87], features an expanded bead set that covers a broader range of chemical space, refined mapping rules that consider size and shape, and a reparameterized interaction matrix to address overstabilization ("stickiness") issues identified in Martini 2. For proteins, Martini 3 enhances side-chain packing, improves the representation of aromatic residues, and increases compatibility with elastic network models or Gō-like potentials that control backbone flexibility [88].

The computational efficiency of Martini arises from its reduced particle count and smoother potential energy landscape, enabling time steps of up to 20–30 fs, which allows for simulations on the microsecond to millisecond scale [89]. These features make it a standard tool for investigating membrane organization, protein-lipid interactions, nanoparticle self-assembly, and supramolecular complexes [86, 88].

However, the coarse-grained representation inevitably loses fine structural detail, which can distort the enthalpy-entropy balance and limit the accuracy of highly directional interactions, such as hydrogen bonding or π–π stacking [89].

Beyond biomolecules, Martini has also been adapted for synthetic and biological polymers [90]. In this context, monomer units are mapped to beads based on their chemical functionality, polarity, and size, with parameters adjusted to reproduce bulk properties like density, radius of gyration, and solubility parameters. The transferable nature of Martini bead types enables the efficient modeling of copolymers, block copolymers, and polymer blends without requiring full reparameterization. This coarse-grained framework enables the simulation of polymer morphologies over hundreds of nanometers and microseconds, making it possible to study self-assembly, phase separation, and mechanical behavior. Applications in this area include modeling polymer-membrane interactions, drug delivery vehicles, and stimuli-responsive materials. The

expanded bead library in Martini 3 further enhances the description of aromatic, branched, and charged moieties commonly found in functional polymers.

Despite its versatility, CG polymer models face the challenge of accurately capturing chain flexibility and persistence length. In Martini, this is addressed by adjusting bonded parameters (bonds, angles, and dihedrals) as well as nonbonded interactions. For semi-rigid or rod-like polymers, additional constraints or elastic networks are often introduced to enhance their properties. Multiscale workflows are commonly employed: Martini simulations capture mesoscale structure and dynamics, while backmapping to atomistic resolution allows for a detailed assessment of mechanical, spectroscopic, and chemical properties.

In summary, Martini offers a computationally efficient coarse-grained framework applicable to a wide range of biomolecular and soft-matter systems. Its ongoing development, particularly with Martini 3, highlights a growing emphasis on chemical specificity and accuracy, while maintaining the scalability that contributes to its popularity in both biological and materials science simulations.

2.5 Force-field combinations used in this work

The methodological framework adopted in this work integrates multiple levels of molecular resolution, each governed by force fields optimized for different physical regimes. The workflow follows a hierarchical logic that links high-resolution, quantum-derived atomistic models to coarse-grained representations capable of describing mesoscale organization. Such integration ensures that each simulation scale preserves internal consistency while capturing the dominant energetic and structural features relevant to the system.

At the atomistic level, energy minimization of the LRP1 models was performed using the AMBER ff14SB parameter set in implicit solvent, which provides accurate local

geometry and efficient electrostatic screening. Production simulations employed the CHARMM36m force field in explicit TIP3P water, ensuring a balanced representation of folded and intrinsically disordered regions. This combination reproduces both local structural stability and solvent-mediated conformational dynamics with high fidelity. For mesoscale analyses, the Martini 3 coarse-grained model was used to investigate polymeric self-assembly, enabling the sampling of collective motions on the microsecond timescale. Taken together, these methodological combinations established a coherent multiscale framework capable of capturing the atomistic determinants of LRP1 flexibility and the emergent mesoscale organization of polymeric assemblies, bridging chemical precision with large-scale physical behavior.

3 Umbrella Sampling

Having established the theoretical and practical framework of molecular dynamics simulations, we now describe the enhanced-sampling approach used to explore rare conformational events, which typically correspond to high-energy or transition states [91]. It is particularly suited for calculating the free energy profile along a chosen reaction coordinate (or collective variable, see Section 3.3), known as the potential of mean force (PMF).

3.1 Theoretical Background

In standard MD, rare events may not be observed due to high energy barriers, limiting the estimation of equilibrium properties such as free energy differences. Umbrella sampling overcomes this limitation by introducing a biasing potential that enhances sampling in specific regions of the reaction coordinate [91, 92].

Let $q(\mathbf{r})$ be a reaction coordinate, defined as a function of atomic coordinates $\mathbf{r}$. The free energy profile along q is related to the probability distribution $P(q)$ through:

$$A(q) = -k_B T \ln P(q) + C \tag{2.8}$$

where $A(q)$ is the PMF, k_B is Boltzmann's constant, T is the absolute temperature, and C is an arbitrary constant.

To enhance sampling, an external biasing potential is applied. Typically, this takes the form of a harmonic restraint:

$$U_{\text{bias}}(q) = \frac{k}{2}(q - q_0)^2 \tag{2.9}$$

where k is the force constant and q_0 is the center of the restraint. The total potential energy during the biased simulation is:

$$U_{\text{total}}(\mathbf{r}) = U(\mathbf{r}) + U_{\text{bias}}(q(\mathbf{r})) \tag{2.10}$$

Multiple such biased simulations (or "windows") are performed at different values of q_0, ensuring sufficient overlap of sampled regions.

3.2 Free Energy Reconstruction with WHAM

The Weighted Histogram Analysis Method (WHAM) is commonly used to reconstruct the unbiased PMF from multiple biased simulations [93]. The probability distribution $P(q)$ is estimated using:

$$P(q) = \frac{\sum_{i=1}^{N} n_i^{-1} H_i(q)}{\sum_{i=1}^{N} n_i^{-1} e^{-\beta(U_{\text{bias}}^i(q) - f_i)}} \tag{2.11}$$

$$e^{-\beta f_i} = \int dq P(q) e^{-\beta U^i_{\text{bias}}(q)} \tag{2.12}$$

where:

- $H_i(q)$ is the histogram from window i,
- n_i is the number of samples in window i,
- $U^i_{\text{bias}}(q)$ is the biasing potential in window i,
- f_i is the free energy offset for window i,
- $\beta = \frac{1}{k_B T}$.

Once $P(q)$ is obtained, the PMF is recovered as:

$$A(q) = -k_B T \ln P(q) + C \tag{2.13}$$

3.3 Collective variables in the umbrella sampling

Enhanced-sampling techniques accelerate the exploration of configurational space by applying external biasing potentials along selected collective variables (CVs), or order parameters, that describe the slow modes of a molecular process. The success of these methods depends on identifying low-dimensional variables that capture the essential physics of the transition while preserving a thermodynamically meaningful connection to the underlying free-energy surface.

In umbrella sampling and related enhanced approaches (for example, metadynamics [94, 95]), the accuracy and convergence of the reconstructed free-energy profile depend critically on the definition of the CV. An appropriate variable must capture the dominant slow degrees of freedom governing the transition of interest while remaining sufficiently

simple to be exhaustively sampled in each biased window. If the chosen coordinate fails to represent the true reaction pathway, unaccounted orthogonal motions can hinder overlap between sampling windows, introducing hysteresis and slow convergence. Conversely, physically meaning- ful coordinates, such as distances, dihedral angles, coordination numbers, or composite path-based descriptors, provide continuous sampling along the transition path and allow the free-energy estimator to recover a smooth and thermodynamically consistent potential of mean force. Careful identification and validation of CVs are therefore essential prerequisites for reliable free-energy calculations.

In this work, the definition of the collective variables was guided by the structural and mechanistic features of the LRP1 domains. For the separation of the α and β chains of the β-propeller domain, the interchain distance was selected as the reaction coordinate, as it directly quantifies the slow structural transition associated with cleavage and mechanical dissociation. For calcium release, the distance between the ion and the centroid of its coordinating acidic residues was used, representing the primary coordinate governing binding and unbinding. Both variables isolate the key motions responsible for the observed transitions while minimizing coupling to orthogonal degrees of freedom, ensuring adequate overlap between biasing windows and efficient convergence of the potential of mean force. These definitions illustrate the principle that physical interpretability and kinetic relevance of the chosen CVs are decisive for the reliability of biased-sampling simulations.

4 Molecular Docking

4.1 Global docking using HPEPDOCK

HPEPDOCK [96] is a computational method developed for the blind docking of peptides to proteins, explicitly accommodating the intrinsic flexibility of peptide chains. The approach operates through a hierarchical pipeline that systematically refines peptide-protein complexes, transitioning from coarse, global searches to fully flexible, energy-minimized models. Notably, this methodology is adept at predicting binding modes without requiring prior knowledge of the binding site.

The process commences with the preparation of input structures. Users can provide the receptor and peptide in PDB format or supply the peptide sequence alone. In cases where only the peptide sequence is provided, initial three-dimensional conformations of the peptide are constructed utilizing backbone modeling tools such as MODRefiner. To effectively address receptor flexibility, HPEPDOCK can generate an ensemble of receptor conformations derived either from experimentally resolved structures or through computational approaches, such as normal mode analysis.

The docking procedure is organized into three primary stages. The first stage involves a rigid-body search in which thousands of peptide conformations are systematically evaluated across the receptor surface. This is executed using a fast Fourier transform-based algorithm, which adeptly assesses shape complementarity and preliminary interaction energies. In the subsequent stage, the top-scoring candidates undergo a coarse refinement process, during which limited backbone adjustments are made to the peptide to alleviate steric clashes and optimize hydrogen bonding patterns. The final stage introduces full flexibility to both the peptide and key receptor residues, permitting side-chain and backbone minimization under the guidance of a composite energy function that incorporates van der Waals forces, electrostatic interactions, and statistical potentials.

Following refinement, each predicted complex is appraised using a weighted scoring function. The resulting energy is a linear combination of the van der Waals term, the electrostatic term, and a statistical term based on established peptide–protein interaction patterns. The top-scoring models are then ranked, clustered, and presented to the user, accompanied by coordinate files and visualization tools for examining the predicted binding modes.

HPEPDOCK's hierarchical strategy offers several significant advantages. By delineating the global search from the fully flexible refinement process, it effectively reduces computational demands without compromising accuracy. The inclusion of receptor ensembles ensures the accommodation of conformational changes in the target protein, while full peptide flexibility captures a comprehensive range of potential binding conformations. These features render HPEPDOCK particularly useful in contexts where structural information is limited, such as epitope mapping, peptide drug design, and the characterization of short linear motifs involved in protein–protein signaling.

4.2 Local docking using HADDOCK2.4

HADDOCK (High Ambiguity Driven protein–protein Docking) [97] is an advanced information-driven docking platform that facilitates the prediction of three-dimensional structures of biomolecular complexes. This platform integrates experimental data and bioinformatics-based interface information directly into the docking process, diverging from traditional geometric or energy-based approaches. This capability enables HADDOCK to prioritize biologically relevant interaction modes, even when partial knowledge of the binding interface is available.

The docking process initiates with the atomic coordinates of the individual binding partners, which may include proteins, nucleic acids, or other biomolecules. Users are required to provide a set of residues categorized as "active" or "passive". Active residues

are those for which there is direct experimental evidence of their participation in binding, such as residues identified through NMR chemical shift perturbations, mutagenesis, or cross-linking techniques. In contrast, passive residues are spatial neighbors to the active ones and may play an indirect role in the interaction. The designations of these residues are then translated into ambiguous interaction restraints (AIRs), which guide the docking process without rigidly constraining it.

The docking algorithm operates in three primary stages. The first stage involves rigid-body energy minimization, during which the binding partners are positioned relative to each other without altering their internal conformations. The AIRs guide this initial search, ensuring that only poses are considered that are compatible with the established interface information. In the second stage, the best solutions from the initial search undergo refinement through semi-flexible simulated annealing. This stage permits the movement of both side chains and backbone segments at the interface, allowing for the adaptation of the complex, the resolution of steric clashes, and improved accommodation between interacting surfaces. The final stage entails refinement in an explicit water environment, where solvation effects, hydrogen-bond networks, and side-chain orientations are optimized, culminating in the generation of physically realistic models of the complex.

Scoring in HADDOCK is determined by a weighted combination of van der Waals and electrostatic terms derived from molecular mechanics, alongside an empirical desolvation term and penalties for any violations of the applied restraints. The solutions are clustered based on interface root mean square deviation (RMSD), with the clusters ranked according to the average HADDOCK score of their members.

By employing a systematic integration of experimental information into its search algorithm, HADDOCK provides a robust approach for modeling biomolecular complexes that may pose significant challenges for purely ab initio docking methods. Its flexibility,

multi-stage refinement process, and capability to support a diverse array of biomolecular systems, including protein-protein, protein-DNA, and multi-component assemblies, establish HADDOCK as an essential tool in structural biology and drug discovery.

The OZ equation has various applications in liquid state theory. For instance, Baxter [98] demonstrated that for hard-sphere fluids, if the direct correlation function $c(r)$ vanishes beyond a finite range R, the OZ equation can be reformulated to involve integrals only over the range $[0, R]$. This reformulation greatly simplifies both numerical and analytical solutions, particularly within the PY framework. Baxter's method yielded an exact analytical solution for the PY closure in hard-sphere systems, serving as a benchmark in theoretical fluid studies.

To extend the OZ equation to more complex molecular systems, Blum and Torruella [99] devised a strategy utilizing invariant expansions of the two-body correlation functions. This approach is particularly efficient for nonspherical molecules, enabling the OZ equation to be expressed as a set of uncoupled matrix equations through the use of group-theoretical techniques. These advancements broaden the application of the equation to fluids with internal degrees of freedom, including rotational symmetry and anisotropic interactions like dipoles and quadrupoles, providing compact and computationally efficient formulations of correlation functions.

The study of inhomogeneous fluids, especially those in contact with surfaces, has also benefited from the OZ framework. Henderson, Abraham, and Barker [100] adapted the OZ equation for these scenarios by modeling the surface as a second species in a highly dilute binary mixture. This adaptation led to a modified OZ equation that accounts for the geometry and boundary conditions imposed by external walls or interfaces, offering insights into phenomena such as adsorption, wetting, and density oscillations near surfaces [100].

Høye and Blum [101] extended the Baxter method for fluids that interact through Yukawa potentials, applying closures that assume a form for $c(r)$ as a sum of exponentially decaying terms, appropriate for screened interactions typically found in electrolytes. They utilized Fourier and Laplace transform techniques to derive analytical and semi-analytical expressions for the pair correlation function $g(r)$. Their results not only reproduced existing findings but also provided a systematic approach for extending the method to mixtures and more complex interactions.

From a computational perspective, Kelley and Pettitt [102] introduced a fast and robust solver for the OZ equation, integrating a multilevel solver with Newton–GMRES iterations while leveraging the Atkinson–Brakhage framework. Their method is designed to work with smooth, short-range interaction potentials, such as the Lennard-Jones potential, significantly enhancing the speed and accuracy of OZ equation solutions. Their implementation outperformed traditional Picard iteration methods by a factor of 40 in specific scenarios, making it a valuable tool for analyzing fluid structure and thermodynamics.

In terms of critical phenomena, Martynov (2008) examined how closure relations influence the analytical properties of the OZ equation. He categorized closures into two types: analytical and non-analytical. Analytical closures, such as PY and HNC, are capable of determining the critical point and other thermodynamic properties of fluids. In contrast, non-analytical closures are necessary for accurately reproducing the scaling behavior and critical exponents observed in real systems [103].

5 The Ornstein-Zernike equation

The Ornstein–Zernike (OZ) equation is a fundamental concept in the statistical mechanics of fluids. It relates the total correlation function $h(r)$, which captures both direct and indirect correlations between particles, to the direct correlation function $c(r)$, which

reflects only the direct influence one particle has on another. For a homogeneous and isotropic fluid, the equation can be expressed as:

$$h(r) = c(r) + \rho \int c(r')h(|r - r'|)\, d^3r' \tag{2.14}$$

Here, $h(r) = g(r) - 1$, where $g(r)$ is the radial distribution function and ρ is the number density of the fluid. Since this equation alone does not suffice for solving the correlation functions, it must be accompanied by a closure relation. Two commonly utilized closures are the Percus–Yevick (PY) approximation and the Hypernetted Chain (HNC) approximation. The PY closure is defined as:

$$c(r) = [1 - e^{-\beta\phi(r)}]g(r) \tag{2.15}$$

while the HNC closure takes the form:

$$c(r) = h(r) - \ln g(r) - \beta\phi(r) \tag{2.16}$$

where $\phi(r)$ represents the pair potential between particles and $\beta = k_B T$.

5.1 Iterative solution of the Ornstein–Zernike equation

The RDF $g(r)$ is used to extract interaction potentials by solving the OZ equation with the HNC closure:

$$g(r) = \exp[-\beta\Phi(r) + h(r) - c(r)]. \tag{2.17}$$

.

Since both $\Phi(r)$ and $c(r)$ are unknown, we implemented an iterative solution. We first assume that

$$g(r) = \exp[-\beta U_{tot}(r)] \tag{2.18}$$

An initial potential $u_0(r)$, approximated by a Lennard–Jones form, is used to compute the indirect RDF $g_{ind_0}(r) = g(r)\exp[-\beta u_0(r)]$. The direct correlation function is then rewritten as

$$c_0(r) = g(r) - g_{ind,0}(r) \tag{2.19}$$

because $c = h - \theta$, where θ is the indirect correlation; hence $c = g - (1+\theta) \sim g - \exp(\theta)$, being θ small. Then, we identify by hypothesis $\exp(\theta) = g_{ind}$.

The potential is iteratively refined according to:

$$u_i(r) = g(r) - 1 - \ln g(r) - c_{i-1}(r). \tag{2.20}$$

Next, we calculate $g_{ind,i} = g(r)\exp[-\beta u_i(r)]$c c_i and u_{i+1} until convergence is reached with $u_i(r) \simeq u_{i-1}(r)$.

The final potential is obtained as:

$$\Phi(r) = h(r) - \ln g(r) - c_i(r), \tag{2.21}$$

expressed in units of $k_B T$.

This iterative framework enabled the decomposition of hydration free energy into direct and indirect correlation terms, identifying distinct hydration shells around conserved residues.

Chapter 3

The LRP1 structure

1 Introduction

Because the experimental structures of LRP1 are incomplete, computational modeling is crucial for exploring its architecture. Neural network (NN) predictors, such as AlphaFold2 and RoseTTAFold, have demonstrated remarkable accuracy in capturing local protein folding. However, global structural predictions of very large, modular proteins, such as LRP1, remain challenging. In this chapter, we present the construction of monomeric and dimeric models of LRP1, integrating NN-based motif predictions with homology modeling against its close homolog LRP2.

2 Monomeric LRP1 model generation

After benchmarking the performance of neural-network–based structure prediction algorithms (Appendix A), the LRP1 monomer was reconstructed by combining the individually modeled extracellular domains. The assembled architecture was refined to generate alternative conformations, capturing both extended and coiled arrangements of

the R clusters (Figures 3.1 and 3.2). Among these, the conformation shown in Figure 3.2 represents the most physically consistent model obtained after structural refinement, and this final version has been deposited in ModelArchive to ensure public accessibility and long-term preservation (https://modelarchive.org/doi/10.5452/ma-k8036).

Calcium-binding motifs were consistently occupied by Ca^{2+} ions, and disulfide bridges linked the conserved cysteine residues characteristic of the LDL receptor family. Fifty-two predicted N-linked glycosylation sites were decorated with high-mannose glycans, yielding a glycosylation pattern consistent with the expected extracellular maturation of LRP1 (Figure 3.2.b).

The resulting monomeric model reproduces the modular organisation and flexibility of LRP1, offering a structural basis for the analyses presented in the following sections.

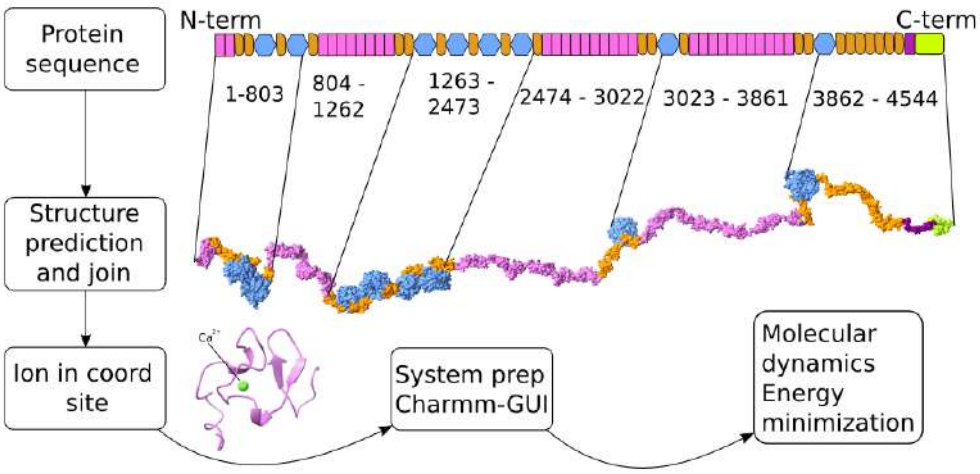


Figure 3.1: Protocol for the production of the LRP1 monomeric model. We first separated the protein sequence into six portions and generated a predicted structure for each using RoseTTAFold. The fragments were then joined and adjusted to the conformation shown here and in Figure 3.2.b. We further added calcium ions to the coordination sites of the calcium-binding motifs and finally prepared the glycoprotein for molecular dynamics simulations using the Charmm-GUI web-server [104].

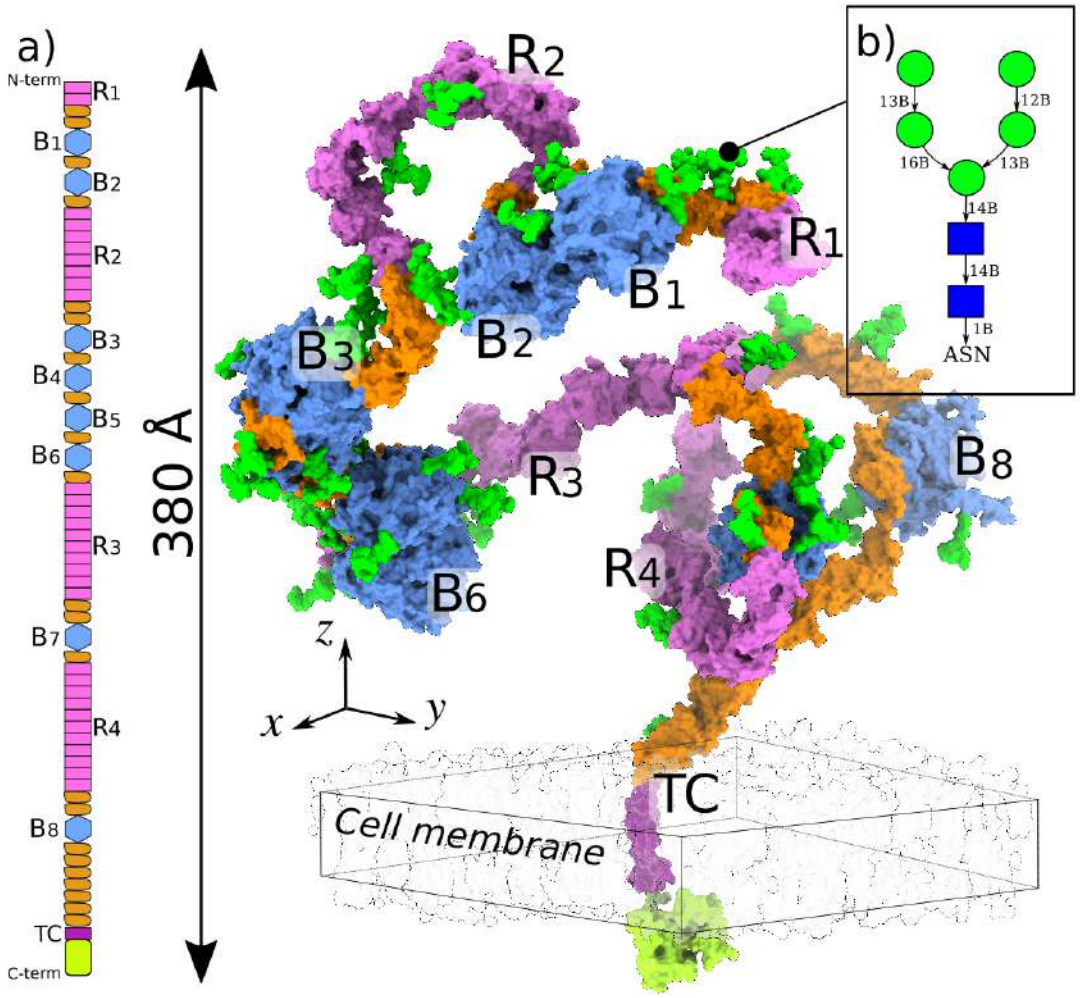


Figure 3.2: a. The proposed structure of the LRP1 monomer inserted into the cell membrane is represented by a vdW surface. Each motif follows the sequence scheme on the left. Glycans are represented in green surface. b. 2D representation of the GlcNAc(2)Man(5) glycans used in this work.

LRP1 N-glycan sites	114, 136, 185, 239, 274, 357, 446, 729, 928, 1050, 1154, 1155, 1195, 1218, 1511, 1558, 1575, 1616, 1645, 1723, 1733, 1763, 1825, 1933, 1995, 2048, 2117, 2127, 2472, 2502, 2521, 2539, 2601, 2620, 2638, 2815, 2905, 3048, 3089, 3264, 3333, 3488, 3662, 3788, 3839, 3953, 4075, 4125, 4179, 4278, 4279, 4364

Table 3.1: N-glycosylation sites (Asparagines) in LRP1.

3 Dimeric LRP1 model generation

To extend to the dimeric state, we employed the cryo-EM structure of murine LRP2 (PDB 8EM4) at pH 7.5 as a template.

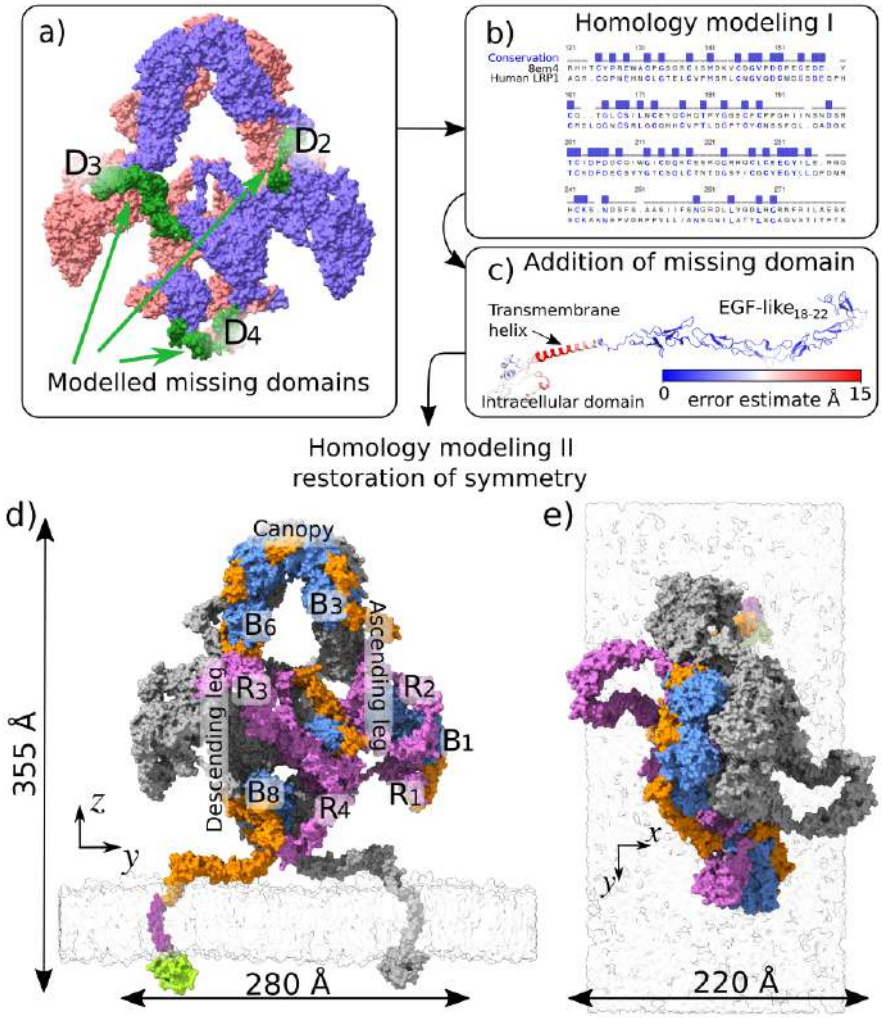


Figure 3.3: a-c. Protocol steps for the dimeric structure prediction of LRP1. The steps are explained in the text. d. Structure of the LRP1 dimer obtained by homology modelling from PDB 8EM4. The forefront monomer follows the representation in Figure 1.1, while the background monomer is in grey. Bn and Rn indicate the β-propeller and the CB domains, respectively. e. Dimer view from the z-axis perspective.

Missing regions of LRP2 were reconstructed with RoseTTAFold and AlphaFold2 (evaluation in Appendix A), then integrated into the structure (Figure 3.3.a). Homology modeling aligned LRP1 and LRP2 sequences (Figure 3.3.b), and with the addition of other missing protein portions (Figure 3.3.c) produced a complete dimeric assembly (Figures 3.3.d,e). The resulting structure reveals a bent conformation spanning $\sim 220 \times 280 \times 355$ Å³, with the β-propellers B3–B6 forming a central arc. The buried interface (Figure 3.4.a) accounts for 5951 Å² (1.23% of total surface area), with main contacts localized at the canopy and secondary interactions at the B8 interface (Figure 3.4.b). The structure is available in ModelArchive at https://modelarchive.org/doi/10.5452/ma-ubwf7.

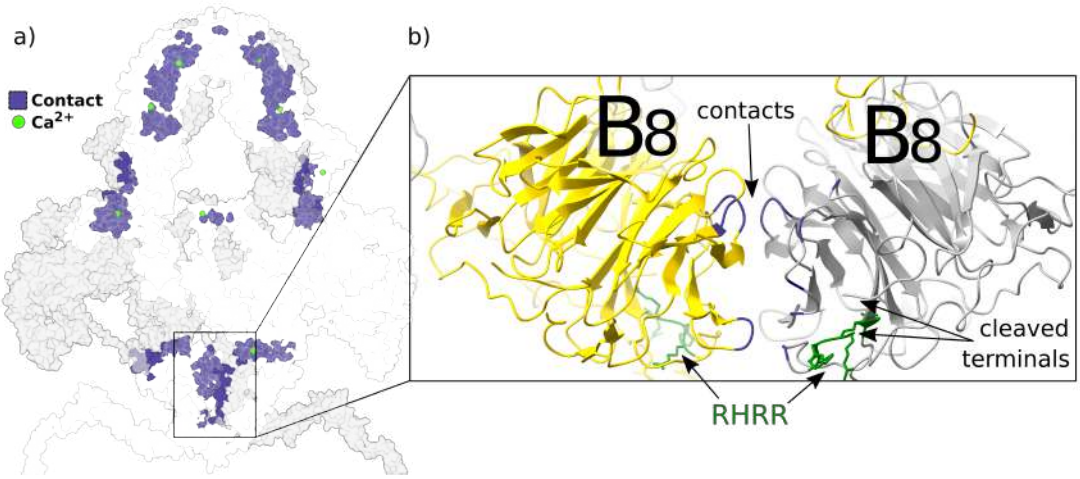


Figure 3.4: a. Dimeric LRP1 shown as vdW surface; the front monomer is transparent white and the back is light grey. The transparent surfaces allow for the visualization of hidden contact surfaces (in violet) between the two monomers. b. Zoom in on the lower intra-dimer interface between the two B8, in yellow and grey cartoon, respectively. In green, the two subsequences RHRR after furin cleavage. The interactive amino acids at the interface are colored in violet.

4 Methods

Six extended fragments of the LRP1 sequence were predicted with RoseTTAFold and assembled into a continuous structure using the Build Structure → Join Models tool of

UCSF Chimera [105]. Fragments were joined pairwise by creating new C–N peptide bonds at their termini. This operation introduces a default local geometry: the new C–N distance is set to 1.33 Å, and the associated dihedral angles assume the canonical values $\omega = 180^\circ$ and $\varphi = -120^\circ$. Only the junctional bond and dihedral parameters were defined by this procedure; all other internal coordinates were retained from the RoseTTAFold predictions.

Because automated joining occasionally placed atoms in close proximity, a small number of backbone dihedral angles near the junctions were manually adjusted to remove steric clashes and ensure chemically reasonable orientations prior to relaxation. These manual edits were minimal and restricted to local torsions.

Calcium ions were inserted using the Build Structure → Start Structure function of UCSF Chimera. For each calcium-binding motif, the ion was placed at the center of mass of the three Asp residues forming the acidic coordination cage (see Figure 1.1). Disulfide bonds were assigned between conserved cysteine pairs characteristic of the LDL receptor family.

Prediction of N-linked glycosylation sites was based on UniProt annotations, which report experimentally supported [106, 107] or computationally inferred N-glycosylation motifs following the canonical N–X–S/T sequon (evidence code ECO:0000255, indicating automated homology-based prediction rather than direct experimental validation). Fifty-two sites annotated in the LRP1 extracellular portion were included in the model. Glycans were added using the CHARMM-GUI Glycan Modeler [104], which constructs the canonical N-linked glycosidic bond between the side-chain nitrogen of each Asn residue and the anomeric carbon of the reducing-end GlcNAc unit, and generates the corresponding carbohydrate topology and parameters. The high-mannose glycoform GlcNAc(2)Man(5) was selected because it represents the conserved precursor added co-translationally in the endoplasmic reticulum, provides a chemically homogeneous

description of early-stage N-glycosylation, and avoids the structural variability introduced by hybrid or fully processed complex-type glycans.

After assembly, the complete model was energy-minimized in OpenMM 8.0 using the AMBER ff14SB and GLYCAM06j-1 force fields. All bond lengths, angles, and dihedral conformations were allowed to relax under the force-field potential, ensuring that any strain introduced during assembly was resolved. Implicit solvation was described by the GB-Neck2 model with a non-bonded cutoff of 1 nm, and calcium ions were treated with TIP3P-compatible parameters. Minimization proceeded until convergence below 1 kJ/mol.

The dimeric model was obtained using murine LRP2 (PDB 8EM4) as a structural template. Missing regions of LRP2 were predicted with RoseTTAFold and AlphaFold2 and then incorporated into the structure. Sequence alignment between LRP1 and LRP2 was performed using the Needle algorithm [108], and homology models were constructed with Modeller [109]. This step also reintroduced disulfide bonds and enforced homodimeric symmetry. The C-terminal transmembrane domain was predicted separately using RoseTTAFold and then confirmed with AlphaFold2, before being oriented to facilitate insertion into the lipid bilayer. The two domains were joined to the extracellular portion of each monomer using USFC Chimera, as described for the monomer assembly. USFC Chimera was finally employed to add calcium ions to each calcium coordination site, as already done for the monomer. In the dimer case, no glycans were added due to the extensive atomic clashes at the canopy interface. This problem is solved in section 4. The entire dimer was minimized with OpenMM using the same protocol as for the monomer.

5 Conclusions

This chapter presents an integrative structural model of LRP1 in both monomeric and dimeric states. Neural network predictors proved highly reliable for capturing local motif folding, although they remain insufficient for predicting the full-length conformation of a large, modular protein such as LRP1. The monomeric model highlights the importance of glycosylation and calcium-binding motifs in maintaining structural stability, while the dimeric model, constructed by homology to LRP2, emphasizes canopy-mediated contacts as the principal interface region, complemented by smaller interactions at the B8 motifs. Together, these models provide a robust foundation for the dynamical and functional analyses presented in the following chapters.

Chapter 4

The dynamical features of LRP1

1 Furin's shedding side belongs to a β-propeller motif

The furin enzyme cleaves LRP1 in correspondence of B8 (Figure 4.1.a), generating two chains, α and β, that are non-covalently bonded. A more detailed description of the phenomenon can be found in the Introduction. To provide information on the shedding mechanism, we investigated the stability of the noncovalent interactions, specifically the hydrophobic effective interactions between adjacent β-sheets (Figure 4.1.b,c), which hold the resulting chains together using molecular dynamics (MD) simulations.

We performed a series of umbrella sampling simulations (see General Methods) to estimate the energy required to separate the α and β chains, going from the conformation shown in Figure 4.1.b to the final conformation shown in Figure 4.1.d. The potential of mean force (PMF) shown in Figure 4.1.e reveals that the energy required to break the interactions between the two chains is approximately 180±2 kT.

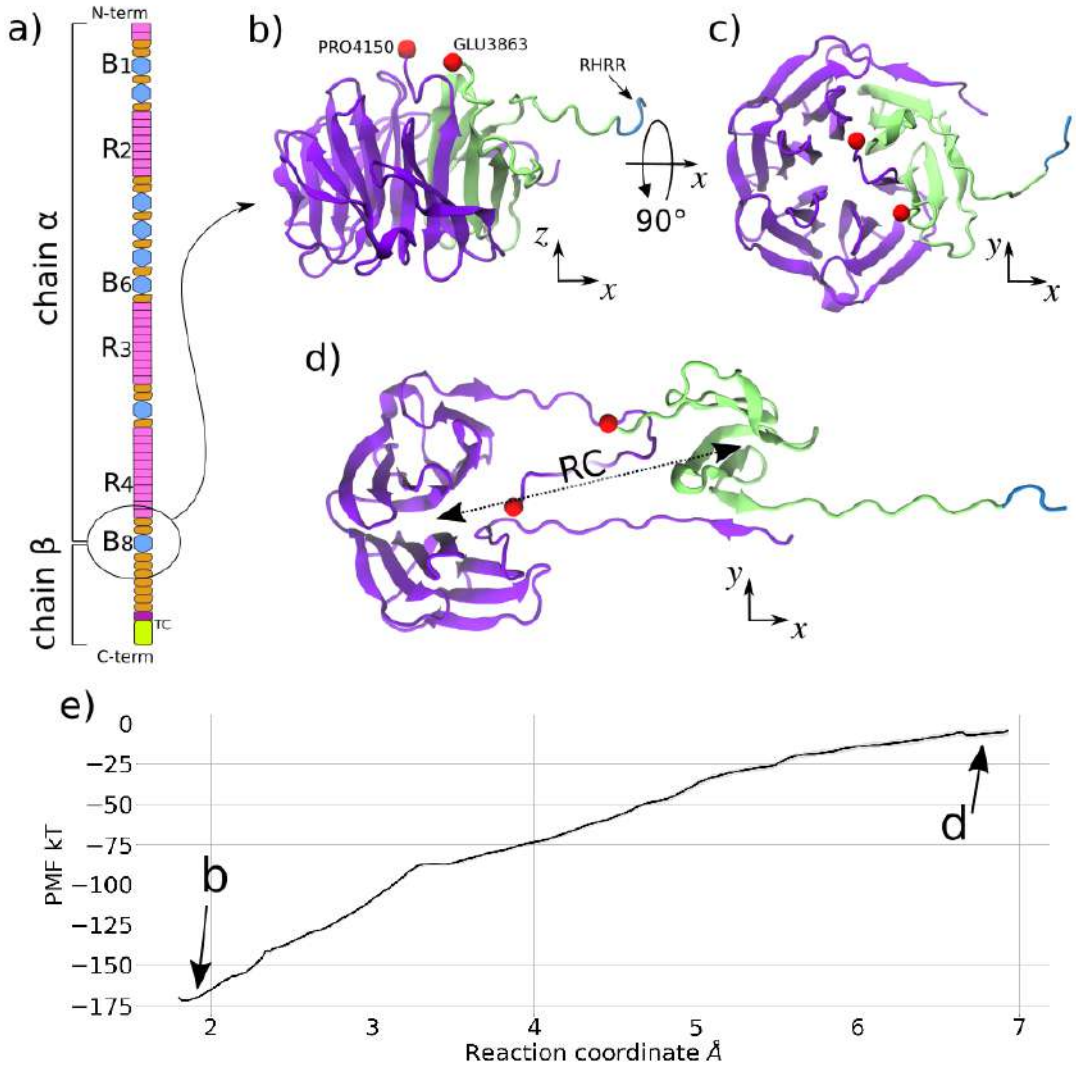


Figure 4.1: a. The scheme illustrates the location of β-propeller 8 (B8) along the LRP1 motif sequence. In accordance with this motif, the protein is composed of α and β chains. b,c. B8 structure prediction from AlphaFold2, depicted as a cartoon in lime (α chain) and violet (β-chain) with side (b) and top (c) perspectives. The RHRR sequence is depicted in blue; the extreme amino acids of the unit are depicted in red spheres. d. B8 at the extreme distance of α and β centers of mass. e. The potential of mean force for the α and β centers of mass distance.

2 Time evolution of the flexible domains

The repetition of CB motifs forms three long chains, R2, R3, and R4, as shown in Figure 1.1. The flexible behavior of the R domains was observed in the crystal structure of LRP2, where some units of the corresponding clusters have not been resolved with sufficient resolution (Figure S5). We evaluated the temporal conformational changes of the three LRP1 R clusters, isolated by the rest of the protein, with MD simulations (see General Methods).

In all systems, we observed a tendency to coil, as measured by the end-to-end distance (d_{EE}). The distributions of d_{EE} over the last 200 ns of simulations are shown in Figure 4.2.b. Here, it is visible the reduction of d_{EE} with respect to the initial conformation (dotted black vertical line). We also compared these distributions to the dimeric conformation (red vertical line, Figure 4.2.b and c).

Flexibility was further quantified by calculating the RMSF of α-carbons with respect to the initial structures, shown in Figure 4.2.d. The results highlight heterogeneous flexibility across domains, with some CB motifs more prone to large deviations than others.

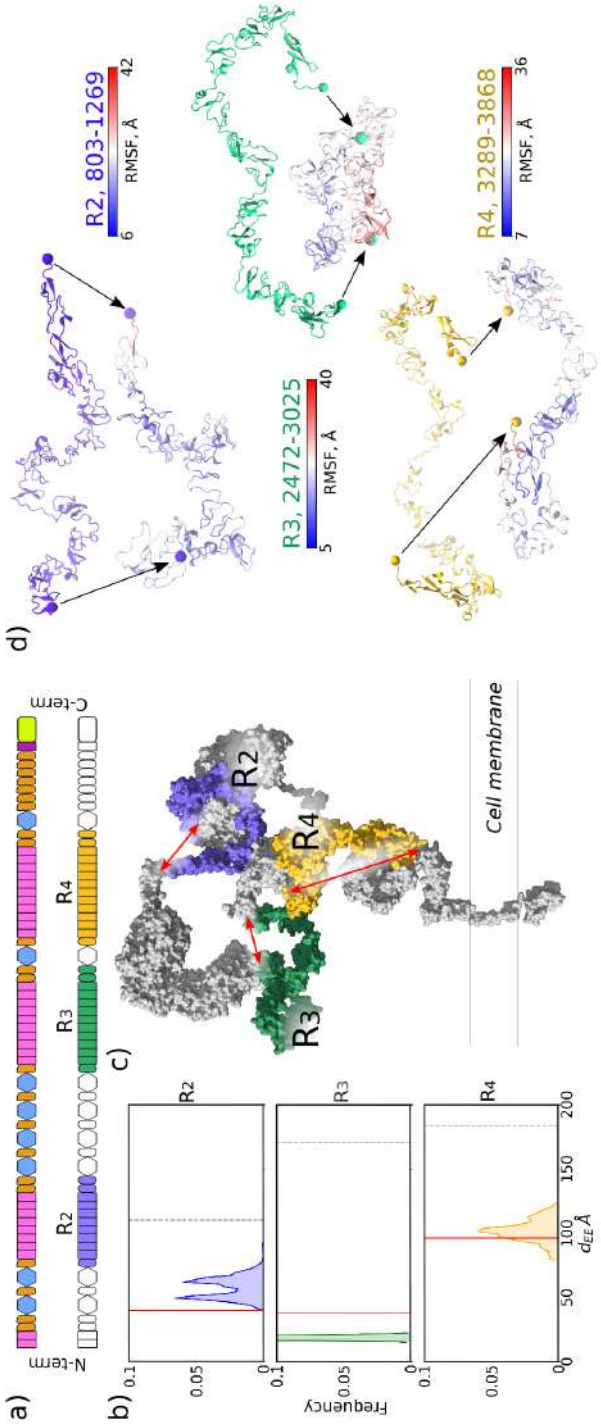


Figure 4.2: a. Schemes of LRP1 motif sequence. Top: same scheme as Figure 1.1. Bottom: same scheme, highlighting macrodomains R2, R3, and R4. b. Normalized frequency distribution of the Rn domain end-to-end distance d_{EE} during the last 200 ns of MD simulations. The vertical dotted line represents the d_{EE} value in the initial conformation; the red solid line is the d_{EE} value for the R domain in the dimeric conformation. c. Visual representation of LRP1 monomer in vdW surface. d. Initial and final conformations of the R domains. The final conformations are colored according to the RMSF per amino acid.

3 The calcium-binding motif coordination site and the ion binding intensity

We did not observe any divalent calcium ions leaving their coordination sites during the simulations of the flexible domains. To quantify the stability of these interactions, we performed umbrella sampling simulations of CB7 (see General Methods).

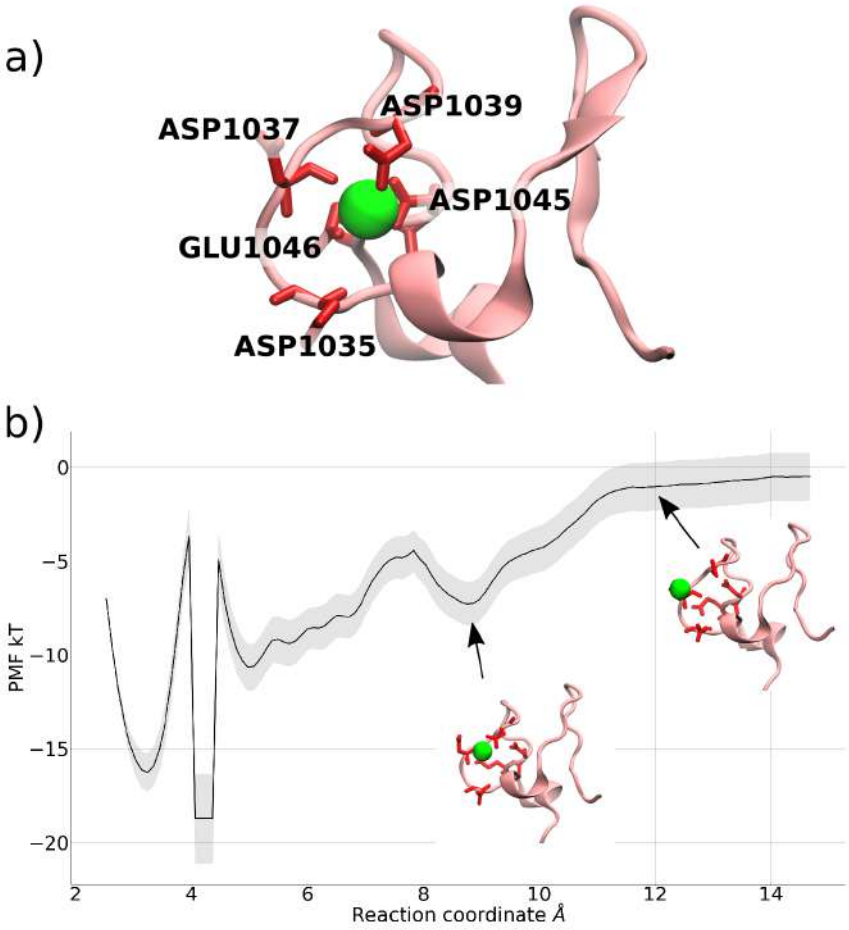


Figure 4.3: a. CB7 from R2 displayed as pink cartoon. Key amino acids are shown as red sticks. The calcium ion is a green sphere. b. Energy profile from the umbrella sampling protocol. The shaded area shows the uncertainty of the measure. The inset cartoons show local configurations of the ion.

The potential of mean force (Figure 4.3.b) shows a deep equilibrium well corresponding to the bound state. As the reaction coordinate increases, transient metastable states appear before full dissociation. The energy required to remove the ion was estimated at 16±2 kT, consistent with the rarity of spontaneous calcium release events.

4 The glycosylation affects LRP1's dimer conformation

An interesting feature of LRP1 dimers is the large interaction surface between the two monomers in correspondence with the canopy area (Figure 4.4.a). Since this area is densely glycosylated, we investigated the role of N-glycans in stabilizing the dimeric interaction. Systems were modeled in four variants: fully glycosylated, asymmetrically glycosylated (two cases), and non-glycosylated (see General Methods).

The temporal evolution of the monomer–monomer distance (d_{MM}, Figure 4.4.c) and RMSD (Figure 4.4.d) highlighted a clear discrepancy. Fully glycosylated systems maintained a "closed" conformation throughout the simulations, while partially or non-glycosylated systems transitioned to and stabilized in an "open" conformation (Figures 4.4.e,f).

Hydrogen bond analysis (Figures 4.5.b,c) revealed that the stability of the closed conformation is mainly supported by protein–glycan interactions, whereas the open conformation relies exclusively on protein–protein interactions. Table 4.1 summarizes the most stable hydrogen bonds in the glycosylated and non-glycosylated systems.

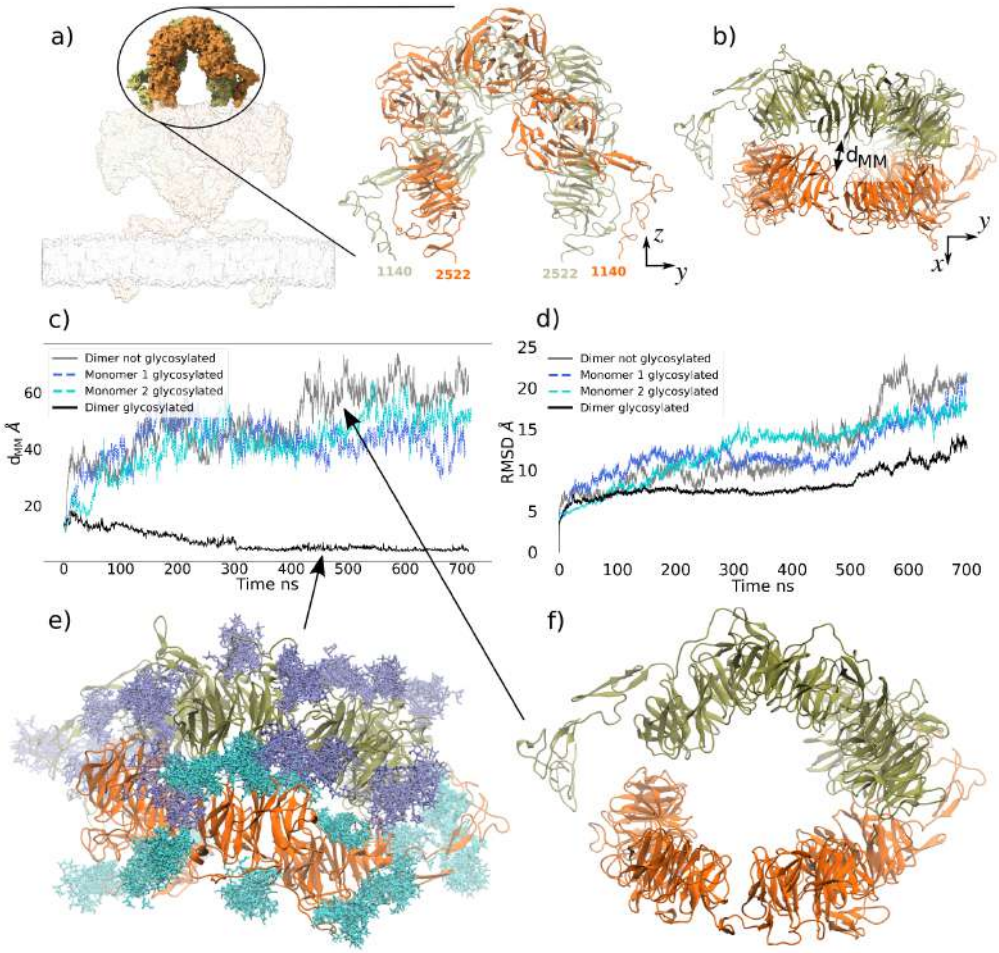


Figure 4.4: a. Detail of the dimeric canopy structure. The two monomers are colored orange and khaki. b. The xy view of the dimeric canopy highlights the protein vicinity, quantified by the distance d_{MM}. c. d_{MM} in function of time for the four systems. d. Root Mean Square Distance of heavy atoms for the four systems as a function of time. e. Glycosylated canopy structure after 500ns. f. Non-glycosylated canopy structure after 500ns.

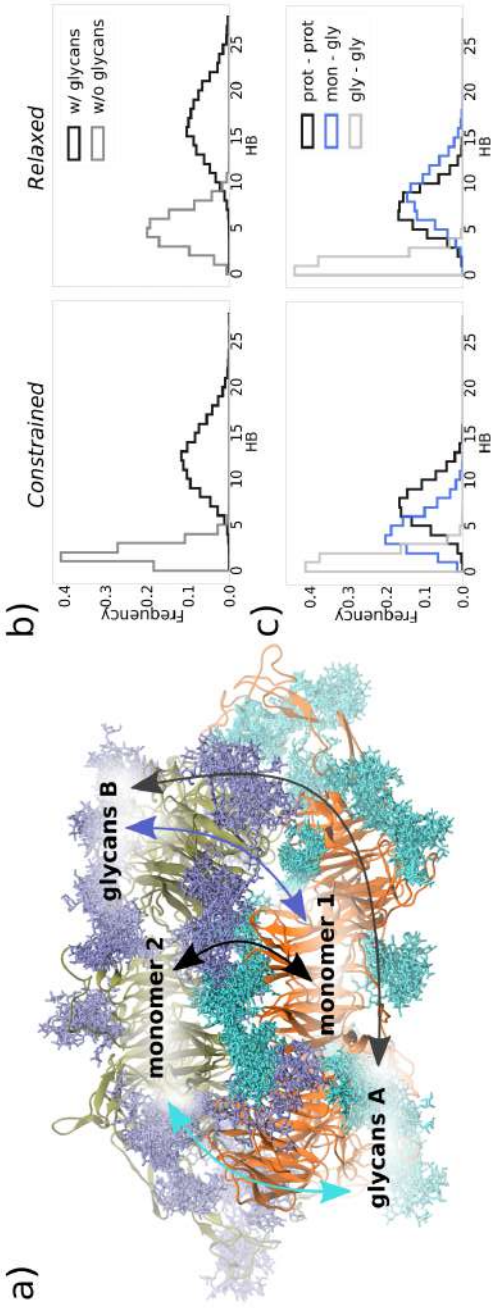


Figure 4.5: a. Closed dimeric canopy structure at the end of 700 ns. Monomers are colored orange and khaki, glycans in cyan/ice blue. b. Distribution of hydrogen bonds in the constrained and relaxed setups. c. Distribution of hydrogen bonds by type (protein–protein, protein–glycan, glycan–glycan) for the fully glycosylated system.

Glycosylated		Glycosylated - relaxed	
HB	Occ %	HB	Occ %
ARG1497 GLU2418	93.2	ARG1966 ASP1407	87.6
ARG1517 GLU2418	87.9	TYR2030 PRO1411	61.5
TYR2030 PRO1411	61.8	TYR2030 SER1409	54.9
TYR2030 SER1409	57.1	LYS1788 GLU1899	53.3
Not glycosylated		Not glycosylated - relaxed	
HB	Occ %	HB	Occ %
ARG1517 GLU2418	39.5	ARG1517 GLU2418	59.1
ARG1517 ASP2399	21.2	ARG1517 ASP2399	41.6
LYS2454 GLU1476	19.6	LYS2416 PHE1477	30.2
ARG1517 PRO2419	15.2	ARG1454 ASP2008	29.9

Table 4.1: First four intra-dimer hydrogen bonds, ordered by stability, for the glycosylated and non-glycosylated canopy, before (400–500 ns) and after (600–700 ns) the relaxation of terminal constraints.

5 General Methods

All molecular dynamics (MD) simulations were performed with GROMACS [110], using the CHARMM36m force field for proteins [80]. Systems were prepared with the CHARMM-GUI web server [104], solvated with TIP3P water [82], and neutralized with 0.15 M KCl. The LINCS algorithm was applied to constrain bond lengths, while long-range electrostatics were treated with the particle-mesh Ewald (PME) method. Van der Waals interactions used a cutoff of 1.2 nm with force-switch activated at 1.0 nm. Temperature was maintained at 303.15 K using either the velocity rescale or Nose–Hoover thermostat, and pressure was controlled with the Parrinello–Rahman barostat (1 bar, compressibility 4.5×10^{-5} bar^{-1}).

Energy minimization was performed with the steepest descent algorithm until forces fell below 1000 kJ mol^{-1} nm^{-1}. Following minimization, systems underwent NVT equilibration (1–10 ns depending on system size) with gradually released positional restraints, and NPT equilibration (0.5–50 ns) when necessary. Production runs were carried out with a 2 fs timestep. Trajectory analyses were conducted with the MDAnalysis Python package [111] and VMD [112].

Umbrella Sampling Protocols

Umbrella sampling was employed to calculate the potential of mean force (PMF) for specific structural processes. Configurations along reaction coordinates were extracted at intervals of 2 Å. Each window was equilibrated (100–500 ps) before 10 ns production runs, followed by PMF reconstruction with the Weighted Histogram Analysis Method (WHAM) [93]. Uncertainties were estimated by Bayesian bootstrap analysis with 200 resamplings [113].

System-Specific Setups

- **Furin cleavage site (B8)**: Residues ASP3779–GLN4235 were isolated to represent β-propeller 8. The α- and β-chains were pulled apart with a constant force of 300 kJ mol^{-1} nm^{-2} over 25 ns. The PMF was calculated as a function of the center-of-mass distance between the two chains.
- **Flexible domains (R2, R3, R4)**: The macrodomain fragments were simulated: R2 (CB3–10, residues 803–1269), R3 (CB11–20, residues 2472–3025), and R4 (CB21–31, residues 3289–3868). Simulations started from stretched conformations and were followed until equilibrium. Root mean square fluctuations (RMSF) were calculated after aligning trajectories to the initial structure, and end-to-end distances were measured on α-carbon terminals of CB sequences.
- **Calcium-binding site (CB7)**: Residues ASP1035–GLU1046 and all six conserved cysteines were included with one bound Ca^{2+} ion. The ion was pulled from its coordination site with a harmonic potential (1000 kJ mol^{-1} nm^{-2}). The PMF was reconstructed to evaluate binding strength.
- **Canopy dimer with glycans**: Residues 1140–2522 of both monomers (B3–B6 and EGF-like5–10) were simulated in four systems: fully glycosylated, asymmetrically glycosylated (two cases), and non-glycosylated. Glycosylation sites were taken from GlyGen [114] and modeled with Glycosylator [115]. N-glycans were added as high-mannose chains. Each system was simulated for 500 ns with restrained terminals, followed by 200 ns without restraints. Hydrogen bonds were analyzed between protein–protein, protein–glycan, and glycan–glycan pairs.

6 Conclusions

The simulations presented in this chapter reveal that the dynamical properties of LRP1 are tightly linked to its structural organization and post-translational modifications. The furin cleavage site within the B8 β-propeller is stabilized by hydrophobic effective interactions within the β-sheet, necessitating a significant energy barrier to dissociate the α and β chains. This finding helps rationalize how proteolytic shedding preserves non-covalent contacts even after cleavage.

The flexible R domains exhibited marked coiling behavior, with their end-to-end distances decreasing from the stretched initial states to their equilibrium conformations. The observed fluctuations underscore the dynamic nature of CB motifs when arranged in tandem, reinforcing the notion that local rigidity coexists with large-scale structural plasticity. This property may underlie the capacity of LRP1 to adapt to diverse ligands and mechanical constraints.

At the local level, calcium coordination cages exhibited high binding energies, with calcium ions firmly retained in their acidic environments. This highlights the stabilizing role of divalent ions in CB motifs, underscoring their importance for maintaining structural integrity and signaling.

Finally, glycosylation was found to exert a profound influence on dimer stability. Fully glycosylated dimers remained locked in a compact “closed” conformation, while partially or non-glycosylated systems adopted an “open” state stabilized by protein–protein hydrogen bonds alone. These results underscore the dual structural and regulatory role of glycans, which not only impose steric hindrance but also provide specific stabilizing interactions at the dimer interface.

Altogether, the analyses converge on a picture of LRP1 as a highly dynamic receptor in which proteolytic sensitivity, domain flexibility, ion coordination, and glycosylation

jointly define its conformational landscape. This dynamic versatility is likely essential for its ability to interact with a wide range of ligands and participate in diverse physiological contexts.

From a broader perspective, the interplay between structural dynamics and biochemical modifications provides a framework to understand how LRP1 balances stability with adaptability. Such a balance is crucial to its roles in endocytosis, signaling, and disease-related processes, including cancer, neurodegeneration, and vascular pathology. Future work could focus on how external cues, such as ligand binding or membrane mechanics, modulate these intrinsic dynamic features to drive context-specific receptor functions.

Chapter 5

The binding activity of LRP1

1 Searching the binding sites

The binding activity of LRP1 and other members of the LDL receptor family has been described in the Introduction (Section 6.2). Among LRP1 ligands, Angiopep2 is of particular interest. This nineteen–amino acid peptide (sequence: TFFYGGSRGKRNNF-KTEEY) was first identified for its ability to cross the blood–brain barrier (BBB) [51, 50] and has been shown to shuttle diverse molecular cargos, including anticancer and analgesic agents, RNA, DNA, and bacteriophages [116, 117, 118, 51, 119]. More recently, it has been demonstrated that the LRP1–Angiopep2 interaction can modulate the trafficking of molecules across the BBB [45, 44]. This modulation occurs through multiple, weak interactions between LRP1 and Angiopep2 molecules displayed on nanoparticle surfaces, a regime known as mid-avidity or Goldilocks avidity [44]. The range of biological and pharmaceutical applications associated with this phenomenon has driven growing interest in the molecular characterization of the interaction.

The specific binding sites of LRP1 have not yet been conclusively identified. Given the large dimensions of the receptor, it is reasonable to hypothesize that it contains multiple

binding regions. The most frequently described motif, the calcium-binding (CB) domain, is generally considered to display the highest affinity for ligands. Quantitative studies have reported binding affinities for CB5, CB6, and CB17 [49]. However, recent work has suggested that ligands may also bind to β-propeller motifs [120].

In this chapter, the interaction between LRP1 and Angiopep2 is examined using a multi-step docking approach. Two molecular docking strategies were employed: blind docking to identify potential binding regions and local docking to refine specific interaction hypotheses.

2 Angiopep2 structure

For the LRP1–Angiopep2 docking task, a structural model of the peptide was first generated. As reported by di Polidoro et al. [49], Angiopep2 displays pronounced conformational flexibility. This property is illustrated in Figure 5.1.a, where each amino acid is represented according to its probability of adopting an α-helical (green), β-sheet (red), or other (blue) secondary structure. The probabilities shown in Figure 5.1.a were obtained with PEP-FOLD3, a structural prediction algorithm designed explicitly for peptides [121]. A representative structural model derived from the prediction is shown in Figure 5.1.b. The N-terminal residue, threonine at position 1, serves as the attachment point to the polymer when Angiopep2 is used to decorate nanoparticles, a detail that will be relevant in Section 4. In contrast, the C-terminal end is free to move and represents the most hydrophilic region of the peptide, enriched in polar and charged residues. At physiological pH, Angiopep2 carries a net positive charge of +2e.

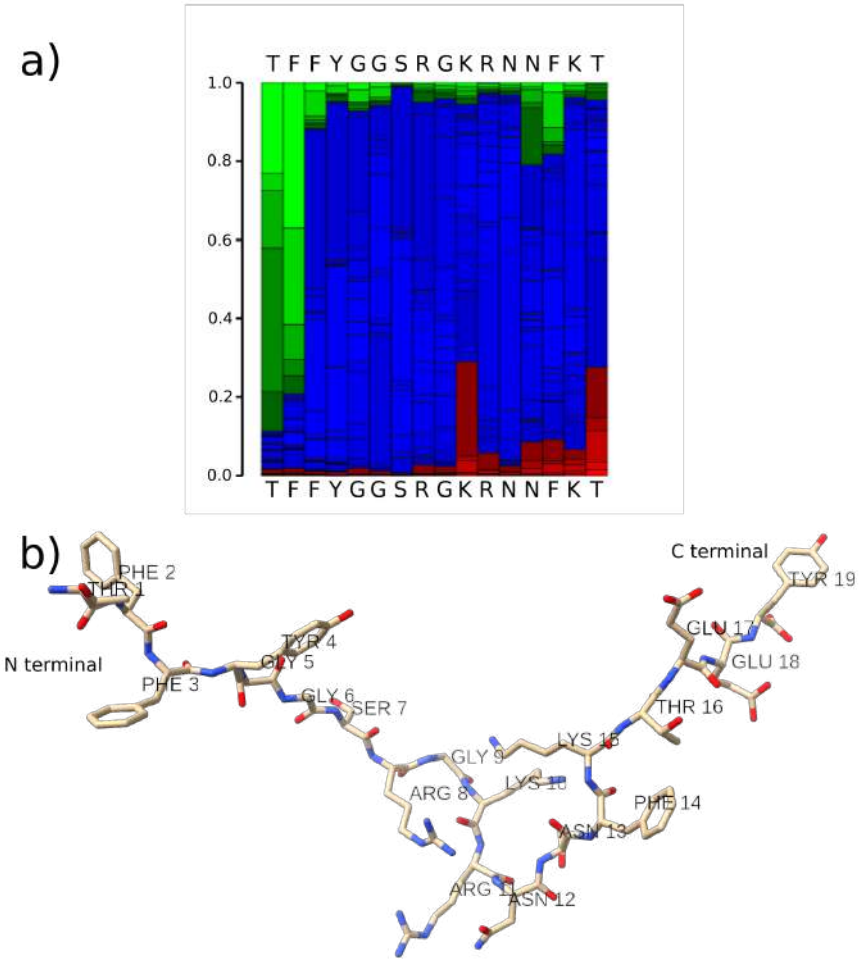


Figure 5.1: a. Secondary structure folding probability of Angiopep2 per residue. Probabilities are assigned to α-helical (green), β-sheet (red), or other (blue) conformations. b. Representative 3D structure of Angiopep2 predicted by PEP-FOLD3, shown as sticks. Atoms are colored by type: nitrogen (blue), oxygen (red), and carbon (beige).

3 Global docking on the LRP1 dimer

We performed an initial blind docking to explore the possible binding sites of Angiopep2 on the LRP1 protein. The Procedure was carried out with HPEPDOCK [96], as described in the corresponding methods section.

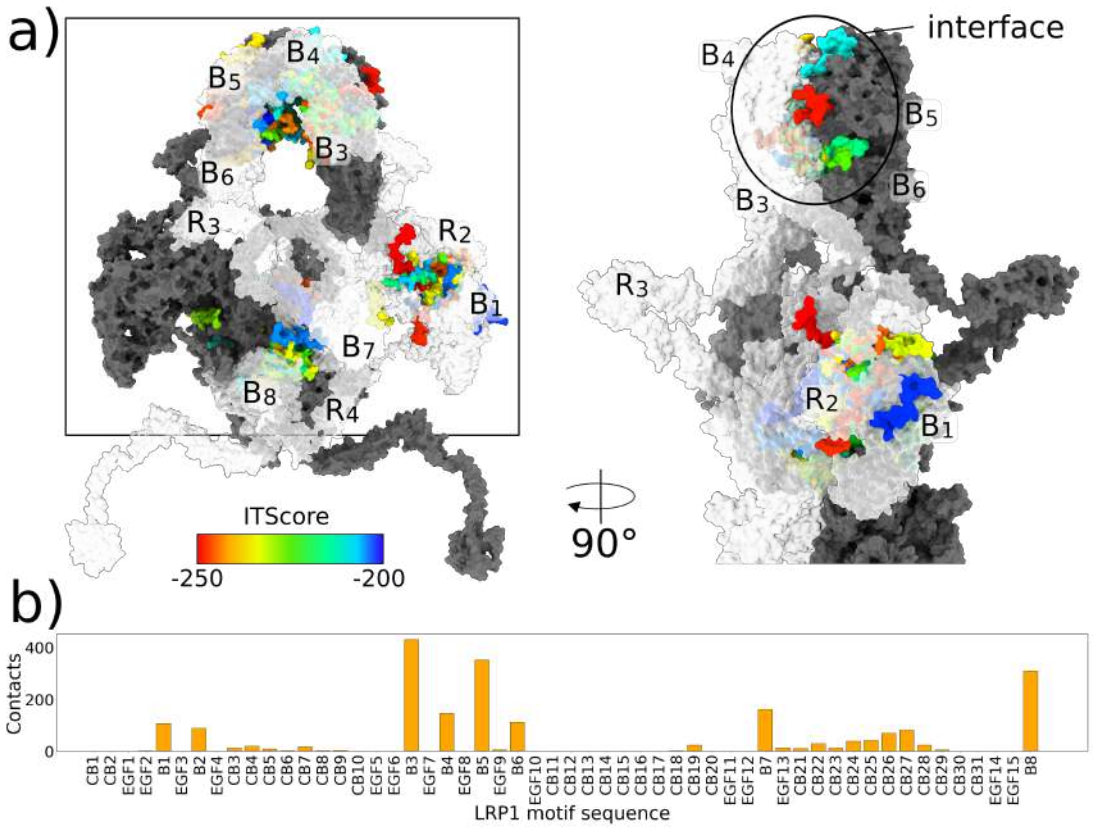


Figure 5.2: a. The top one hundred Angiopep2 poses obtained from HPEPDOCK blind docking on the LRP1 dimer. The protein is shown as a van der Waals (vdW) surface, with the two monomers in white (front) and grey (back). Ligand poses are displayed in a rainbow color scheme. Two perspectives are shown: an overall view (left) and a zoomed lateral view (right). The latter highlights the presence of poses at the dimeric interface. b. Histogram reporting the number of protein–ligand contacts for each LRP1 motif.

The web tool requires as input the structure of the protein receptor and the sequence or structure of the peptide ligand. Accordingly, we uploaded the LRP1 dimer, as obtained in Section 3, together with the Angiopep2 sequence. HPEPDOCK generated 1,000 peptide conformations, which were docked across the surface of the receptor. From these, the top 100 models were ranked according to the ITScore-PP scoring function [122] and selected for further analysis. The 100 poses generated by HPEPDOCK are displayed in Figure 5.2.a. Each pose is calculated independently of the others; their simultaneous

representation serves only for graphical comparison. The poses, shown as vdW surfaces in rainbow colors, are distributed across multiple domains of the receptor.

The global docking results allow discrimination between CB, EGF, and B motifs. To quantify this distribution, we counted the number of contacts between each ligand pose and each LRP1 motif, as introduced earlier. A contact was defined as an interatomic distance below 3.5 Å. Because the receptor is dimeric, each amino acid is present twice; to avoid double counting, equivalent residues from the two monomers were considered only once in the analysis.

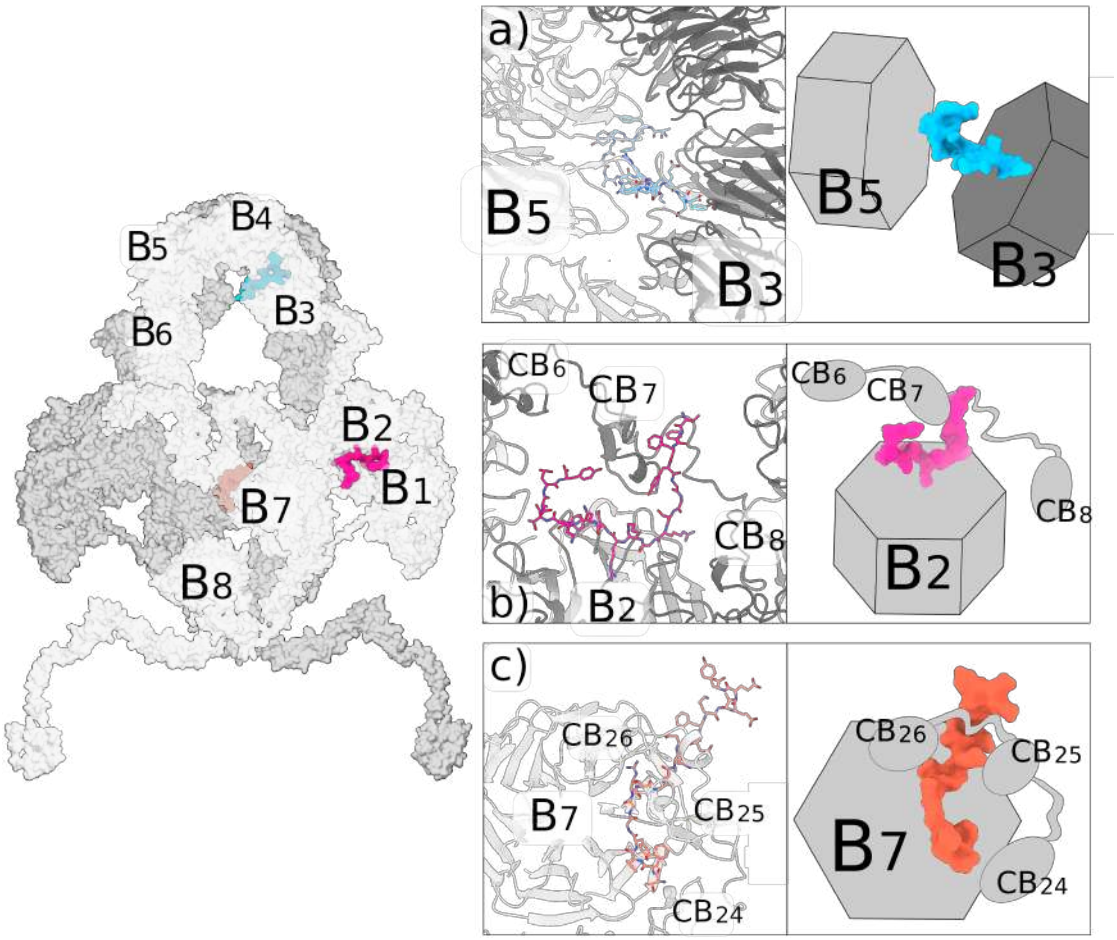


Figure 5.3: Representative Angiopep2 poses on the LRP1 dimer. The protein is depicted as a van der Waals (vdW) surface, with the two monomers represented in different shades of grey. Peptides are displayed in colored vdW surfaces. a. A pose located between the two monomers, interacting with B3 of one monomer and B5 of the other. b. A pose positioned between B2 and CB6–8 within the R2 domain, all belonging to the same monomer. c. A pose similar to (b), involving B7 and CB24–26 within the R4 domain.

The histogram in Figure 5.2.b highlights a preference for Angiopep2 to interact with the β-propellers, particularly B3, B5, and B8. As shown in Figure 5.2.a, many poses are located in the canopy region, at the interface between the two monomers. In this area, complexes often assume configurations such as that in Figure 5.3.a, where the peptide is enclosed between two propellers. A similar situation occurs for the B8 motifs, which are positioned closely together in the dimer. This monomer–monomer interface thus appears to represent a recurrent binding hotspot for Angiopep2. Additional hotspots include the regions between B1–2 and R2 (Figure 5.3.b) and between B7 and R4 (Figure 5.3.c), where the peptide establishes contacts with both β-propellers and CB domains.

From this analysis, we identified the specific protein residues involved in the interaction with the ligand. This information provides the basis for the second docking Procedure, which is presented in the following section.

4 A local docking based on previous evidence

The global molecular docking with HPEPDOCK provided an initial screening of possible LRP1–Angiopep2 binding sites. However, this method does not account for the specific electrostatic contributions of calcium ions in the CB pockets. To incorporate this effect and to perform a more focused exploration, we used HADDOCK2.4. This tool enables refined local docking and, through the Prodigy-LIG module (PROtein binDIng enerGY prediction for LIGands), allows estimation of binding affinities for the predicted complexes [123].

We performed two local docking Procedures, each defined by constraints on active residues. In the first run, the active residues were those identified in the global docking analysis with HPEPDOCK (Section 3). In the second run, the active residues were set to the calcium-binding cages of each of the 31 CB motifs. These motifs are frequently implicated in ligand recognition, and the conservation of negatively charged residues

within them suggests a strong electrostatic contribution to binding positively charged ligands. For each Procedure, HADDOCK2.4 generated 500 poses, resulting in a total of 1,000 structures. Both runs used the minimized LRP1 dimer structure (Section 3) and the Angiopep2 structure predicted by PEP-FOLD3 (Section 2). The workflow is summarized in Figure 5.4.

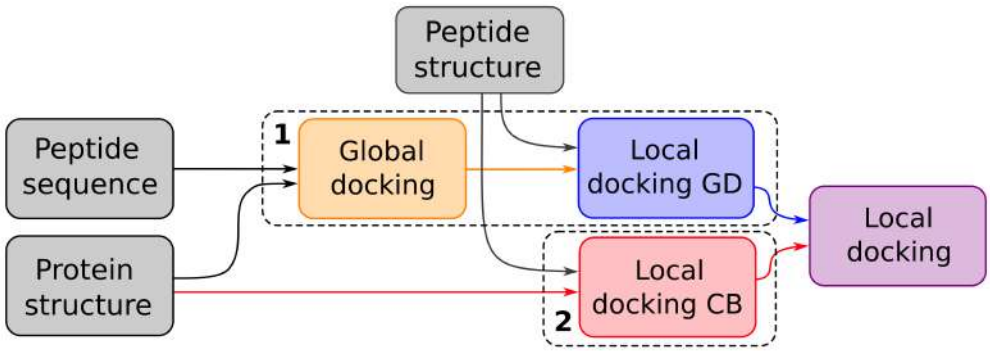


Figure 5.4: Schematic overview of the two local docking protocols, or Procedures. Procedure 1: Active residues defined from the results of the global docking with HPEPDOCK (Section 3). Procedure 2: Active residues are defined as the calcium-binding cages of all CB motifs.

Docked structures were downloaded and analyzed locally. The HADDOCK refinement pipeline includes three levels: (i) rigid docking; (ii) flexible refinement of protein and ligand backbones and sidechains at the interface; and (iii) final refinement with explicit-water molecular dynamics. Due to the size of the LRP1 dimer, we employed level (ii), which includes backbone and sidechain flexibility but excludes explicit water.

Prior to analysis, we removed all poses in which Angiopep2 interacted through its N-terminus, since this residue is covalently attached to polymers in nanoparticle-decorated systems (Section 2). Contacts were defined as hydrogen bonds within 2.5 Å; longer-range interactions were excluded. This filtering removed 117 poses, leaving 883 for analysis: 429 from Procedure 1 and 454 from Procedure 2. We then quantified the number of contacts per residue of Angiopep2 (Figure 5.5).

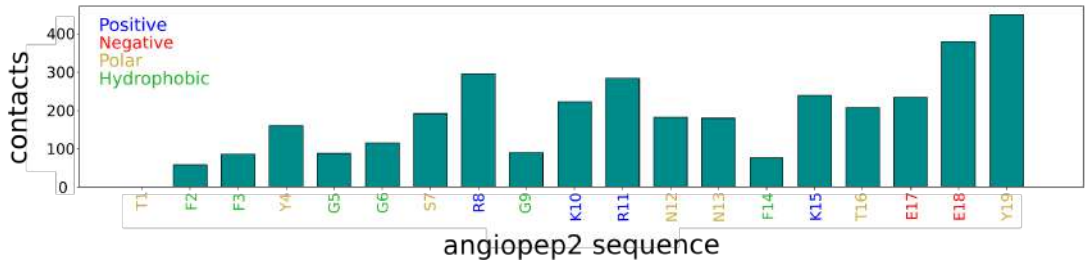


Figure 5.5: Residue-level analysis of Angiopep2 contacts with LRP1 for the two local docking Procedures. Histograms report the number of protein–ligand contacts for each amino acid in Angiopep2. Colors indicate amino acid type: polar, hydrophobic, positively charged, and negatively charged.

The histogram highlights the prominent role of polar and charged residues in binding. The C-terminal region contributes most strongly, particularly tyrosine 19 and glutamic acids 17 and 18. In addition, positively charged residues arginine 8, lysine 10, arginine 11, and lysine 15 show high contact counts. Interestingly, residues of opposite charge contribute comparably, suggesting that recognition depends more on spatial arrangement than on electrostatic polarity alone.

We next analyzed the two docking Procedures separately. Figure 5.6 shows the results of Procedure 1, where active residues were derived from HPEPDOCK.

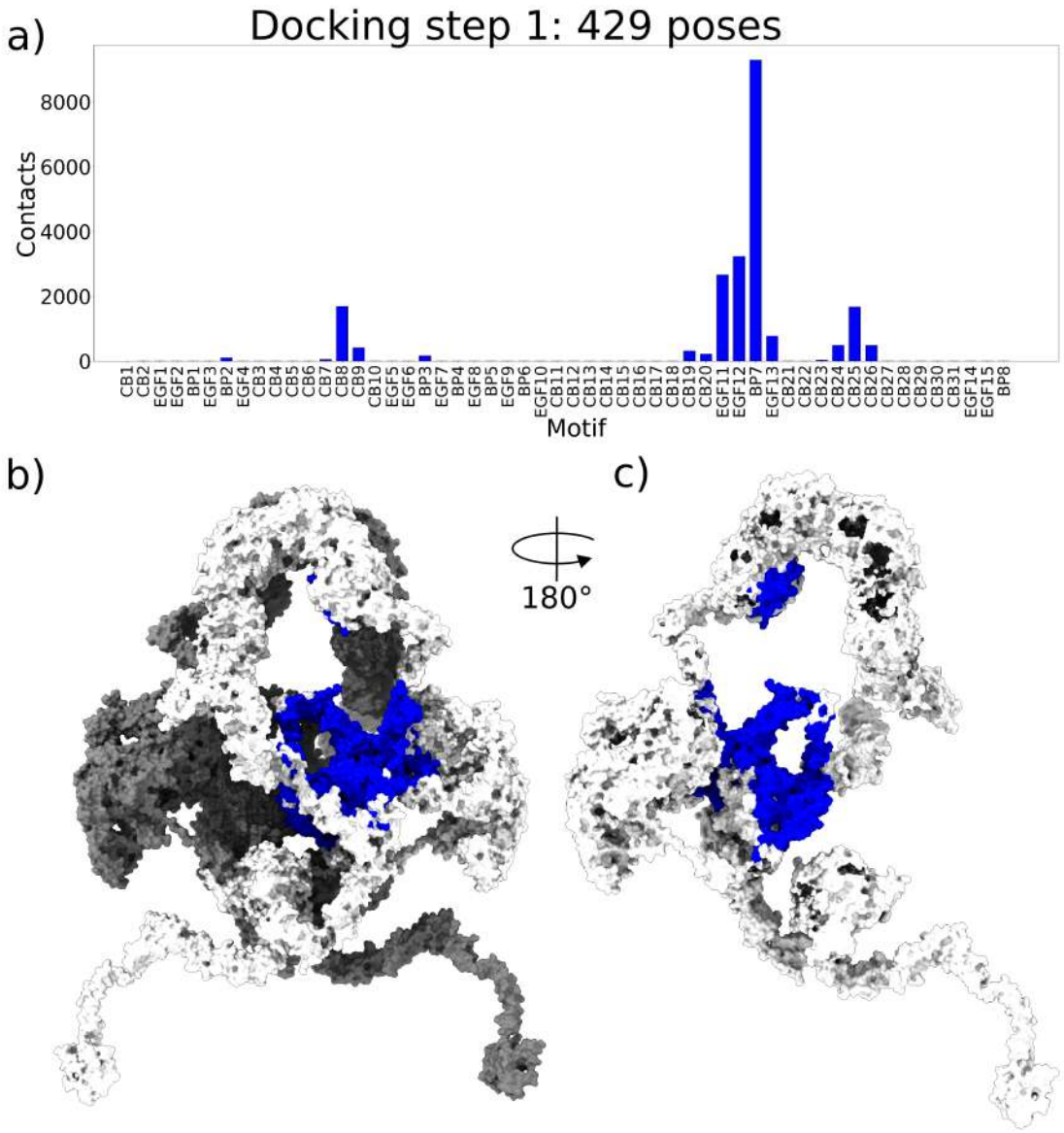


Figure 5.6: Results of local docking Procedure 1, constrained by residues from the HPEPDOCK global docking. a. Histogram reporting the number of protein–ligand contacts per LRP1 motif. b. LRP1 dimer in van der Waals (vdW) representation, with the two monomers in white and gray. Blue shading indicates residues contacting one or more Angiopep2 poses. c. Alternative perspective of the front monomer in (b), highlighting interfacial residues contacting Angiopep2.

Contacts were defined as interatomic distances below 8 Å, following Duarte et al. [124]. The analysis reveals a predominance of interactions with B7, the central propeller of the dimer, consistent with the global docking (Figure 5.3.a). Many of the 429 poses interact with both copies of B7 at the monomer–monomer interface (Figures 5.6.b,c). Additional contacts involve CB8–9 (R2), CB19–20 and EGF11–12 (R3), and CB24–26 with EGF13 (R4). In all cases, binding is centered on the geometric core of the receptor, where R2, R3, and R4 coil around B7.

Procedure 2, in which active residues were restricted to CB calcium cages, produced 454 poses. These were distributed over multiple CBs, particularly those surrounding B7. As shown in Figure 5.7.a, the most frequent contacts involve CB17–20 (R3) and CB21–29 (R4), with nearly the entire R4 domain engaged in recognition. Comparing Figures 5.6.a and 5.7.a, it is clear that in Procedure 1, B7 shares contacts with neighboring CBs, whereas in Procedure 2 the CB motifs themselves account for the majority of interactions.

The combined dataset of 883 poses demonstrates that neither Procedure identified binding at the canopy region, which had been favored in the global HPEPDOCK analysis. Instead, both approaches converge on the geometric center of the monomer, where B7 and surrounding CB motifs form an extended surface that accommodates the flexible Angiopep2 peptide. This arrangement permits simultaneous interactions with motifs that are distant in the primary sequence. Importantly, calcium ions appear to play a decisive role in shaping the electrostatic environment of CB motifs, thereby promoting Angiopep2 binding, a feature absent in the initial global docking. Although no single, well-defined binding pocket was identified, the results point to an extended and continuous recognition surface on LRP1 that supports multivalent peptide interactions.

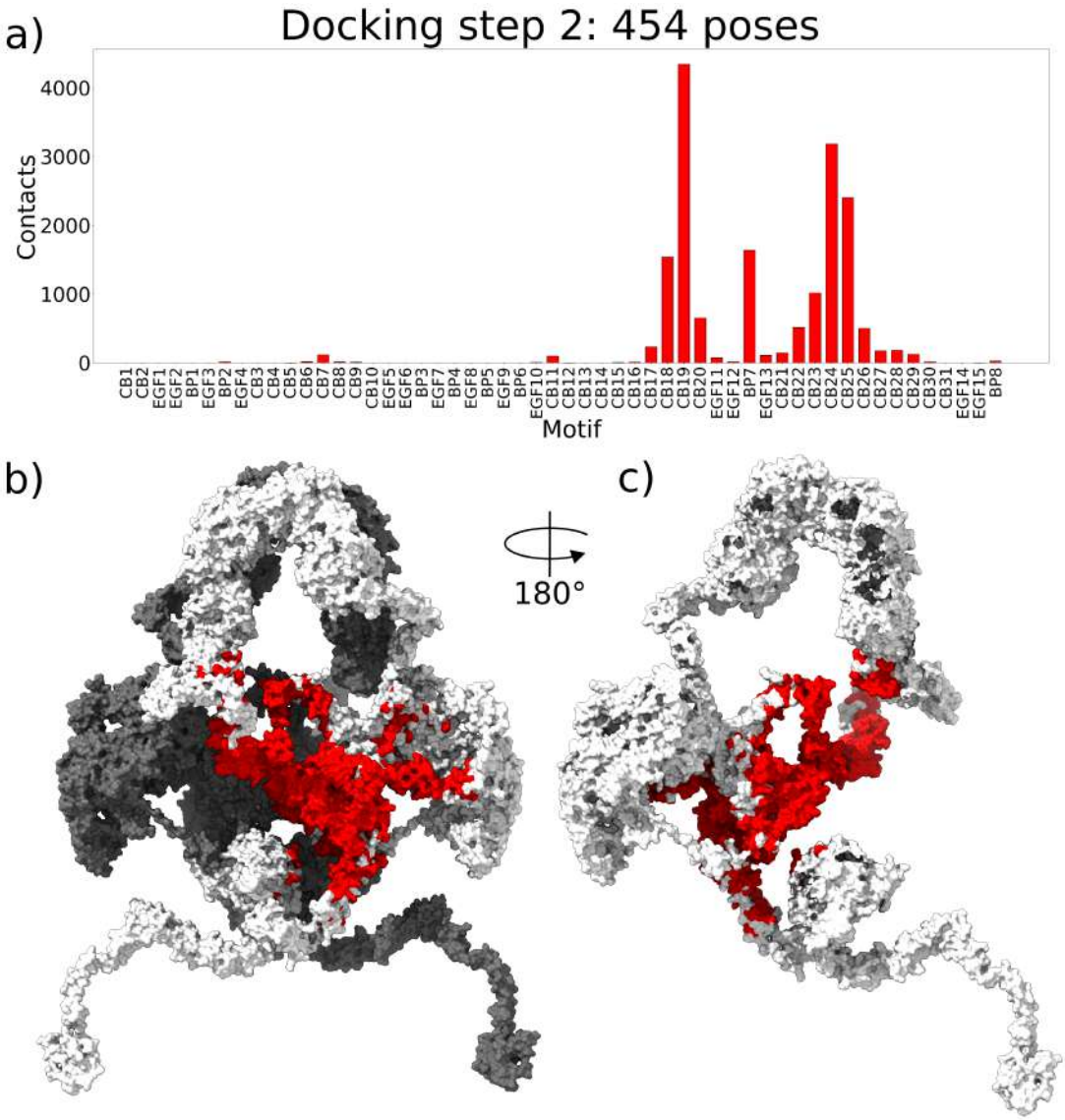


Figure 5.7: Results of local docking Procedure 2, constrained by CB calcium-binding motifs. a. Histogram reporting the number of protein–ligand contacts per motif. b. LRP1 dimer in vdW representation, with the two monomers in white and gray. Red shading highlights residues in contact with Angiopep2. c. Alternative perspective of the front monomer in (b), emphasizing interfacial residues at the dimer interface.

Taken together, the local docking results suggest that Angiopep2 does not rely on a single well-defined binding pocket, but instead engages an extended recognition surface on LRP1, centered around B7 and the surrounding CB motifs. This multivalent mode of interaction, distributed across structurally distant motifs yet spatially converging in the dimeric architecture, is consistent with the notion of mid-avidity or "Goldilocks avidity," whereby multiple weak interactions collectively stabilize the complex without saturating the receptor surface. Such a binding regime may underlie the ability of Angiopep2-decorated nanoparticles to modulate transcytosis across the BBB, providing a mechanistic explanation for the observed trafficking behavior.

In the following section, we refine these observations by mapping the LRP1 binding free energy surface for Angiopep2, which enables a quantitative assessment of the preferred binding regions and their relative stability.

5 The definition of an LRP1–Angiopep2 free energy density map

The results of the previous section highlight the difficulty of defining a single binding pocket for the LRP1 dimer. Instead of a canonical site, the interaction involves a multiplicity of overlapping binding regions formed by combinations of LRP1 motifs. To capture this phenomenon, we defined a binding surface corresponding to regions of the protein van der Waals (vdW) surface that exhibit affinity for Angiopep2, as identified from the docking poses. In this section, we describe the Procedure used to calculate this surface.

We first estimated the binding free energy of each pose. Prodigy-LIG predicts the binding energy ΔG using a linear function of electrostatic contributions and atomic contacts:

$$\begin{aligned}\Delta G_{\text{predicted}} &= 0.0115148\,E_{\text{elec}} - 0.0014852\,AC_{CC} \\ &\quad + 0.0057097\,AC_{NN} - 0.1301806\,AC_{XX} - 5.1002233.\end{aligned} \tag{5.1}$$

Here, E_{elec} is the electrostatic energy obtained from HADDOCK docking; AC_{CC}, AC_{NN}, and AC_{XX} represent the number of carbon–carbon, nitrogen–nitrogen, and polar hydrogen–heavy atom contacts, respectively. The coefficients were derived by regression against a dataset of 200 protein–ligand complexes [123]. For our dataset of 883 LRP1–Angiopep2 complexes, the predicted binding energies range from -14.5 to $-8\,kT$.

Atomic contacts between protein and ligand were then computed locally, with a contact defined as an interatomic distance below 7 Å. For each pose n, we counted the number of contacts $m(n)$ and distributed the total binding energy $\varepsilon(n)$ equally among them. If the i-th atom of amino acid k (position $\vec{r}_i$) is within 3.5 Å of a ligand atom j (position $\vec{r}_j$), the contribution to the interaction energy of residue k in pose n is given by:

$$E_{k,n}(\vec{r}_i,\vec{r}_j) = \begin{cases} \dfrac{\varepsilon(n)}{m(n)} & \text{if } \|\vec{r}_{ij}\| \leq 3.5\,\text{Å} \\ 0 & \text{if } \|\vec{r}_{ij}\| > 3.5\,\text{Å} \end{cases} \tag{5.2}$$

Summing over all docking poses yields the total enthalpic contribution of amino acid k:

$$E_k = \sum_{n=1}^{N=883} E_{k,n}(\vec{r}_i,\vec{r}_j). \tag{5.3}$$

This energy, ΔU_k, represents the average enthalpic contribution of residue k to peptide binding in the gas-phase approximation, mainly arising from hydrogen bonding and electrostatics. It does not define a discrete binding site but rather the cumulative contribution of each residue to the extended binding surface.

Assuming solvation enthalpy to be negligible, the residue-level free energy can be written as:

$$\Delta G_k = \Delta H_k - T\Delta S_k \approx E_k - T\Delta S_k, \tag{5.4}$$

where T is the temperature and ΔS_k is the entropic term associated with residue k.

The entropic component was estimated by treating each docking pose contacting residue k as a binding conformer. Using the Gibbs entropy, this contribution is:

$$\Delta S_k = -k_B \sum_{n=1}^{N} p \ln p, \tag{5.5}$$

with k_B the Boltzmann constant and p the probability of residue k being in contact with pose n. This probability can be expressed as a function of the interatomic distance:

$$p_{k,n}(\vec{r}_i,\vec{r}_j) = \begin{cases} \frac{1}{N} & \text{if } \|\vec{r}_{ij}\| \leq 3.5\,\text{Å} \\ 0 & \text{if } \|\vec{r}_{ij}\| > 3.5\,\text{Å} \end{cases} \tag{5.6}$$

Finally, combining Equations 5.4–5.6, the free energy contribution of residue k is obtained as:

$$\Delta G_k \approx \sum_{n=1}^{N=883} \left(E_{k,n}(\vec{r}_i,\vec{r}_j) + k_B T\, p_{k,n}(\vec{r}_i,\vec{r}_j) \ln p_{k,n}(\vec{r}_i,\vec{r}_j)\right). \tag{5.7}$$

This formulation defines a residue-level free energy surface, considering both enthalpic and entropic contributions, and provides a quantitative map of the LRP1–Angiopep2 binding interface.

The final step of this analysis is to define a free energy density map, which can be interpreted as a surface tension of the form $T = E/A$, where E is an energy and A is a surface area. In practical terms, this corresponds to assigning a free energy density T_k to the k amino acid by normalizing its free energy contribution ΔG_k over the solvent-exposed surface area A_k.

A difficulty arises for residues with very small solvent-accessible surface areas, which would artificially inflate the density. To avoid such singularities, we smoothed the surface area term by averaging over local sequence neighbors. The effective surface area for residue k is therefore defined as:

$$A_k^{av} = \frac{A_{k-2} + A_{k-1} + A_k + A_{k+1} + A_{k+2}}{5}, \tag{5.8}$$

which distributes the normalization over five consecutive residues.

The residue-level free energy density function can then be written as:

$$\Delta T_k = \frac{\Delta G_k}{A_k^{av}}. \tag{5.9}$$

This formulation projects the free energy contributions onto the protein van der Waals surface, thereby providing a continuous representation of the binding landscape in terms of energy density.

It is important to note that this formulation relies on several simplifying assumptions. First, the repartition of binding energy equally across all contacts in a pose neglects the possible heterogeneity of specific interactions, such as hydrogen bonds or salt bridges, which may contribute disproportionately. Second, solvation enthalpy is neglected, meaning that desolvation effects and water-mediated interactions are not explicitly captured. Third, the entropy term is estimated from the distribution of docking poses and thus reflects conformational heterogeneity at the docking level rather than from explicit dynamical sampling. Finally, the probability p was approximated as uniform for all valid contacts, which simplifies the treatment but may underestimate differences in residue-specific binding propensities. Despite these limitations, the approach provides a coherent and interpretable free energy surface that captures the combined enthalpic and entropic contributions to LRP1–Angiopep2 recognition.

6 The steric hindrance of glycans causes a repulsion field

One detail that usual docking algorithms are not able to define, at least directly, is the protein-ligand repulsive component of the interaction. We can assume that the surface not explored by the ligand poses is indeed repulsive, but we cannot quantify the repulsion strength. With repulsion, we consider the hindrance due to amino acids in energetically unfavorable configurations, such as those caused by protein unfolding. Another cause of steric hindrance can be the presence of post-translational modifications on the surface of

the protein receptor, such as glycans. In this section, we evaluate the effect of glycans as a source of steric hindrance on the surface of LRP1.

In regions occupied by the glycans, the protein is shielded by interactions with the ligands. For the ligand to approach the protein, it must squeeze the glycan, thereby reducing the volume of space the sugars can occupy. This mechanism, explored in Acosta-Gutierrez et al. [125], represents the glycans as small polymers and their repulsive effect as the energy necessary to compress the sugar's occupied spherical volume from a radius r_0 to a compressed radius r_b.

If the glycan is a polymeric chain made of N_G monomers of Kuhn length b_G, the mean end-to-end distance of a glycan is

$$\langle r_0^2 \rangle = N_G b_G^2 \tag{5.10}$$

and the partition function for a given glycan branch is

$$q_j(r) = \frac{3\pi}{4N_G b_G^2} erf\left(\sqrt{\frac{3}{2N_G b_G^2}} r\right). \tag{5.11}$$

With B glycan branches, the repulsive potential of a single glycan is

$$u_{gly}(\langle r_0 \rangle, \langle r_b \rangle) = \ln\left[\frac{\sum_{j=1}^{B} q_j(\langle r_0 \rangle)}{\sum_{j=1}^{B} q_j(\langle r_B \rangle)}\right] = \ln\left[\sum_{j=1}^{B} \frac{erf\left(\sqrt{\frac{3}{2}}\right)}{erf\left(\sqrt{\frac{3}{2}}\frac{\langle r_B \rangle}{\langle r_0 \rangle}\right)}\right] \tag{5.12}$$

with $\langle r_0 \rangle = \sqrt{\langle r_0^2 \rangle}$ for simplicity.

We applied this potential to all the LRP1 N-glycans reported in 3, and calculated the repulsive effect for all amino acids within $\langle r_0 \rangle$ measured from the N-glycan insertion site.

We consider the repulsive contribution for the k-th amino acid, identified by the vector $\vec{r}_k$, from a single glycan as:

$$u_{k,g} = u(\vec{r}_k, \vec{r}_g, \langle r_0 \rangle, \langle r_b \rangle) = \begin{cases} \ln \left[\sum_{j=1}^{B} \frac{erf\left(\sqrt{\frac{3}{2}}\right)}{erf\left(\sqrt{\frac{3}{2}} \frac{\langle r_B \rangle}{\langle r_0 \rangle}\right)} \right] & \text{if } \|\vec{r}_{kg}\| \leq \langle r_0 \rangle \\ 0 & \text{if } \|\vec{r}_{kg}\| > \langle r_0 \rangle \end{cases} \tag{5.13}$$

with $\vec{r}_{kg} = \vec{r}_g - \vec{r}_k$; $\vec{r}_g$ identifies the glycan insertion site.

When summing over all the glycans N_G, we have

$$u_k = \sum_{g}^{N_G} u_{k,g}, \tag{5.14}$$

and thus we can rewrite Equation 5.7 as

$$\Delta G_k \approx \sum_{n}^{N=883} \left(E_{k,n}(\vec{r}_i, \vec{r}_j) + k_B T \, p_{k,n}(\vec{r}_i, \vec{r}_j) \ln p_{k,n}(\vec{r}_i, \vec{r}_j) \right) + k_B T \sum_{g}^{N_G} u_{k,g}. \tag{5.15}$$

The Equation 5.15, inserted in Equation 5.9, gives

$$\Delta T_k \approx \frac{1}{A_k^{av}} \left[\sum_{n}^{N=883} \left(E_{k,n}(\vec{r}_i, \vec{r}_j) + k_B T \, p_{k,n}(\vec{r}_i, \vec{r}_j) \ln p_{k,n}(\vec{r}_i, \vec{r}_j) \right) + k_B T \sum_{g}^{N_G} u_{k,g} \right]. \tag{5.16}$$

Equation 5.16 was implemented in an ad hoc Python program, as we set the parameters for the repulsive potential as follows: $N_G = 5$, $b_G = 5$, $B = 2$, and the compressed radius $r_B = 10$nm. The resulting free energy density surface is shown in Figure 5.8.

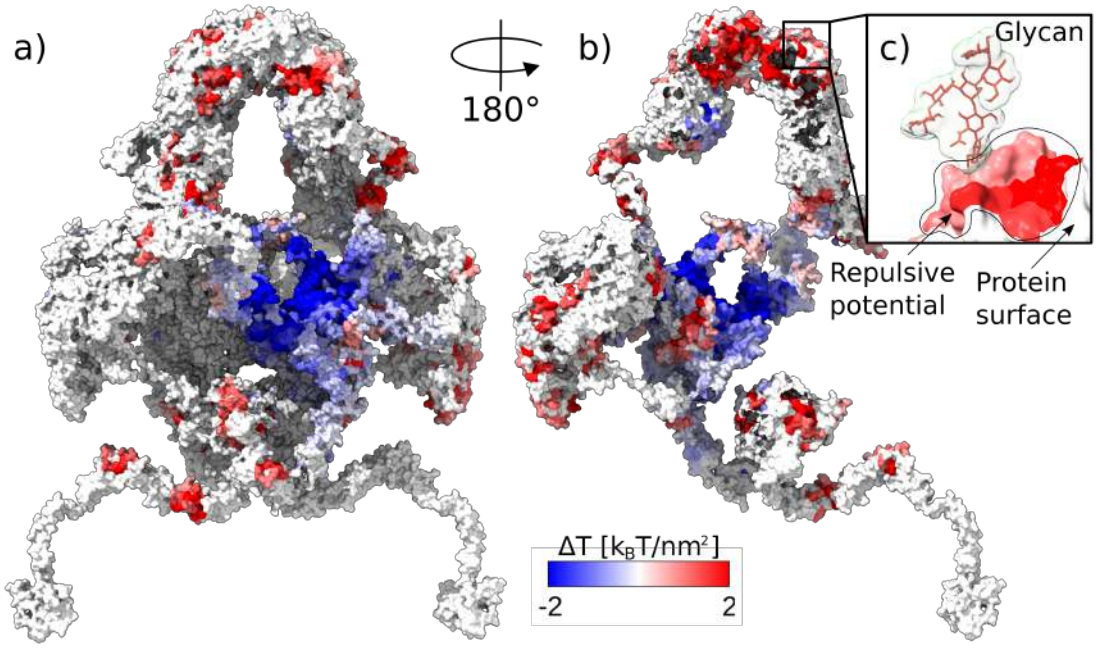


Figure 5.8: a. Vision of the dimer in vdW surface. The front monomer is colored following the color scheme and represents the surface tension of Equation 5.16. b. The front monomer rotated by 180 degrees. c. Detail of the glycan and the surrounding repulsive potential on the protein vdW surface.

It is interesting to note that the presence of glycans is particularly strong at the canopy interface, something that we analyze in Chapter 4. The detail of a glycan, with the corresponding repulsive potential density, is shown in Figure 5.8.

It should be emphasized that this formulation treats glycans as ideal polymeric chains with uniform Kuhn length and isotropic volume exploration. Such simplifications neglect the intrinsic conformational heterogeneity of glycan branches, as well as their potential to form specific hydrogen bonds or engage in long-range electrostatic interactions. Moreover, the use of a spherical approximation for the accessible volume may underestimate the anisotropy of glycan shielding. Despite these limitations, the approach provides a tractable means of quantifying the steric hindrance exerted by glycans and highlights their role in shaping the effective binding surface of LRP1.

7 Conclusions

The analyses presented in this chapter highlight the LRP1–Angiopep2 interaction as an extensive and dynamic recognition surface, rather than just a single binding site. By employing docking ensembles alongside residue-level energy repartitioning and entropy estimation, we have established a quantitative free energy surface that emphasizes the diverse contributions of multiple binding motifs. This methodology encapsulates the essence of multivalent binding, aligning with the mid-avidity regime observed in Angiopep2-decorated nanoparticles.

The role of glycan steric hindrance further enhances our understanding of this interaction. N-glycans are shown to be crucial modulators of the accessible binding landscape, functioning as repulsive fields that obscure certain areas of the protein surface, thereby directing ligand recognition toward specific sites. Notably, the canopy interface is significantly influenced by glycan shielding, indicating its importance in regulating dimer stability and accessibility.

While our models depend on certain simplifying assumptions, such as the equal partitioning of binding energy among contacts, the omission of solvation effects, and the idealized treatment of polymers as glycans, the results yield a clear and interpretable framework. Collectively, these findings deepen our understanding of LRP1–Angiopep2 interactions by integrating enthalpic, entropic, and steric factors into a cohesive binding surface map. This comprehensive perspective sets the stage for future investigations into protein dynamics and trafficking, where factors like flexibility, glycosylation, and multivalent binding will likely further inform the recognition mechanism.

Chapter 6

LRP1 and water: how tertiary structure affects solvation

Protein hydration plays a central role in shaping structural stability, flexibility, and recognition capacity. In globular proteins, conserved amino acids are often assumed to share comparable solvation properties, independent of their specific location. However, in large multidomain receptors such as LRP1, the situation is less straightforward. The flexible CB motifs and the tertiary arrangement of the Rn domains create heterogeneous microenvironments, where identical residues may experience distinct levels of water exposure.

The dynamics of the Rn domains described in Chapter 2 highlighted coiling patterns that could not be explained solely by intramolecular contacts such as hydrogen bonds or salt bridges. This observation prompted us to investigate whether differences in hydration could contribute to domain rearrangements. Specifically, we aimed to determine whether conserved amino acids (Table B.1), present in multiple positions along the CB motifs (see Figure 6.1), display similar hydration behavior or whether their solvation is instead shaped by local tertiary context.

Addressing this question is important for two reasons. First, it challenges the traditional assumption that the type of amino acid alone dictates hydration behavior, suggesting instead that the local structural environment can override the intrinsic polarity or hydrophobicity of the residue. Second, hydration patterns may influence the energetics of domain flexibility, thereby contributing to LRP1's ability to adapt its conformation during ligand binding and dimerization.

In this Chapter, we develop a residue-specific description of hydration in LRP1. We use radial distribution functions *corrected* for excluded protein volume. The volume correction is essential to account for the fact that some of the environment surrounding an amino acid is inaccessible to water because of the presence of another amino acid. Next, we derive interaction potentials from the Ornstein–Zernike equation with hypernetted-chain closure. Together, these analyses provide quantitative insight into how water interacts with conserved amino acids across flexible domains, and how solvation heterogeneity contributes to the structural adaptability of the receptor.

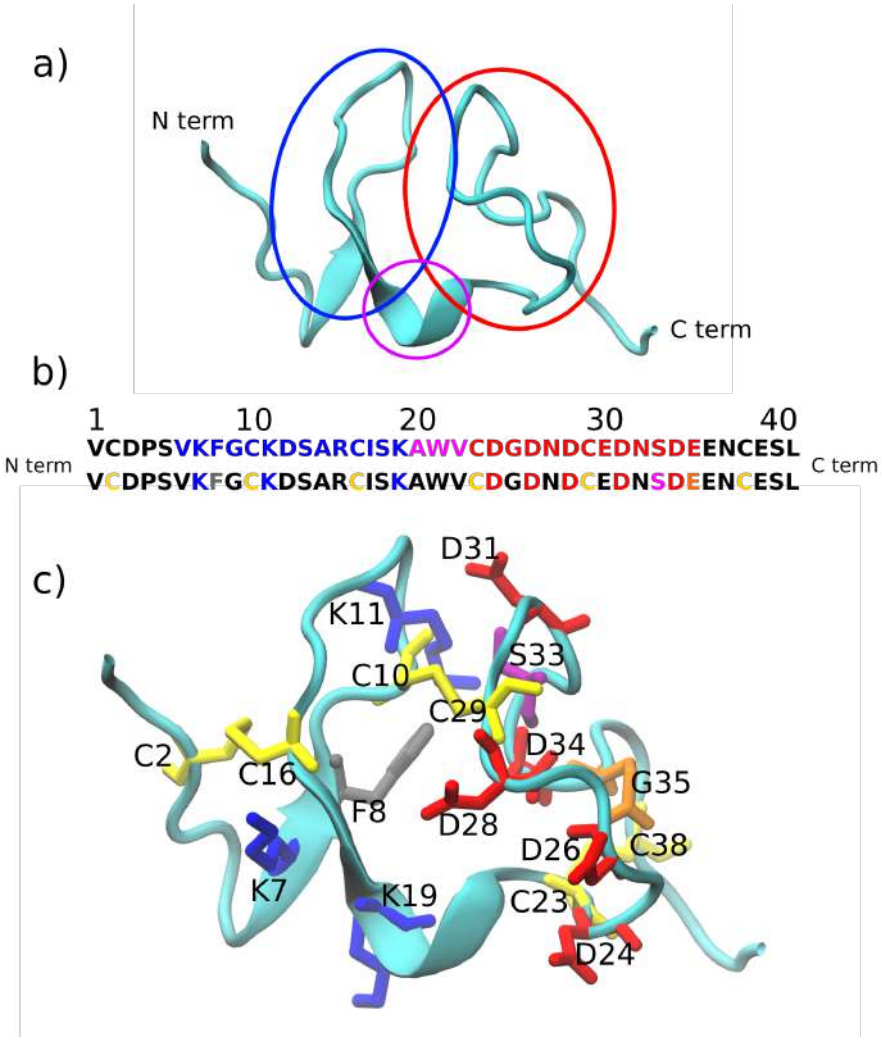


Figure 6.1: a. Graphic representation (cyan cartoon) of the CB motif structure, highlighting the structural units of the beta loop (blue), a small helix (purple) and calcium cage (red). b. Sequence of the CB highlighting the three structural units composing the motif (top) and the highly conserved amino acids (bottom). c. Graphic representation of the conserved amino acids.

1 Amino acid-water potential of mean force for the conserved amino acids

Before describing how we corrected the RDF and derived the effective interaction potentials of the amino acids with water, we present here the results of our analysis that provide residue-specific hydration signatures across the CB motifs. To organize the analysis, we grouped amino acids into families and compared their behavior. Representative plots are reported here as multi-panel figures, while the full residue-by-residue dataset is provided in Appendix B.

Figures B.1-B.5 show the interaction potentials $\Phi(\sigma)$ with water for the five conserved aspartic acids. Aspartic acid (D, Asp) 2 and 3 exhibit stable hydration, characterized by consistently negative interaction energies and minor fluctuations. Asp1 and Asp4 exhibit intermediate stability, with larger standard deviations and occasional anomalies where interaction energies become slightly positive, suggesting reduced solvent accessibility. Asp5 is the most variable case, with large fluctuations and multiple instances of residues hidden from water, particularly in R2. This last case is shown as an example in Figure 6.2. This result suggests that the absence of glycans in R2 makes it more susceptible to coiling, which can shield Asp5 from hydration.

The six cysteines (C, Cys)(Figures B.6-B.11) show highly heterogeneous hydration behaviors. Cys3 is the most consistently solvated, although still less than Asp residues. By contrast, Cys2 is almost entirely hidden from solvent, with strongly positive $\Phi(\sigma)$ values, consistent with its role in forming disulfide bonds. Cys5 shows mixed behavior, with CB9 hidden and CB27 highly hydrated, reflecting strong local fluctuations. Cys1, Cys4, and Cys6 present mostly hydrophobic signatures, punctuated by isolated hydrophilic peaks. The hydration of cysteine side chains is not generally expected, as disulfide bridges

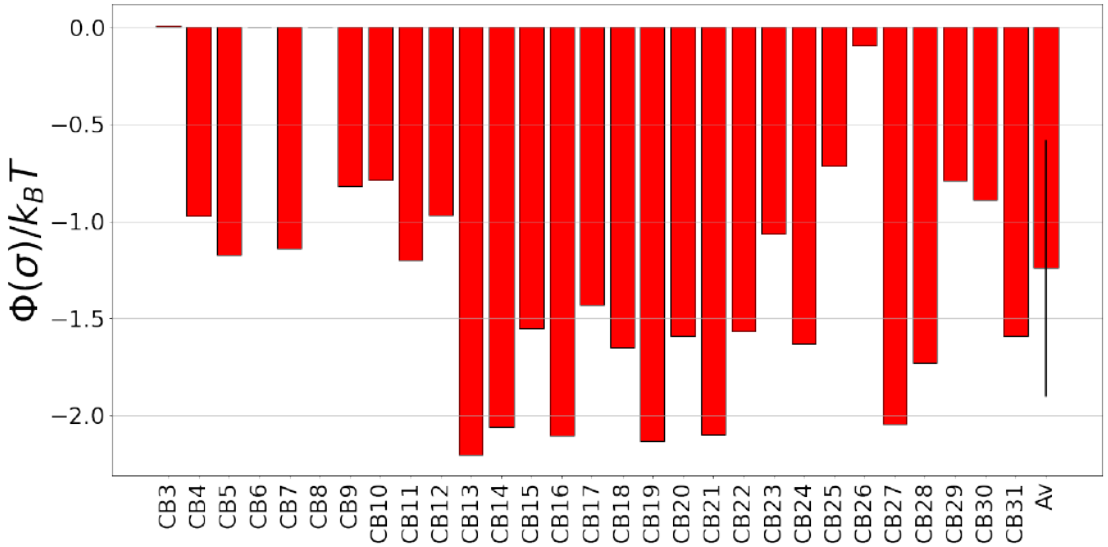


Figure 6.2: Interaction potential $\Phi(\sigma)$ for amino acid Asp5 in CB position 34.

are usually buried; these fluctuations may reflect the non-globular architecture of CB motifs.

Lysine (K, Lys) 1–3 (Figures B.12-B.14) consistently show strong hydration, with the most negative $\Phi(\sigma)$ values among all residues analyzed. Their strong solvent exposure aligns with the basic nature of lysine. Two anomalies are notable: Lys2 in CB30 and Asp5 in the same region both show reduced hydration, suggesting anomalous coiling of CB30; and Lys1 in CB9 displays a deviation without clear correspondence in other residues.

Glutamic acid (E, Glu) residues (Figure B.15) behave similarly to Asp, with stable hydration and negative $\Phi(\sigma)$ values. Serine (S, Ser) (Figure B.17) residues are also consistently solvated, as expected for a polar amino acid. Phenylalanine (F, Phe) (Figure B.16), however, is more surprising: although aromatic and generally considered hydrophobic, the conserved Phe residues are partially hydrated. This behavior is consistent with the disordered, non-globular architecture of CB motifs, which contrasts with the compact protein cores where Phe residues are typically buried.

The average interaction potentials $\Phi(\sigma)$ across all amino acid families are compared in Figure 6.3. Asp, Glu, Ser, and Lys show consistently negative hydration energies, reflecting their polar character. Cys residues fluctuate between solvated and buried states, while Phe shows unexpected partial hydration. These results demonstrate that the hydration of conserved amino acids depends strongly on structural context, rather than amino acid identity alone.

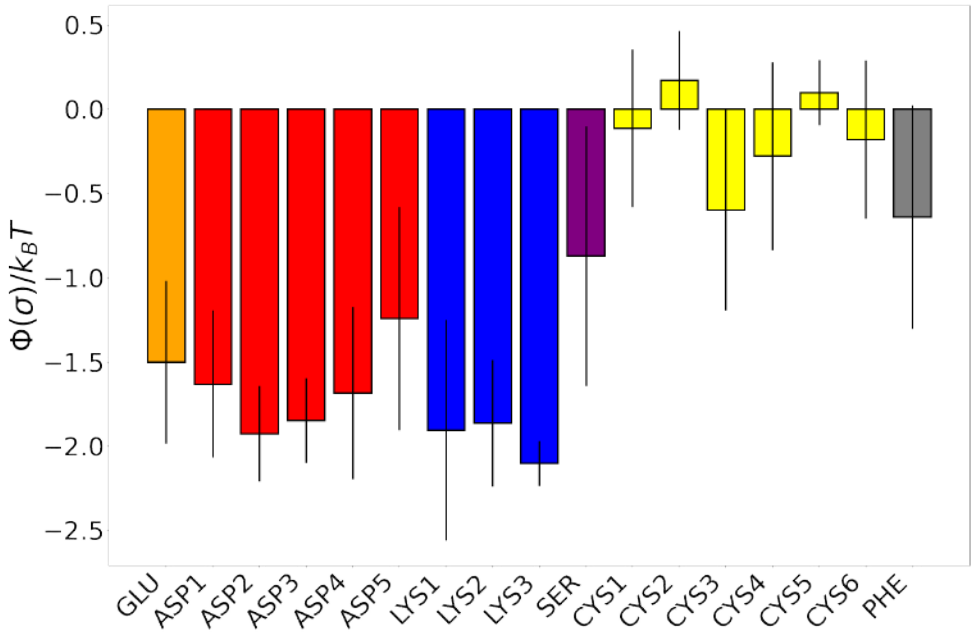


Figure 6.3: Average interaction potential $\Phi(\sigma)$ for the studied amino acid. Each color corresponds to an amino acid.

2 Estimation of the desolvation enthalpy and further evaluation of docking poses

From the previous Section, we understood how similar amino acids, despite their discrepancies, exhibit similar features regarding their interactions with water. We

then decided to extend the analysis to the rest of the amino acid spectrum, studying non-conserved amino acids along the R domains. Using the same method, we obtained the histogram in Figure 6.4.

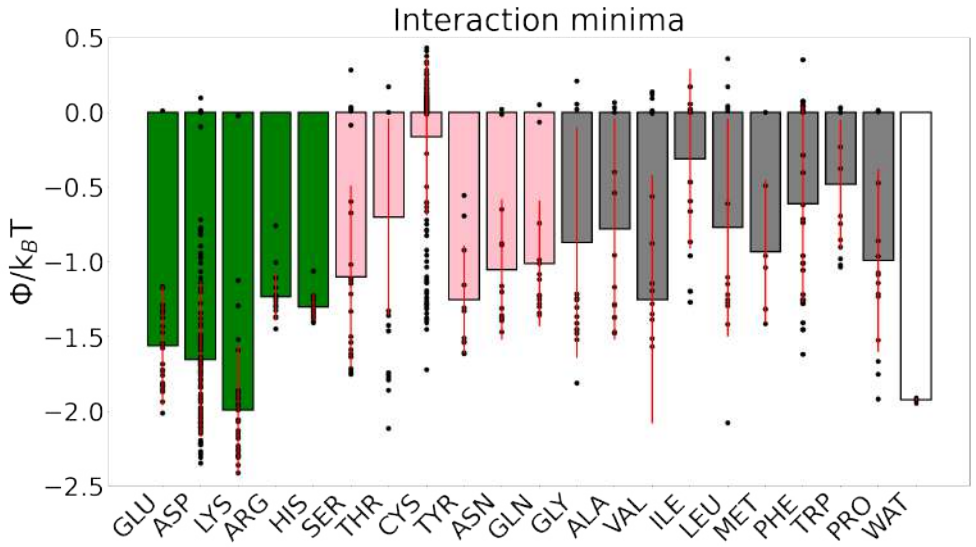


Figure 6.4: Average interaction potential $\Phi(\sigma)$ for all amino acids in the R domains. Colors represent the types of amino acids: charged (green), polar hydrophilic (pink), and nonpolar hydrophobic (gray). In white, the water-water interaction potential in bulk. The points represent the measurements for each amino acid; the standard deviation is shown in red.

Here, we observe an overall high exposure of charged amino acids (in green) to water, as well as a smaller exposure for polar (pink) and non-polar (grey) amino acids. It is instructive to observe that the potential of mean force minimum $\Phi(\sigma)$ for water is very high for bulk water, defining how generally water prefers to stay in the bulk hydrogen bond network rather than on the surface of the protein. However, charged amino acids have very similar interaction energies, especially Lysine and, to a lesser extent, Aspartic and Glutamic acid. If we interpret these results from a different point of view, we need to apply more energy to extract a water molecule from the bulk water rather than the protein hydration layer. In this sense, we can interpret $\Phi(\sigma)$ as the amino acid desolvation

enthalpy. When two molecules (in this case, proteins or peptides) come into contact, we can estimate the total enthalpy barrier ΔG_{desol} that they need to overcome by removing water from the interaction site; ΔG_{desol} is the sum of the desolvation enthalpies for the amino acids involved in the binding interaction (Figure 6.5.a).

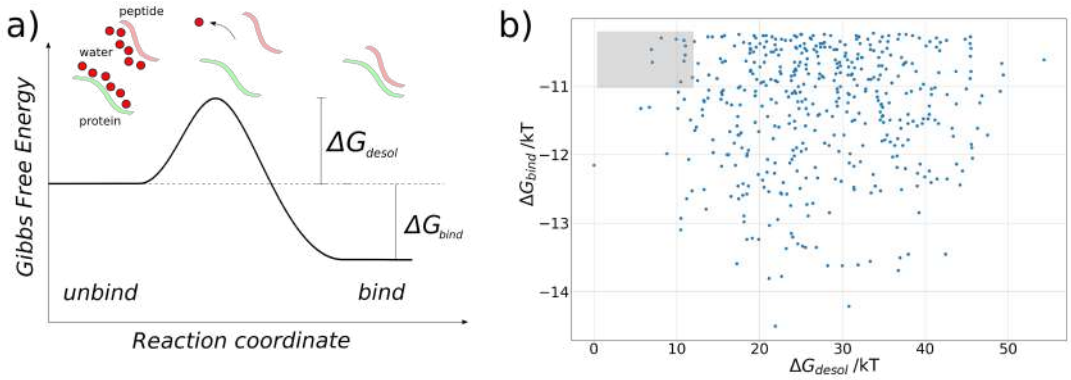


Figure 6.5: a. The binding approximation follows an activation rule: an enthalpy barrier ΔG_{desol} for interface desolvation prevents the system from transitioning freely from an unbound to a bound conformation. b. Each pose from Section 4, Procedure 2, is identified by the combination of ΔG_{bind} and ΔG_{desol}. The grey area identifies the best poses (see text for details)

We calculated the ΔG_{desol} for each of the 454 LRP1-Angiopep2 docking poses of Procedure 2 (aiming to the CB motifs) analyzed in Section 4, and considered the docking results $\Delta G_{predicted}$ as binding energy ΔG_{bind}. Each pose is then assigned to the two Gibbs Free Energy barriers and final states. We could then identify each pose in the bidimensional plot of Figure 6.5.b. We assume that the best LRP1-Angiopep2 for our interests are the ones with a small ΔG_{bind}, to ensure the Goldilocks zone (see Section 6.1), and at the same time, a small energy barrier ΔG_{desol} which allows fast binding (see the following Section 2.1). This region is highlighted in grey in the plot of Figure 6.5.b.

The top ten poses are highlighted in Figure 6.6; all of them primarily interact with CBs. The results still do not allow for finding a well-defined binding site; however, they reduce the binding region defined in the previous Chapter 5 confirming what we observed in

Figure 5.7: the main targets for Angiopep2 on the LRP1 surface are CBs 7, 19, 22, 23, 24, and 25.

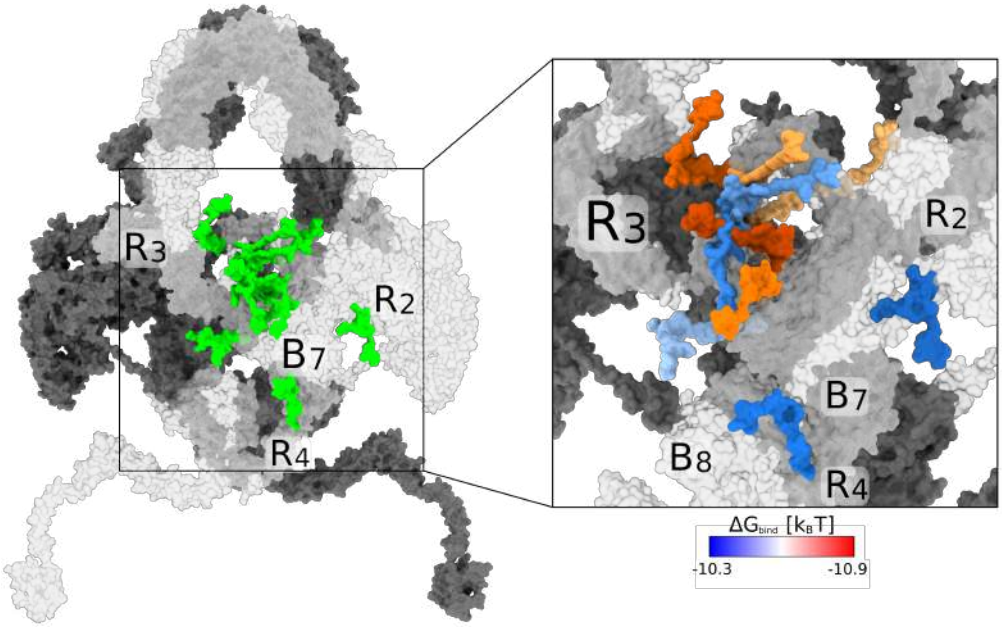


Figure 6.6: Top ten LRP1-Angiopep2 poses identified by the evaluation of ΔG_{bind} and ΔG_{desol} (see text for details).

2.1 Estimation of the binding time

Knowing the free energy of desolvation as an enthalpic barrier, it is possible to estimate the binding time. We consider one of the cases analyzed in the previous Section 5, illustrated in Figure 6.7.

In this specific pose, we observe that seven of the nineteen amino acids from Angiopep2 are interacting with eight LRP1 amino acids from CB19 and CB25, which belong to the macrodomains R3 and R4, respectively. Hence, for the pose to exist, the two molecules need to remove water from the surface of fifteen amino acids. Following the values of

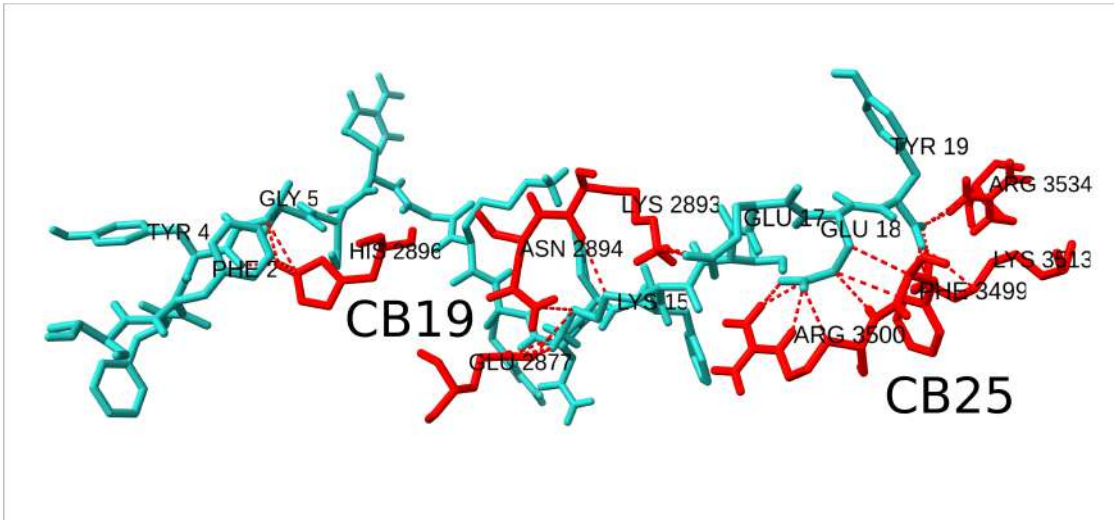


Figure 6.7: Pose of Angiopep2 interacting with CB units 19 (from R3 domain) and 25 (from R4 domain). The peptide is colored in cyan; LRP1 amino acids interacting with the peptide are colored in red.

Figure 6.4, we estimate that

$$\Delta G_{desol} = 20.2\ kT$$

is the enthalpy barrier to overcome for the creation of the bond. We can then calculate the time needed for the bond to occur, as

$$t_{bond} = \tau_0 \exp(\Delta G_{desol}) \tag{6.1}$$

where τ_0 is the scale time for the atomic interaction in this event, assumed to be $\sim 10^{-12}\ s$. Substituting the values in the Equation 6.1, we obtain that the time scale for the system to overcome the enthalpy barrier is $0.6\ ms$.

3 Methods

Molecular dynamics trajectories and RDF computation

The conserved amino acids in the CB domains were used as reference sites to calculate the RDF of water. The calculations were performed using GROMACS [110] with the standard RDF analysis tool. In this setup, water oxygen atoms were selected as the solvent reference, and sidechain heavy atoms of the conserved amino acids were used as solute centers. The RDF was sampled every 0.02 Å over a 40 Å interval and averaged across the final 200 ns of equilibrated trajectories.

Correction of the excluded protein volume

Because protein atoms exclude part of the solvent shell around the reference amino acid, the raw RDF $g(r)$ underestimates the local water density. To account for this, we introduced a renormalization factor based on the excluded volume $V_{aa}(r)$ at each radial distance r, yielding the corrected RDF:

$$g'(r) = g(r)\frac{4\pi\Delta r^3/3}{4\pi\Delta r^3/3 - V_{aa}(r)}. \tag{6.2}$$

Two complementary strategies were used to estimate $V_{aa}(r)$.

Surface element method. For short distances ($r < 15$ Å), the shell surface was discretized into elements $dS = r^2 \sin\theta\, d\theta\, d\phi$, as shown in Figure 6.8. The center of each element was tested against nearby atoms. If the distance d_{sa} between a surface element center c_{el} and an atom center C_{at} was smaller than the atomic van der Waals radius R^{at}_{vdW},

the element was considered occupied. This can be written as:

$$V_{aa}(r,\theta,\phi) = r^2 \Delta r \sum_{\theta} \sum_{\phi} \sin\theta \Delta\theta \Delta\phi \; H(R^{at}_{vdW} - d_{sa}), \tag{6.3}$$

where H is the Heaviside step function.

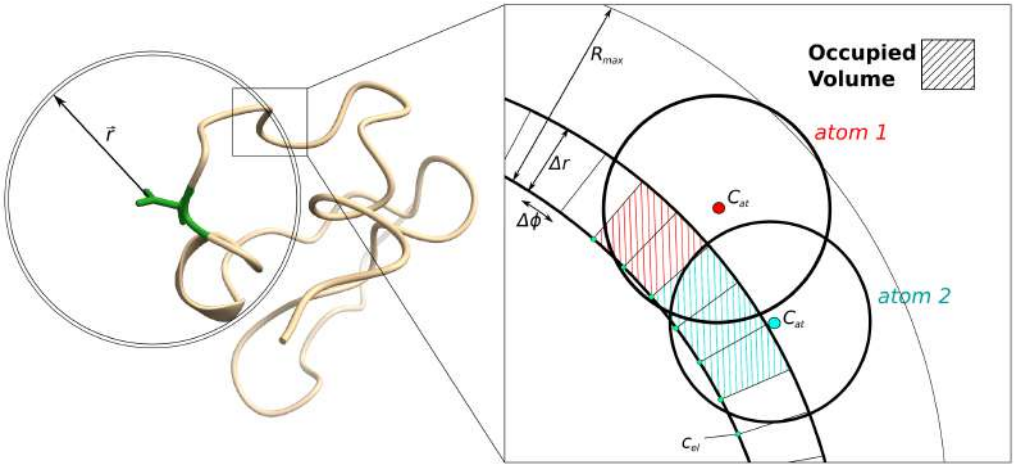


Figure 6.8: Surface element method for the calculation of the excluded volume V_{aa}. If the center of a surface element falls inside the van der Waals sphere of a protein atom, that element is flagged as occupied and removed from the hydration volume.

Analytical method. For larger distances ($r > 15$ Å), the excluded volume was computed by approximating the intersection of the solvent shell with protein atoms as spherical segments, as shown in Figure 6.9. The occupied volume contribution from each atom i was expressed as

$$V_{sz} = \sum_i \pi \Delta r \left[(R^{at}_{vdW})^2 - |\vec{d}_i - \vec{r}|^2 \right] + \frac{\pi \Delta r^3}{6}, \tag{6.4}$$

where $|\vec{d}_i - \vec{r}|$ is the distance between the atom center and the shell. Although this method can slightly overestimate $V_{aa}(r)$ due to overlaps, the error is negligible at large r.

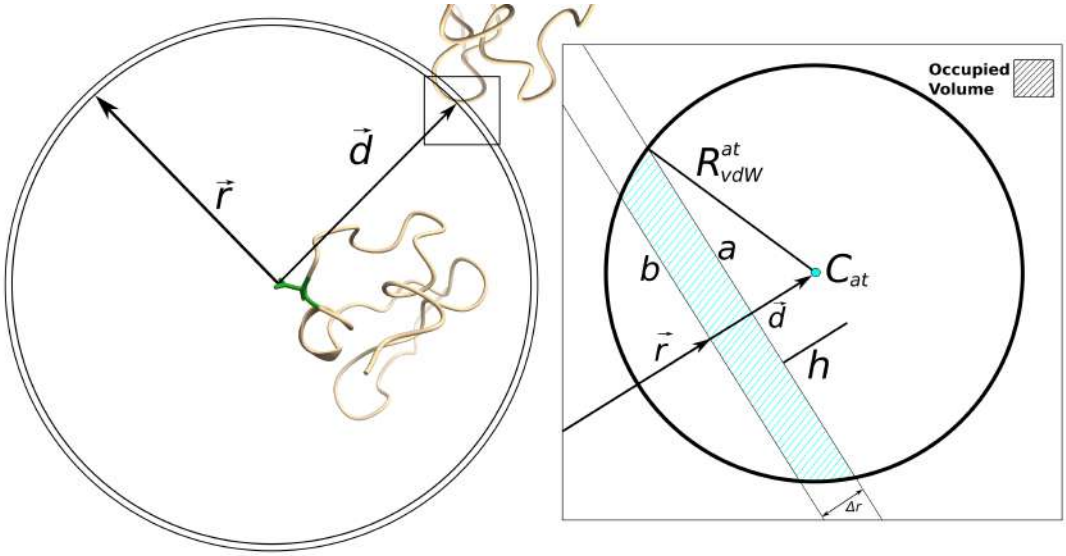


Figure 6.9: Analytical method for the excluded volume. For $r > 15$ Å, the overlap of the shell with an atom's van der Waals sphere is treated as a spherical segment of thickness Δr.

Construction of the correction factor $A(r)$

To apply the excluded volume correction efficiently, we did not compute $V_{aa}(r)$ at every sampling point. Instead, a correction factor $A(r)$ defined below was calculated on a discrete set of radii $x \in X$, combining the two complementary strategies described above. For the first interval (1–15 Å), the surface element method was applied with high resolution (0.5 Å spacing), capturing fine details of the first hydration shells. For the second interval (16–40 Å), the analytical method was used with a coarser sampling (2 Å spacing), which is sufficient for long-range corrections where $V_{aa}(r)$ is small.

At each sampled distance x, the corrected RDF $g'(x)$ was computed according to Eq. 6.2, and the ratio

$$A(x) = \frac{g'(x)}{g(x)} \tag{6.5}$$

was evaluated. The resulting discrete values of $A(x)$ were linearly interpolated over the entire r domain, yielding a continuous function $A(r)$.

The final corrected RDF was obtained as

$$g'(r) = A(r)\, g(r), \tag{6.6}$$

which preserves the peak positions of the raw RDF $g(r)$ while restoring the correct peak intensities by accounting for the excluded protein volume.

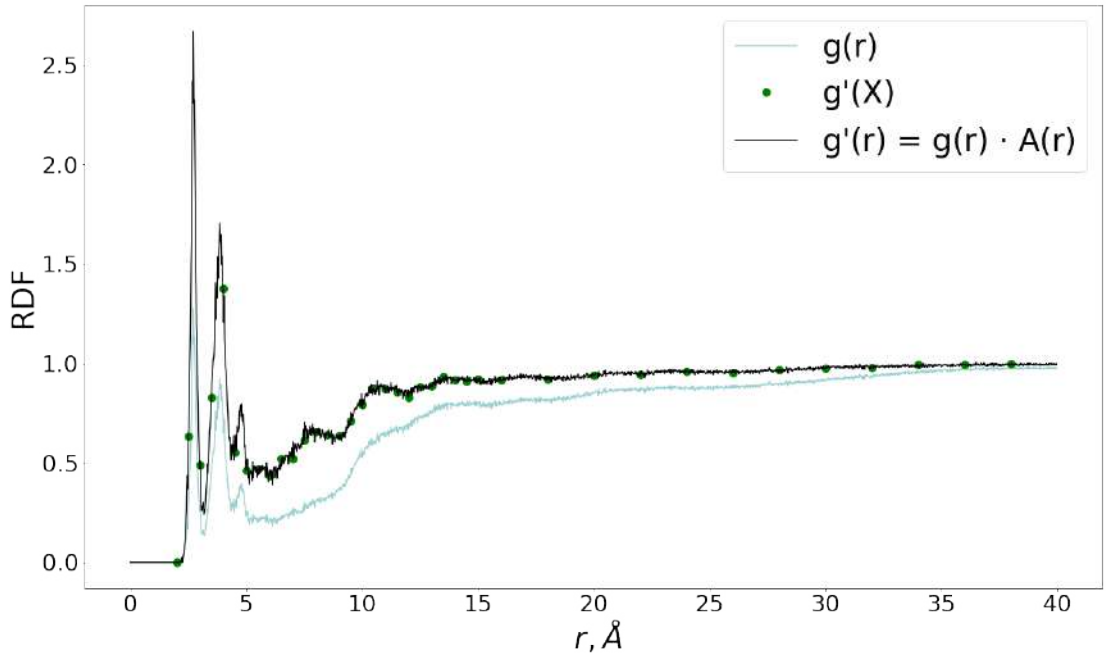


Figure 6.10: Example of amino acid-water RDF renormalization by removal of the protein-occupied volume.

The pair potential is then found iteratively as discussed in the Methodologies, Equations 2.17-2.21 in Section 5.1. An example of the potential for the RDF in Figure 6.10 is shown in Figure 6.11.

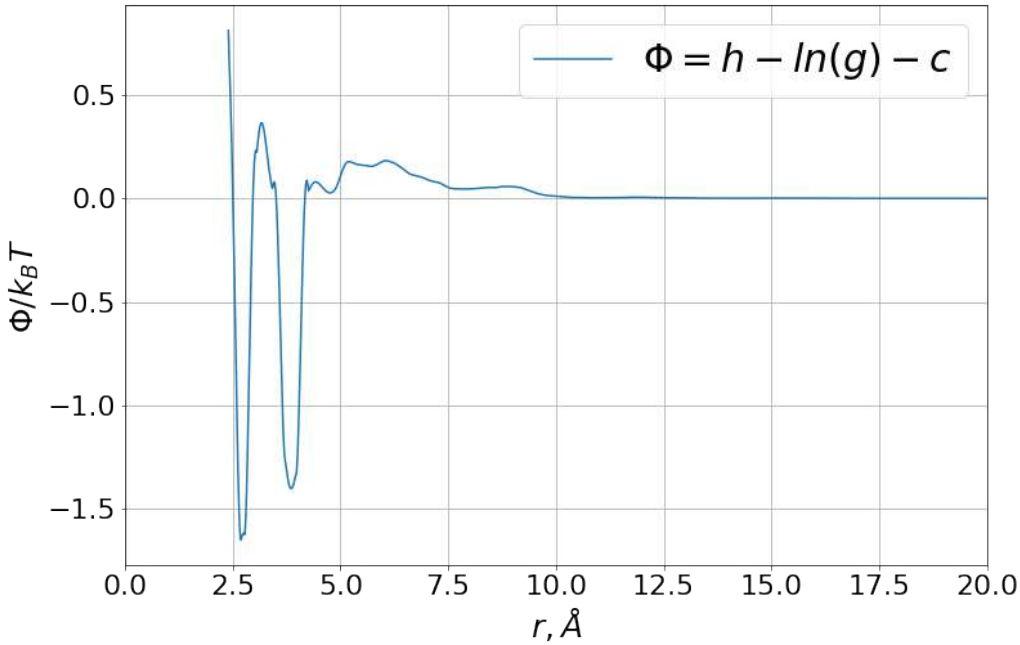


Figure 6.11: Amino acid-water interaction potential $\Phi(r)$ calculated from the corrected RDF in Figure 6.10 using Equation 2.21. Two distinct basins are visible at approximately 2.6 Å and 3.7 Å, corresponding to the first and second hydration shells around the reference amino acid sidechain. The depth of these minima reflects the strength of water–residue interactions in each shell.

4 Conclusions

The hydration analysis presented in this chapter underscores the importance of solvent accessibility as an additional layer of structural organization in LRP1. Rather than behaving as isolated chemical entities, conserved residues are embedded within a dynamic tertiary framework that shapes their interactions with water. This explains why amino acids of identical types may exhibit markedly different hydration profiles depending on their structural context.

These findings reinforce the notion that solvation is closely linked to protein flexibility. In LRP1, the extended and non-globular arrangement of CB motifs generates heterogeneous hydration landscapes, where local coiling or glycan shielding can alter solvent exposure in ways that are not predictable solely from the sequence. Hydration, therefore, emerges as both a consequence of structural dynamics and a potential driver of conformational adaptation, as it stabilizes or destabilizes particular local states.

From a functional perspective, this coupling between hydration and flexibility is highly relevant for receptor activity. Water contributes directly to the electrostatic environment that mediates ligand recognition, and fluctuations in solvation may help explain the receptor's capacity to engage multiple ligands with different affinities.

The extension of the analysis to all amino acid types demonstrated that solvation energetics can be interpreted in terms of desolvation enthalpies. This approach allowed us to quantify the energetic barrier associated with removing water molecules from binding interfaces. In the specific case of Angiopep2 binding to CB19 and CB25, we estimated a desolvation enthalpy of approximately $20kT$, which corresponds to a binding timescale of ~ 0.6 ms. This timescale lies within the microsecond–millisecond window typically observed for biomolecular recognition events [126], underscoring that desolvation directly governs the kinetics of ligand recognition at the BBB.

More generally, the results highlight that hydration patterns should not be regarded as passive background features but rather as active components of the protein's free energy landscape.

In summary, solvation in LRP1 illustrates a broader principle: the behavior of conserved residues cannot be understood in isolation but must be interpreted within the dynamic structural and solvent environment of the receptor. This perspective complements the analyses of domain flexibility and glycan sterics developed in previous chapters. It

now extends to docking energetics, providing a more integrated view of how structure, hydration, and dynamics converge to shape the function of LRP1.

Chapter 7

The therapeutic nanoparticles

As discussed throughout the thesis, LRP1 is central to ligand recognition and transport across the BBB. Once its structure is established, a natural next step is to examine the therapeutic nanoparticles designed to target this receptor.

Polyethylene oxide-block-polylactide (PEG-b-PLA) diblock copolymers are widely used in soft-matter systems, exhibiting diverse self-assembled morphologies [127]. Their popularity in nanomedicine stems from the high biocompatibility and low toxicity of both blocks, together with the degradability of PLA [128]. These properties have supported FDA approvals of PEG-b-PLA for multiple medical applications [129]. As a result, PEG-b-PLA micelles are extensively explored as drug delivery systems, with substantial innovation in cancer therapy [129, 130, 131].

In this chapter, we characterize PEG_{46}-PLA_{16} micelles from two complementary perspectives. First, we analyze their dynamical features, including the average micelle radius and the distribution of polymers within the spherical assembly. Second, we investigate their electrostatic properties through the calculation of the zeta potential, a key descriptor of charged interfaces. The zeta potential provides insight into colloidal stability and

nanoparticle interactions with the biological environment, linking the intrinsic chemistry of the PEG-PLA copolymer to the long-term electrostatic behavior of the assembled micelles. In this way, both structural and electrostatic aspects are addressed to better understand how PEG-PLA micelles may exploit LRP1-mediated pathways for therapeutic delivery across the BBB.

1 Building the micelle

We first modeled a single PEG_{46}–PLA_{16} polymer. The scheme is shown in Figure 7.1.

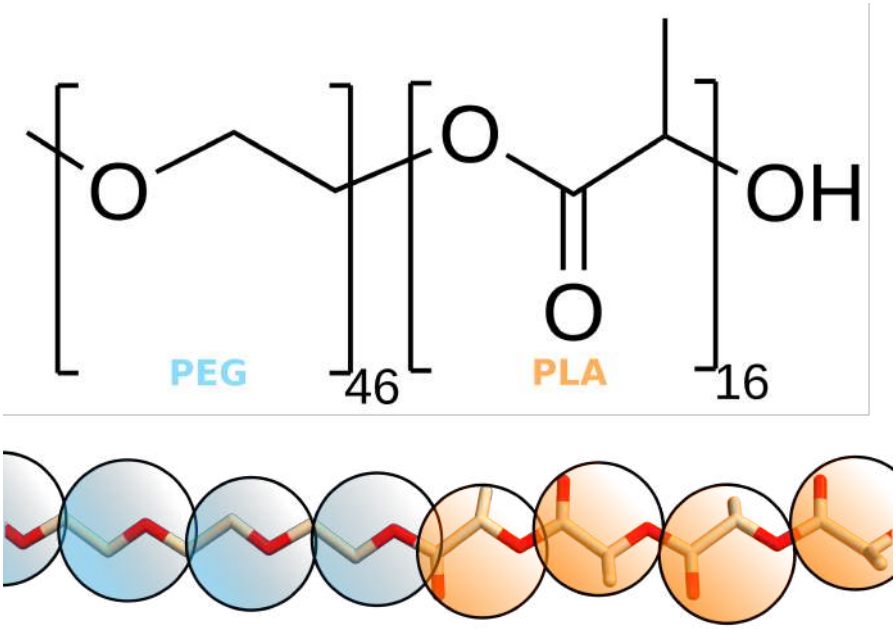


Figure 7.1: a. Polymer scheme of PEG_{46}–PLA_{16}. b. Stick representation of an atomic PEG–PLA junction (hydrogens excluded). Colored spheres highlight each monomer.

We employed coarse-grained molecular dynamics with the MARTINI 2.2 force field [132]. Coarse-graining reduces the number of particles (beads), enabling larger systems and longer trajectories. We generated an initial linear topology by placing beads at fixed

neighbor-dependent distances and saving the configuration as a `pdb`. These distances are nominal and chosen to be close to equilibrium. The initial polymer length is 22.8 nm.

We first evolved the isolated polymer in vacuo to promote compaction. To preserve chain integrity, we used only bond terms in this step and omitted angles and dihedrals. All bonded parameters are reported in Table C.1 in Appendix C, following López-Rios de Castro [133].

The compact conformation is shown in Figure 7.2. Two features are clear. First, the coiled polymer spans roughly one tenth of the nominal length. Second, the chain segregates into a PLA-rich core (orange) and a PEG-rich corona (blue). This arrangement follows from the parameters and is consistent with experiments [133].

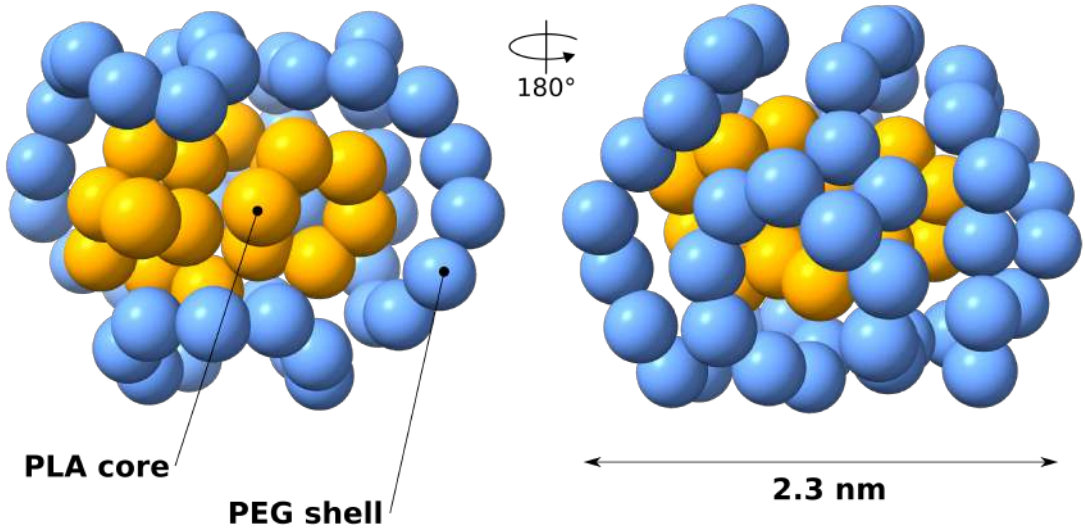


Figure 7.2: PEG_{46}–PLA_{16} copolymer after initial MD. The polymer contracts to $\sim$ 10% of its nominal length and segregates into a PLA core and PEG corona.

We replicated the conformation in Figure 7.2 45 times and placed the replicas on a cubic lattice in a 24.4 nm box. The system was solvated with refined polarizable MARTINI water (refPOL) [134] and NaCl [132]. The copolymers aggregated spontaneously into two micelles of 23 chains each, reproducing the PLA-core/PEG-corona scheme. Because

experiments indicate an average aggregation number of 46, we applied a mild external driving to overcome PEG–PEG repulsion between nascent micelles. Further MD yielded a single micelle, shown in Figure 7.3.a.

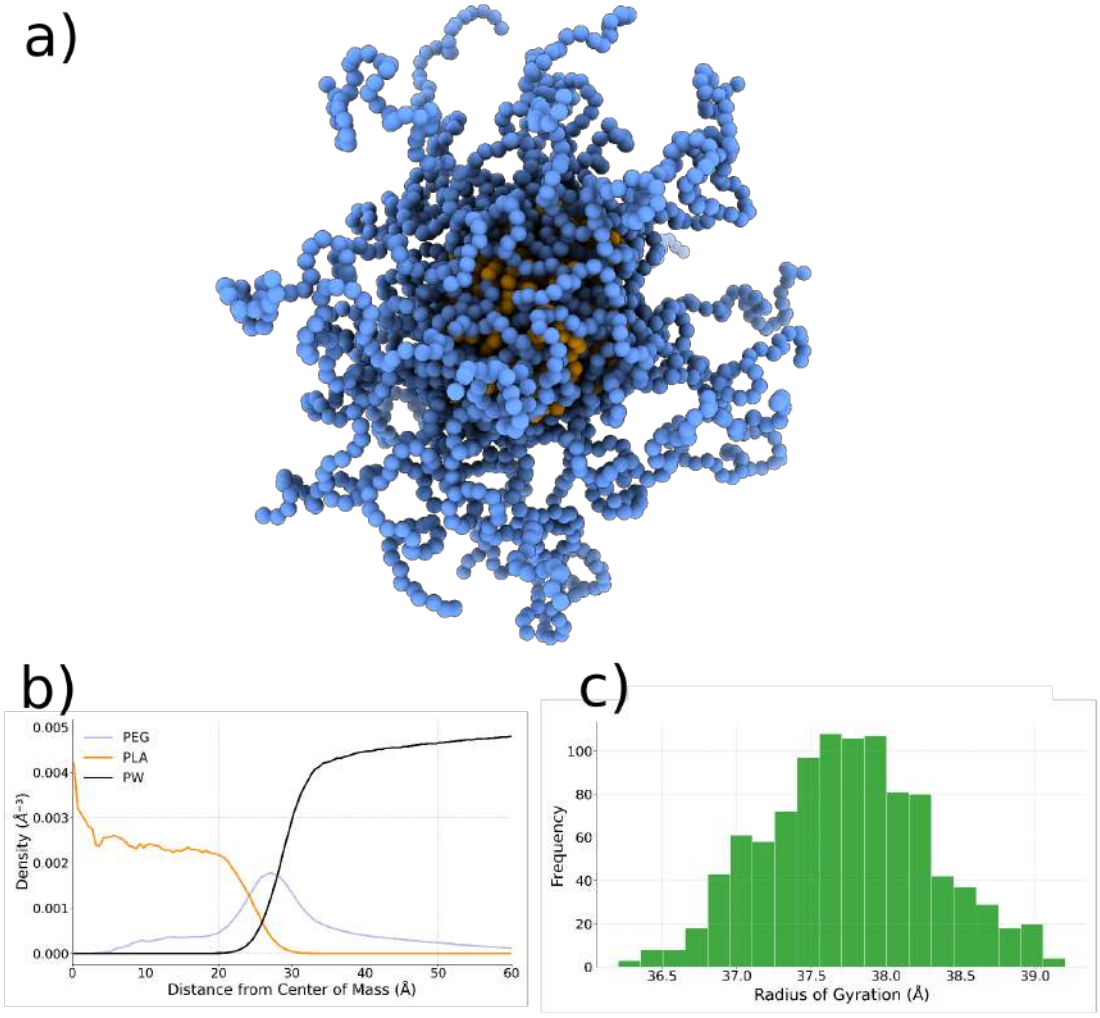


Figure 7.3: a. Equilibrated PEG_{46}–PLA_{16} micelle of 46 copolymers. b. Radial density profile of PEG, PLA, and polarizable water (PW). c. Radius of gyration over the final 20 ns.

The micelle presents a clear PLA core and a PEG brush. Figure 7.3.b shows a PLA-dense interior and a PEG density peak at ~ 2.8 nm, marking the interface. At this radius, water hydrates the surface, as indicated by the PW profile. PEG density remains lower than

PLA in the core because PEG prefers water over self-contact. The brush extends beyond the density peak, as seen in the radius of gyration in Figure 7.3.c, in agreement with Molecular Bionics' experiments.

2 The micelle's zeta potential

The zeta potential is the electrostatic potential at the shear plane separating mobile liquid from the stagnant fluid at an interface. It is used to infer colloidal stability, characterize charged nanoparticles, and interpret electrokinetic phenomena such as electrophoresis and electroosmosis [135, 136]. In biological and nanotechnological contexts, it is a practical proxy for surface charge density, informing interactions of membranes, polymers, and nanocarriers with their environment [137, 138].

Experimentally, the zeta potential is inferred from electrophoretic mobility or streaming potential and related to the interfacial potential through continuum models such as the Helmholtz–Smoluchowski equation [135, 136]. Molecular dynamics complements these approaches by providing atomistic estimates and highlighting the roles of interfacial water structure, ion adsorption, and dipoles [139, 140, 141].

The concept is not universally applicable. It presumes a well-defined shear plane, which may fail for rough, heterogeneous, or molecularly complex surfaces such as lipid bilayers [142, 143]. Continuum models can also underestimate interfacial effects, neglecting ion-specific adsorption, solvent structuring, or local viscosity changes [144, 145]. In such cases, the zeta potential becomes an effective descriptor rather than an absolute state variable.

Thus, while powerful, the zeta potential must be interpreted in the context of the system and the assumptions used to obtain it.

Results

The presence of charged species within the micelle is supported by experimental measurements of a negative zeta potential in an almost pure-water environment. Moreover, the zeta potential is not stable in time, but it shows a decrease. As the zeta potential in this condition is due principally to the micelle charge, we claim that the micelle charge is decreasing in time. A plausible molecular origin is a degradation of the PEG-PLA co-polymer, where the bond between the two monomer types is spontaneously hydrolyzed, leaving residual PEG^+ and PLA^- fragments; the chemical phenomena have been described by Shokri and colleagues [146]. The bond break allows PEG^+ to be dispersed in solution, while PLA^- is retained at the micelle core, as its behavior is still overall hydrophobic. The concentration of charged PLA monomers at the micelle core surface generates a negatively charged micelle.

To test this hypothesis, we modified the micelle from Section 1 to include a controlled fraction of charged components. We removed a selected percentage of PEG from the PLA core to generate micelles with a tunable net charge. Table 7.1 reports the absolute charges used. All systems were prepared as detailed in the Methods.

PLA^- charge %	**Absolute charge (e^-)**
0	0
5	2
10	5
15	7
20	9
25	12
30	15
35	16
40	18
45	21
50	23

Table 7.1: Percentage and absolute micelle charge used in this work.

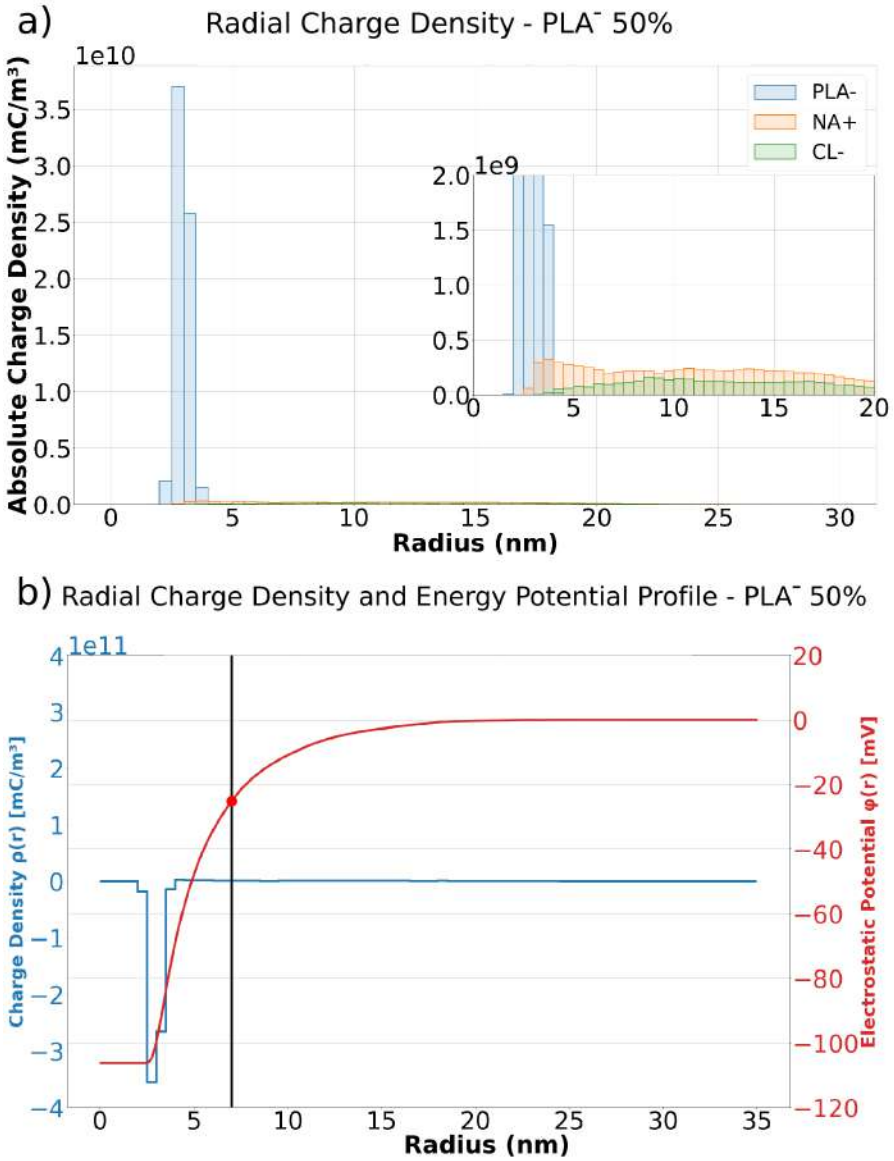


Figure 7.4: a. Absolute radial charge density for each charged species as a function of distance from the micelle core. b. Radial charge density of all charged species and electrostatic potential. The black vertical line marks the slip plane; the red dot indicates the zeta potential ζ.

We first computed the absolute radial charge density. In the representative case of Figure 7.4.a, PLA^- charges remain confined within the micelle radius of ~ 3.7 nm, consistent with the previous section. Na^+ shows a mild accumulation near the surface,

yet trajectories reveal no persistent trapping within the core or the PEG brush. Beyond ~ 5 nm, the ion density approaches bulk values, indicating electroneutrality.

We then evaluated the electrostatic potential profile as a function of radial distance using the formalism presented in Appendix C. Figure 7.4.b shows a nearly constant potential within the micelle and a gradual decay outside.

Estimating the zeta potential requires knowledge of the slip-plane location; in our case, the slip plane is a spherical surface centered in the core of the micelle, due to obvious symmetry reasons. We identified the slip plane at the radius where the net charge density approaches zero; a value common for all the systems is ~ 7 nm (Figure 7.4). This definition of zeta potential follows Melcr *et al.* [143], although no long-term ionic adsorption was observed in our trajectories.

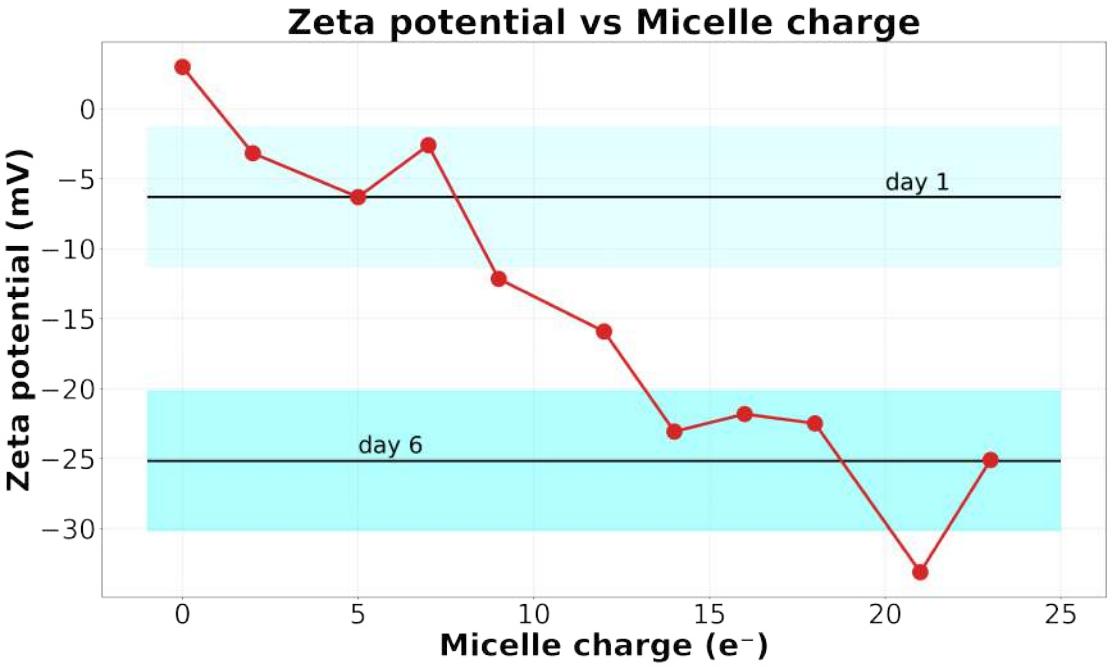


Figure 7.5: Zeta potential calculated for all the micelles of Table 7.1 as a function of the micelles' charge. Black horizontal lines indicate the experimental zeta potential with the relative uncertainty (cyan bands).

Using Eq. C.7, we obtain $\zeta \simeq -25$ mV for the micelle carrying 23 e^- of net charge. This estimate is marked by the red dot in Figure 7.4.b. All the charged species were subjected to the same calculations; all the results are listed in Appendix C.

The calculation of ζ for all charge populations yields the plot shown in Figure 7.5. Here, ζ is plotted as a function of the micelle charge. The tendency to have a lower zeta potential as the micelle charge decreases is evident. The main result of this analysis, however, is the superimposition of the plot with the experimental results. We observe that the reduction in the micelles' zeta potential over time is consistent with an effective decrease in the micelle's charge. The hypothesis of spontaneous hydrolysis explains this observation.

3 Methods

Molecular dynamics simulations were performed using GROMACS [110] with an MD integrator and a timestep of 0.020 ps. Each trajectory was propagated for 250 ns. We used the Verlet cutoff scheme with periodic boundary conditions in all directions. Electrostatics employed the reaction-field method with dielectric constant 2.5 and real-space cutoff 1.1 nm. Van der Waals interactions were truncated at 1.1 nm with a potential-shift–Verlet modification. Temperature was maintained at 303.15 K using the v-rescale thermostat (coupling constant 1.0 ps). Pressure was controlled with the C-rescale barostat in isotropic mode (coupling constant 5.0 ps, compressibility 4.5×10^{-5} bar^{-1}).

Each micelle was simulated in a $\sim$36 nm box, solvated with $\sim 3 \times 10^5$ MARTINI polarizable water molecules [134] and 0.001 M NaCl [132]. Na^+ ions compensated the micelle net charge.

Analyses were carried out over the final 20 ns using in-house Python code based on MDAnalysis [111].

4 Conclusions

This chapter establishes a quantitative picture of PEG_{46}–PLA_{16} micelles designed for LRP1 targeting across the BBB. Coarse-grained simulations reproduce the expected self-assembly. The chains compact and segregate into a PLA-rich core and a PEG brush. Radial density profiles and radii of gyration confirm a sharp core–corona interface hydrated by water.

We then linked interfacial charge organization to electrokinetic response. The radial charge analysis and the Poisson reconstruction provide a consistent electrostatic potential profile. Identifying the slip plane where the net charge density vanishes enables a direct estimate of the zeta potential. The resulting ζ values decrease as the micelle charge is reduced. This behavior overlaps with the experimental trend. The agreement supports a mechanistic picture in which copolymer hydrolysis releases PEG^{+} to solution and leaves PLA^{-} at the core interface, lowering ζ over time.

These findings inform the design of LRP1-targeted nanocarriers. A stable PEG brush and a moderately negative ζ favor colloidal stability, reduce nonspecific adsorption, and help tune avidity in the mid-range required for transcytosis. They also suggest that controlling copolymer integrity is essential to maintain electrostatic properties during circulation.

The study has limitations. The MARTINI description coarsens interfacial water and ion specificity. The reaction-field electrostatics and the slip-plane construction introduce model assumptions. Future work should refine the electrostatics at atomistic resolution, probe ion-specific adsorption, and couple simulations with time-resolved measurements of ζ and aggregation. Such integration will improve control of surface charge, stability, and receptor engagement for efficient BBB delivery.

Chapter 8

Conclusions

This thesis has investigated the structural and dynamical features of the low-density lipoprotein receptor–related protein 1 (LRP1), combining molecular modeling, molecular dynamics simulations, docking analyses, and coarse-grained studies of therapeutic nanoparticles. By integrating different levels of resolution and complementary methodologies, we have highlighted how structural motifs, hydration, glycans, and nanocarrier electrostatics converge to shape the function of this receptor and its potential exploitation in brain drug delivery.

Structural Modeling of LRP1

A first major achievement was the generation of reliable structural models of LRP1, a protein for which no complete experimental structure exists (Chapter 3). By benchmarking neural-network–based predictors (AlphaFold2, RoseTTAFold) and integrating them with homology modeling from LRP2, we reconstructed both monomeric and dimeric conformations. These models revealed the modular architecture of LRP1, composed of repeated calcium-binding (CB), β-propeller (B), and EGF-like motifs, and highlighted

structural similarities with its homolog LRP2. Although global prediction of such a large receptor remains limited by clashes and uncertainties in domain orientations, the domain-by-domain approach demonstrated that meaningful large-scale models can be obtained by careful integration of local predictions.

Protein Flexibility, Hydration, and Free Energy Landscapes

Molecular dynamics simulations (Chapter 4) showed that the R clusters of CB motifs display pronounced flexibility, coiling into equilibrium conformations that are not evident from static structures. These rearrangements are not driven by new covalent or hydrogen-bond networks but by solvent accessibility and steric constraints, emphasizing the dynamic nature of LRP1.

Hydration emerged as a key factor shaping this behavior (Chapter 6). By developing a new correction scheme for radial distribution functions and solving the Ornstein–Zernike closure iteratively, we quantified the interaction potential between conserved amino acids and water. The analysis revealed that identical amino acids, depending on their structural context, can exhibit markedly different hydration patterns. Charged residues, such as Lysine and Aspartic acid, consistently showed a strong affinity for water, while Cysteine and aromatic residues were often shielded.

From this data, we constructed free-energy maps and extended the approach to estimate desolvation enthalpies for ligand binding. For example, in representative docking poses of Angiopep2, the removal of hydration shells from multiple amino acids generated enthalpy barriers of tens of k_BT, consistent with millisecond-scale binding times. This analysis provides a direct link between hydration patterns and binding kinetics, showing that desolvation is a fundamental barrier that must be overcome in ligand recognition. The explicit estimation of desolvation enthalpy ($\sim 20\,k_BT$ for Angiopep2–CB19/25 binding) and its translation into a binding timescale (~ 0.6 ms) demonstrates that hydration is

not only a stabilizing factor but also a kinetic regulator of receptor–ligand interactions. These findings align with the microsecond–millisecond window observed in biomolecular binding events [126], reinforcing the predictive power of this framework.

Role of Glycans and Dimerization

Post-translational glycosylation was shown to introduce a strong steric and electrostatic component to the receptor surface (Chapter 5). By implementing polymer-based models of glycans, we estimated their repulsive contribution to the free energy landscape and demonstrated how they shield binding regions.

Simulations of the canopy region in dimeric LRP1 (Chapter 4) further revealed that glycans stabilize the closed dimer conformation. Fully glycosylated dimers maintained compact structures through protein–glycan and glycan–glycan hydrogen bonds, while non-glycosylated systems adopted an open, pore-like arrangement. These results underline that glycosylation is not merely decorative but a structural determinant of receptor organization and possibly trafficking.

LRP1–Angiopep2 Interactions

Docking analyses of Angiopep2 (Chapter 5) highlighted that LRP1 does not present a single, well-defined binding site but rather an extended surface of overlapping binding regions. Global docking suggested multiple propeller and CB contacts, while local docking refined these predictions by incorporating calcium binding sites and electrostatics. The outcome is a picture of “binding regions” rather than “binding pockets”, consistent with the receptor’s ability to interact with diverse ligands.

By projecting docking poses onto a free-energy density surface, we defined a continuum of possible binding configurations shaped by both motif arrangement and solvent

accessibility. This framework provides a structural explanation for the observed "Goldilocks avidity" in nanoparticle trafficking, where multiple weak peptide–receptor interactions optimize transcytosis without triggering degradation.

Therapeutic Nanoparticles and Zeta Potential

Finally, we extended the analysis to the nanoparticles designed to exploit LRP1, focusing on PEG-b-PLA micelles (Chapter 7). Coarse-grained simulations reproduced their self-assembly into PLA-rich cores surrounded by PEG coronas, consistent with experimental data.

We then investigated their electrostatic properties by computing zeta potentials from atomistic simulations. By introducing controlled PEG/PLA charge defects, we showed that micelle zeta potential decreases (increases in absolute value) with effective charge, in line with experimental measurements. This trend supports the hypothesis that spontaneous hydrolysis of the PEG–PLA junction generates charged fragments, thereby modulating micelle electrostatics over time. Such findings link the intrinsic chemical stability of polymers to their biological performance.

Integrated View and Perspectives

The results of this thesis converge on a central idea: the function of LRP1 is defined not by isolated motifs or single interactions, but by the interplay of modular structure, hydration, glycans, and binding energetics. At the receptor level, the interaction with Angiopep2 illustrates how LRP1 achieves molecular recognition through extended binding regions rather than a discrete pocket. Hydration analyses and docking results show that binding is governed by a combination of solvent accessibility, desolvation enthalpies, and the flexible arrangement of CB and β-propeller motifs. This explains how Angiopep2

can engage multiple regions of LRP1 with weak, distributed contacts, giving rise to a continuum of binding configurations that define the receptor's "Goldilocks" avidity. Such mid-avidity interactions enable efficient transcytosis across the BBB while avoiding receptor saturation or degradation. In this sense, the LRP1–Angiopep2 system provides a molecular framework for understanding how dynamic solvation and structural plasticity underlie ligand trafficking.

On the other hand, therapeutic nanoparticles exploit these principles at a higher level of organization. PEG–PLA micelles do not interact with LRP1 directly through a single binding epitope, but through their surface decoration with Angiopep2. The polymeric architecture and electrostatics of the micelles, especially their zeta potential, determine how these ligands are presented and how the receptor perceives the nanocarrier. Our simulations demonstrate that micelle stability and charge distribution depend not only on polymer chemistry but also on dynamic processes such as spontaneous hydrolysis, which modulates the zeta potential over time. Thus, while Angiopep2 defines the molecular interface with LRP1, the micelle environment regulates how that interface is displayed and maintained during trafficking.

Taken together, these two perspectives highlight a multi-scale view of brain delivery through LRP1. At the molecular scale, solvent-mediated recognition and flexible binding regions allow LRP1 to accommodate diverse ligands such as Angiopep2. At the supramolecular scale, the design of nanoparticles must consider polymer stability, glycan shielding, and electrostatics, as these factors ultimately dictate the efficiency of receptor engagement and transcytosis. Future efforts should therefore bridge these scales, integrating atomic-level binding thermodynamics with nanoparticle formulation, to optimize nanocarriers that exploit LRP1 as a gateway to the brain.

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Bibliography

1. May, P., Woldt, E., Matz, R. L. & Boucher, P. The LDL receptor-related protein (LRP) family: An old family of proteins with new physiological functions. *Annals of Medicine* **39,** 219–228. ISSN: 0785-3890. http://www.tandfonline.com/doi/full/10.1080/07853890701214881 (3 Jan. 2007).

2. Strickland, D. K., Gonias, S. L. & Argraves, W. S. Diverse roles for the LDL receptor family. *Trends in Endocrinology and Metabolism* **13,** 66–74. ISSN: 10432760. https://www.cell.com/action/showFullText?pii=S1043276001005264 (2 Mar. 2002).

3. Spuch, C., Ortolano, S. & Navarro, C. LRP-1 and LRP-2 receptors function in the membrane neuron. Trafficking mechanisms and proteolytic processing in alzheimer's disease. *Frontiers in Physiology* **3 JUL,** 29937. ISSN: 1664042X. www.frontiersin.org (July 2012).

4. Boucher, P. & Herz, J. Signaling through LRP1: Protection from atherosclerosis and beyond. *Biochemical Pharmacology* **81,** 1–5. ISSN: 0006-2952. https://www.sciencedirect.com/science/article/pii/S000629521000701X (1 Jan. 2011).

5. Yamamoto, T. *et al.* The human LDL receptor: A cysteine-rich protein with multiple Alu sequences in its mRNA. *Cell* **39,** 27–38. ISSN: 00928674. https://linkinghub.elsevier.com/retrieve/pii/0092867484901880 (1 Nov. 1984).

6. Daly, N. L., Scanlon, M. J., Djordjevic, J. T., Kroon, P. A. & Smith, R. Three-dimensional structure of a cysteine-rich repeat from the low-density lipoprotein receptor. *Proceedings of the National Academy of Sciences* **92,** 6334–6338. https://www.pnas.org/doi/abs/10.1073/pnas.92.14.6334 (14 July 1995).

7. Simonovic, M. *et al.* Calcium coordination and pH dependence of the calcium affinity of ligand-binding repeat CR7 from the LRP. Comparison with related domains from the LRP and the LDL receptor. *Biochemistry* **40,** 15127–15134. ISSN: 00062960. /doi/pdf/10.1021/bi015688m (50 Dec. 2001).

8. Springer, T. A. An extracellular β-propeller module predicted in lipoprotein and scavenger receptors, tyrosine kinases, epidermal growth factor precursor, and extracellular matrix components. *Journal of Molecular Biology* **283,** 837–862. ISSN: 0022-2836. https://www.sciencedirect.com/science/article/abs/pii/S0022283698921155 (4 Nov. 1998).

9. Jeon, H. *et al.* Implications for familial hypercholesterolemia from the structure of the LDL receptor YWTD-EGF domain pair. *Nature Structural Biology* **8,** 499–504. ISSN: 1545-9985. https://www.nature.com/articles/nsb0601_499 (6 June 2001).

10. Beenken, A. *et al.* Structures of LRP2 reveal a molecular machine for endocytosis. *Cell* **186,** 821. ISSN: 10974172. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9993842/ (4 Feb. 2023).

11. Herz, J. *et al.* Surface location and high affinity for calcium of a 500-kd liver membrane protein closely related to the LDL-receptor suggest a physiological role as lipoprotein receptor. *The EMBO Journal* **7,** 4119–4127. ISSN: 02614189. https://onlinelibrary.wiley.com/doi/10.1002/j.1460-2075.1988.tb03306.x (13 Dec. 1988).

12. Davis, C. G. *et al.* Acid-dependent ligand dissociation and recycling of LDL receptor mediated by growth factor homology region. *Nature* **326,** 760–765. ISSN: 1476-4687. https://www.nature.com/articles/326760a0 (6115 Apr. 1987).

13. Russell, D. W., Brown, M. S. & Goldstein, J. L. Different Combinations of Cysteine-rich Repeats Mediate Binding of Low Density Lipoprotein Receptor to Two Different Proteins*. *Journal of Biological Chemistry* **264,** 21682–21688 (36 1989).

14. Neels, J. G. *et al.* The Second and Fourth Cluster of Class A Cysteine-rich Repeats of the Low Density Lipoprotein Receptor-related Protein Share Ligand-binding Properties*. *Journal Lipid Research* **274,** 31305–31311. http://www.jbc.org (1999).

15. Jensen, G. A. *et al.* Binding Site Structure of One LRP–RAP Complex:Implications for a Common Ligand–Receptor Binding Motif. *Journal of Molecular Biology* **362,** 700–716. ISSN: 0022-2836 (4 Sept. 2006).

16. Bu, G. in *International review of cytology* 79–116 (2001). https://linkinghub.elsevier.com/retrieve/pii/S0074769601090118.

17. Prasad, J. M., Migliorini, M., Galisteo, R. & Strickland, D. K. Generation of a Potent Low Density Lipoprotein Receptor-related Protein 1 (LRP1) Antagonist by Engineering a Stable Form of the Receptor-associated Protein (RAP) D3 Domain. *The Journal of Biological Chemistry* **290,** 17262. ISSN: 1083351X. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4498065/ (28 July 2015).

18. May, P., Reddy, Y. K. & Herz, J. Proteolytic processing of low density lipoprotein receptor-related protein mediates regulated release of its intracellular domain. *Journal of Biological Chemistry* **277,** 18736–18743. ISSN: 00219258. http://www.jbc.org/article/S0021925820851540/fulltext (21 May 2002).

19. Sossin, W. S., Fisher, J. M. & Scheller, R. H. Cellular and molecular biology of neuropeptide processing and packaging. *Neuron* **2,** 1407–1417. ISSN: 0896-6273. https://www.cell.com/neuron/abstract/0896-6273(89)90186-4 (5 May 1989).

20. Gutmann, T. *et al.* Cryo-EM structure of the complete and ligand-saturated insulin receptor ectodomain. *Journal of Cell Biology* **219,** e201907210. ISSN: 0021-9525. https://doi.org/10.1083/jcb.201907210 (1 Nov. 2019).

21. Willnow, T. E., Moehring, J. M., Inocencio, N. M., Moehring, T. J. & Herz, J. The low-density-lipoprotein receptor-related protein (LRP) is processed by furin in vivo and in vitro. *The Biochemical Journal* **313 (Pt 1),** 71–76. ISSN: 0264-6021 (Jan. 1996).

22. Ko, K. W. S. *et al.* Mutation at the Processing Site of Chicken Low Density Lipoprotein Receptor-related Protein Impairs Efficient Endoplasmic Reticulum Exit, but Proteolytic Cleavage Is Not Essential for Its Endocytic Functions *. *Journal of Biological Chemistry* **273,** 27779–27785. ISSN: 0021-9258, 1083-351X. https://www.jbc.org/article/S0021-9258(19)59529-1/abstract (43 Oct. 1998).

23. Quinn, K. A. *et al.* Soluble Low Density Lipoprotein Receptor-related Protein (LRP) Circulates in Human Plasma *. *Journal of Biological Chemistry* **272,** 23946–23951. ISSN: 0021-9258, 1083-351X. https://www.jbc.org/article/S0021-9258(19)65877-1/abstract (38 Sept. 1997).

24. Quinn, K. A., Pye, V. J., Dai, Y.-P., Chesterman, C. N. & Owensby, D. A. Characterization of the Soluble Form of the Low Density Lipoprotein Receptor-Related Protein (LRP). *Experimental Cell Research* **251,** 433–441. ISSN: 0014-4827. https://www.sciencedirect.com/science/article/pii/S0014482799945901 (2 Sept. 1999).

25. Liu, Q. *et al.* LRP1 shedding in human brain: roles of ADAM10 and ADAM17. *Molecular Neurodegeneration* **4,** 1–7. ISSN: 1750-1326. https://molecularneuro-

%20degeneration.biomedcentral.com/articles/10.1186/1750-1326-4-17 (1 Dec. 2009).

26. Sizova, O., John, L. S., Ma, Q. & Molldrem, J. J. Multi-faceted role of LRP1 in the immune system. *Frontiers in Immunology* **14,** 1166189. ISSN: 16643224 (Mar. 2023).

27. Yang, L. *et al.* LRP1 modulates the microglial immune response via regulation of JNK and NF-KB signaling pathways. *Journal of Neuroinflammation* **13,** 1–13. ISSN: 17422094. https://link.springer.com/articles/10.1186/s12974-016-0772-7 (1 Dec. 2016).

28. Covarrubias, R., Wilhelm, A. J. & Major, A. S. Specific Deletion of LDL Receptor-Related Protein on Macrophages Has Skewed In Vivo Effects on Cytokine Production by Invariant Natural Killer T Cells. *PLOS ONE* **9,** e102236. ISSN: 1932-6203. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0102236 (7 July 2014).

29. Mishra, A. *et al.* LRP-1 Attenuates House Dust Mite-induced Eosinophilic Airway Inflammation by Suppressing Dendritic Cell-mediated Adaptive Immune Responses. *The Journal of allergy and clinical immunology* **142,** 1066. ISSN: 10976825. https://pmc.ncbi.nlm.nih.gov/articles/PMC6013321/ (4 Oct. 2017).

30. Li, S. S., Liu, Z., Uzunel, M. & Sundqvist, K. G. Endogenous thrombospondin-1 is a cell-surface ligand for regulation of integrin-dependent T-lymphocyte adhesion. *Blood* **108,** 3112–3120. ISSN: 00064971. https://pubmed.ncbi.nlm.nih.gov/16835379/ (9 Nov. 2006).

31. Yang, T.-H. *et al.* Membrane-Associated Proteinase 3 on Granulocytes and Acute Myeloid Leukemia Inhibits T Cell Proliferation. *The Journal of Immunology* **201,** 1389–1399. ISSN: 0022-1767. https://pubmed.ncbi.nlm.nih.gov/30021768/ (5 Sept. 2018).

32. Qu, L., Wang, F., Wang, Y. & Li, Z. The regulation of LRPs by miRNAs in cancer: influencing cancer characteristics and responses to treatment. *Cancer Cell International 2025 25:1* **25,** 1–22. ISSN: 1475-2867. https://link.springer.com/articles/10.1186/s12935-025-03804-z (1 May 2025).

33. Leslie, P. L., Franklin, D. A., Liu, Y. & Zhang, Y. p53 Regulates the Expression of LRP1 and Apoptosis through a Stress Intensity-Dependent MicroRNA Feedback Loop. *Cell Reports* **24,** 1484–1495. ISSN: 22111247. https://pubmed.ncbi.nlm.nih.gov/30089260/ (6 Aug. 2018).

34. Xue, N. *et al.* Secreted HSP90α-LRP1 Signaling Promotes Tumor Metastasis and Chemoresistance in Pancreatic Cancer. *International Journal of Molecular Sciences* **23.** ISSN: 14220067. https://pubmed.ncbi.nlm.nih.gov/35628341/ (10 May 2022).

35. Huang, W. T. *et al.* Evaluation of miR-429 as a novel serum biomarker for pancreatic ductal adenocarcinoma and analysis its tumor suppressor function and target genes. *European Review for Medical and Pharmacological Sciences* **26,** 4638–4653. ISSN: 22840729. https://pubmed.ncbi.nlm.nih.gov/35856354/ (13 2022).

36. Gopal, U. *et al.* A Novel Extracellular Hsp90 Mediated Co-Receptor Function for LRP1 Regulates EphA2 Dependent Glioblastoma Cell Invasion. *PLOS ONE* **6,** e17649. ISSN: 1932-6203. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0017649 (3 2011).

37. He, C. *et al.* LRP1-mediated pH-sensitive polymersomes facilitate combination therapy of glioblastoma in vitro and in vivo. *Journal of Nanobiotechnology* **19,** 1–17. ISSN: 14773155. https://jnanobiotechnology.biomedcentral.com/articles/10.1186/s12951-020-00751-x (1 Dec. 2021).

38. Boyé, K. *et al.* The role of CXCR3/LRP1 cross-talk in the invasion of primary brain tumors. *Nature Communications* **8.** ISSN: 20411723. https://pubmed.ncbi.nlm.nih.gov/29146996/ (1 Dec. 2017).

39. Ma, Q. *et al.* Blood-brain barrier-associated pericytes internalize and clear aggregated amyloid-β42 by LRP1-dependent apolipoprotein e isoform-specific mechanism. *Molecular Neurodegeneration* **13,** 1–13. ISSN: 17501326. https://link.springer.com/articles/10.1186/s13024-018-0286-0 (1 Oct. 2018).

40. Rauch, J. N. *et al.* LRP1 is a master regulator of tau uptake and spread. *Nature 2020 580:7803* **580,** 381–385. ISSN: 1476-4687. https://www.nature.com/articles/s41586-020-2156-5 (7803 Apr. 2020).

41. Qiu, Z., Strickland, D. K., Hyman, B. T. & Rebeck, G. W. α2-macroglobulin exposure reduces calcium responses to N-methyl-D-aspartate via low density lipoprotein receptor-related protein in cultured hippocampal neurons. *Journal of Biological Chemistry* **277,** 14458–14466. ISSN: 00219258. https://pubmed.ncbi.nlm.nih.gov/11839752/ (17 Apr. 2002).

42. Storck, S. E. *et al.* Endothelial LRP1 transports amyloid-β1–42 across the blood-brain barrier. *The Journal of Clinical Investigation* **126,** 123–136. ISSN: 0021-9738 (1 Jan. 2016).

43. Pflanzner, T. *et al.* LRP1 mediates bidirectional transcytosis of amyloid-β across the blood-brain barrier. *Neurobiology of Aging* **32,** 2323.e1–2323.e11. ISSN: 0197-4580. https://www.sciencedirect.com/science/article/abs/pii/S0197458010-%20002393 (12 Dec. 2011).

44. Tian, X. *et al.* On the shuttling across the blood-brain barrier via tubule formation: Mechanism and cargo avidity bias. *Science Advances* **6,** 4397–4424. ISSN: 23752-548. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7695481/ (48 Nov. 2020).

45. Tian, X., Angioletti-Uberti, S. & Battaglia, G. On the design of precision nanomedicines. *Science advances* **6.** ISSN: 2375-2548. https://pubmed.ncbi.nlm.nih.gov/32042891/ (4 Jan. 2020).

46. Guo, Q. *et al.* LRP1-upregulated nanoparticles for efficiently conquering the blood-brain barrier and targetedly suppressing multifocal and infiltrative brain metastases. *Journal of Controlled Release* **303,** 117–129. ISSN: 0168-3659. https://www.sciencedirect.com/science/article/abs/pii/S0168365919302317 (June 2019).

47. Mastrantuono, E., Ghibaudi, M., Matias, D. & Battaglia, G. The multifaceted therapeutical role of low-density lipoprotein receptor family in high-grade glioma. *Molecular Oncology* **18,** 2966–2976. ISSN: 18780261. /doi/pdf/10.1002/1878-0261.13730 (12 Dec. 2024).

48. Ying, X. *et al.* Angiopep-conjugated electro-responsive hydrogel nanoparticles: Therapeutic potential for epilepsy. *Angewandte Chemie - International Edition* **53,** 12436–12440. ISSN: 15213773. /doi/pdf/10.1002/anie.201403846 (46 Nov. 2014).

49. Di Polidoro, A. C. *et al.* Revealing Angiopep-2/LRP1 Molecular Interaction for Optimal Delivery to Glioblastoma (GBM). *Molecules 2022, Vol. 27, Page 6696* **27,** 6696. ISSN: 1420-3049. https://www.mdpi.com/1420-3049/27/19/6696/htm (19 Oct. 2022).

50. Demeule, M. *et al.* Identification and design of peptides as a new drug delivery system for the brain. *Journal of Pharmacology and Experimental Therapeutics* **324,** 1064–1072. ISSN: 00223565. https://jpet.aspetjournals.org/action/showFull-%20Text?pii=S0022356524343307 (3 Mar. 2008).

51. Demeule, M. *et al.* Involvement of the low-density lipoprotein receptor-related protein in the transcytosis of the brain delivery vector Angiopep-2. *Journal of*

Neurochemistry **106,** 1534–1544. ISSN: 1471-4159. /doi/pdf/10.1111/j.1471-4159.2008.05492.x (4 Aug. 2008).

52. Huang, R. *et al.* Angiopep-conjugated nanoparticles for targeted long-term gene therapy of parkinson's disease. *Pharmaceutical Research* **30,** 2549–2559. ISSN: 07248741. https://link.springer.com/article/10.1007/s11095-013-1005-8 (10 Oct. 2013).

53. Shao, K. *et al.* A brain-vectored angiopep-2 based polymeric micelles for the treatment of intracranial fungal infection. *Biomaterials* **33,** 6898–6907. ISSN: 0142-9612. https://www.sciencedirect.com/science/article/abs/pii/S0142961212-%20007004 (28 Oct. 2012).

54. Zhang, W. *et al.* Optimizing Angiopep-2 Density on Polymeric Nanoparticles for Enhanced Blood–Brain Barrier Penetration and Glioblastoma Targeting: Insights From In Vitro and In Vivo Experiments. *Advanced Functional Materials,* 2425165. ISSN: 1616-3028. /doi/pdf/10.1002/adfm.202425165 (2025).

55. Anfinsen, C. B. Principles that govern the folding of protein chains. *Science* **181,** 223–230. ISSN: 00368075. /doi/pdf/10.1126/science.181.4096.223 (4096 July 1973).

56. Huang, B. *et al.* Protein Structure Prediction: Challenges, Advances, and the Shift of Research Paradigms. *Genomics, Proteomics & Bioinformatics* **21,** 913–925. ISSN: 1672-0229. https://www.sciencedirect.com/science/article/pii/S167202292-%203000657 (5 Oct. 2023).

57. Šali, A. & Blundell, T. L. Comparative protein modelling by satisfaction of spatial restraints. *Journal of Molecular Biology* **234,** 779–815. ISSN: 00222836. https://pubmed.ncbi.nlm.nih.gov/8254673/ (3 Dec. 1993).

58. Martí-Renom, M. A. *et al.* Comparative protein structure modeling of genes and genomes. *Annual Review of Biophysics and Biomolecular Structure* **29,** 291–325. ISSN: 10568700. https://pubmed.ncbi.nlm.nih.gov/10940251/ (2000).

59. Fiser, A., Do, R. K. G. & Šali, A. Modeling of loops in protein structures. *Protein Science* **9,** 1753–1773. ISSN: 0961-8368. /doi/pdf/10.1110/ps.9.9.1753 (9 Jan. 2000).

60. Eswar, N. *et al.* Comparative Protein Structure Modeling Using Modeller. *Current Protocols in Bioinformatics* **15.** ISSN: 1934-3396. https://pubmed.ncbi.nlm.nih.gov/18428767/ (1 Sept. 2006).

61. Lin, B., Luo, X., Liu, Y. & Jin, X. A comprehensive review and comparison of existing computational methods for protein function prediction. *Briefings in Bioinformatics* **25.** ISSN: 14774054. https://dx.doi.org/10.1093/bib/bbae289 (4 May 2024).

62. Pereira, J. *et al.* High-accuracy protein structure prediction in CASP14. *Proteins: Structure, Function and Bioinformatics* **89,** 1687–1699. ISSN: 10970134. https://pubmed.ncbi.nlm.nih.gov/34218458/ (12 Dec. 2021).

63. Jumper, J. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature 2021 596:7873* **596,** 583–589. ISSN: 1476-4687. https://www.nature.com/articles/s41586-021-03819-2 (7873 July 2021).

64. Abramson, J. *et al.* Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630,** 493–500. ISSN: 14764687. https://www.nature.com/articles/s41586-024-07487-w (8016 June 2024).

65. Baek, M. *et al.* Accurate prediction of protein structures and interactions using a three-track neural network. *Science* **373,** 871–876. ISSN: 10959203. https://www.science.org/doi/10.1126/science.abj8754 (6557 Aug. 2021).

66. Hollingsworth, S. A. & Dror, R. O. Molecular Dynamics Simulation for All. *Neuron* **99,** 1129–1143. ISSN: 0896-6273. https://www.sciencedirect.com/science/article/pii/S0896627318306846 (6 Sept. 2018).

67. Salo-Ahen, O. M. *et al.* Molecular Dynamics Simulations in Drug Discovery and Pharmaceutical Development. *Processes 2021, Vol. 9, Page 71* **9,** 71. ISSN: 2227-9717. https://www.mdpi.com/2227-9717/9/1/71/htm%20https://www.mdpi.com/2227-9717/9/1/71 (1 Dec. 2020).

68. Nosé, S. A unified formulation of the constant temperature molecular dynamics methods. *JChPh* **81,** 511–519. ISSN: 0021-9606. https://ui.adsabs.harvard.edu/abs/1984JChPh..81..511N/abstract (1 1984).

69. Hoover, W. G. Canonical dynamics: Equilibrium phase-space distributions. *Physical Review A* **31,** 1695. ISSN: 10502947. https://journals.aps.org/pra/abstract/10.1103/PhysRevA.31.1695 (3 Mar. 1985).

70. Schneider, T. & Stoll, E. Molecular-dynamics study of a three-dimensional one-component model for distortive phase transitions. *Physical Review B* **17,** 1302. ISSN: 01631829. https://journals.aps.org/prb/abstract/10.1103/PhysRevB.17.1302 (3 Feb. 1978).

71. Bussi, G., Donadio, D. & Parrinello, M. Canonical sampling through velocity rescaling. *Journal of Chemical Physics* **126.** ISSN: 00219606. /aip/jcp/article/126/1/014101/186581/Canonical-sampling-through-velocity-rescaling (1 Jan. 2007).

72. Berendsen, H. J., Postma, J. P., Gunsteren, W. F. V., Dinola, A. & Haak, J. R. Molecular dynamics with coupling to an external bath. *The Journal of Chemical Physics* **81,** 3684–3690. ISSN: 0021-9606. /aip/jcp/article/81/8/3684/565473/Molecular-dynamics-with-coupling-to-an-external (8 Oct. 1984).

73. Parrinello, M. & Rahman, A. Polymorphic transitions in single crystals: A new molecular dynamics method. *Journal of Applied Physics* **52,** 7182–7190. ISSN: 0021-8979. /aip/jap/article/52/12/7182/9558/Polymorphic-transitions-in-single-crystals-A-new (12 Dec. 1981).

74. Allen, M. P. & Tildesley, D. J. Computer simulation of liquids: Second edition. *Computer Simulation of Liquids: Second Edition,* 1–626. https://academic.oup.com/book/27866 (Nov. 2017).

75. Steinbach, P. J. & Brooks, B. R. New spherical-cutoff methods for long-range forces in macromolecular simulation. *Journal of Computational Chemistry* **15,** 667–683. ISSN: 1096987X. /doi/pdf/10.1002/jcc.540150702 (7 July 1994).

76. Verlet, L. Computer "Experiments" on Classical Fluids. I. Thermodynamical Properties of Lennard-Jones Molecules. *Physical Review* **159,** 98. ISSN: 0031899X. https://journals.aps.org/pr/abstract/10.1103/PhysRev.159.98 (1 July 1967).

77. Essmann, U. *et al.* A smooth particle mesh Ewald method. *The Journal of Chemical Physics* **103,** 8577–8593. ISSN: 0021-9606. /aip/jcp/article/103/19/8577/180219/A-smooth-particle-mesh-Ewald-method (19 Nov. 1995).

78. MacKerell, A. D. *et al.* All-atom empirical potential for molecular modeling and dynamics studies of proteins. *Journal of Physical Chemistry B* **102,** 3586–3616. ISSN: 15206106. https://pubs.acs.org/doi/abs/10.1021/jp973084f (18 Apr. 1998).

79. Cornell, W. D. *et al.* A Second Generation Force Field for the Simulation of Proteins, Nucleic Acids, and Organic Molecules. *Journal of the American Chemical Society* **117.** ISSN: 15205126. https://pubs.acs.org/doi/abs/10.1021/ja00124a002 (19 1995).

80. Huang, J. *et al.* CHARMM36m: An Improved Force Field for Folded and Intrinsically Disordered Proteins. *Nature methods* **14,** 71. ISSN: 15487105. /pmc/articles/

PMC5199616/%20/pmc/articles/PMC5199616/?report=abstract%20https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5199616/ (1 Dec. 2017).

81. Maier, J. A. *et al.* ff14SB: Improving the Accuracy of Protein Side Chain and Backbone Parameters from ff99SB. *Journal of Chemical Theory and Computation* **11,** 3696–3713. ISSN: 15499626. https://pubs.acs.org/doi/abs/10.1021/acs.jctc.5b00255 (8 July 2015).

82. Boonstra, S., Onck, P. R. & Giessen, E. V. D. CHARMM TIP3P Water Model Suppresses Peptide Folding by Solvating the Unfolded State. *Journal of Physical Chemistry B* **120,** 3692–3698. ISSN: 15205207. /doi/pdf/10.1021/acs.jpcb.6b01316?ref=article_openPDF (15 May 2016).

83. Nguyen, H., Roe, D. R. & Simmerling, C. Improved Generalized Born Solvent Model Parameters for Protein Simulations. *Journal of chemical theory and computation* **9,** 2020–2034. ISSN: 1549-9626. https://pubmed.ncbi.nlm.nih.gov/25788871/ (4 Apr. 2013).

84. Kmiecik, S. *et al.* Coarse-Grained Protein Models and Their Applications. *Chemical Reviews* **116,** 7898–7936. ISSN: 15206890. /doi/pdf/10.1021/acs.chemrev.6b00163?ref=article_openPDF (14 July 2016).

85. Marrink, S. J., Vries, A. H. D. & Mark, A. E. Coarse Grained Model for Semiquantitative Lipid Simulations. *Journal of Physical Chemistry B* **108,** 750–760. ISSN: 15206106. https://pubs.acs.org/doi/abs/10.1021/jp036508g (2 Jan. 2004).

86. Marrink, S. J. *et al.* Two decades of Martini: Better beads, broader scope. *Wiley Interdisciplinary Reviews: Computational Molecular Science* **13.** ISSN: 17590884 (1 Jan. 2023).

87. Souza, P. C. *et al.* Martini 3: a general purpose force field for coarse-grained molecular dynamics. *Nature Methods* **18,** 382–388. ISSN: 15487105. https://www.nature.com/articles/s41592-021-01098-3 (4 Apr. 2021).

88. Borges-Araújo, L., Pereira, G. P., Valério, M. & Souza, P. C. Assessing the Martini 3 protein model: A review of its path and potential. *Biochimica et Biophysica Acta (BBA) - Proteins and Proteomics* **1872,** 141014. ISSN: 1570-9639. https://www.sciencedirect.com/science/article/abs/pii/S1570963924000219?via%3Dihub (4 July 2024).

89. Periole, X. & Marrink, S.-J. The Martini Coarse-Grained Force Field, 533–565. ISSN: 1940-6029. https://link.springer.com/protocol/10.1007/978-1-62703-017-5_20 (2013).

90. Rossi, G., Monticelli, L., Puisto, S. R., Vattulainen, I. & Ala-Nissila, T. Coarse-graining polymers with the MARTINI force-field: polystyrene as a benchmark case. *Soft Matter* **7,** 698–708. ISSN: 1744-6848. https://pubs.rsc.org/en/content/articlehtml/2011/sm/c0sm00481b (2 Jan. 2011).

91. Torrie, G. M. & Valleau, J. P. Nonphysical sampling distributions in Monte Carlo free-energy estimation: Umbrella sampling. *Journal of Computational Physics* **23,** 187–199. ISSN: 0021-9991. https://www.sciencedirect.com/science/article/abs/pii/0021999177901218?via%3Dihub (2 Feb. 1977).

92. Kästner, J. Umbrella sampling. *Wiley Interdisciplinary Reviews: Computational Molecular Science* **1,** 932–942. ISSN: 17590884. /doi/pdf/10.1002/wcms.66 (6 Nov. 2011).

93. Kumar, S., Rosenberg, J. M., Bouzida, D., Swendsen, R. H. & Kollman, P. A. The weighted histogram analysis method for free-energy calculations on biomolecules. I. The method. *Journal of Computational Chemistry* **13,** 1011–1021. ISSN: 10969-87X. /doi/pdf/10.1002/jcc.540130812 (8 Oct. 1992).

94. Laio, A. & Parrinello, M. Escaping free-energy minima. *Proceedings of the National Academy of Sciences of the United States of America* **99,** 12562–12566.

ISSN: 00278424. /doi/pdf/10.1073/pnas.202427399?download=true (20 Oct. 2002).

95. Barducci, A., Bussi, G. & Parrinello, M. Well-Tempered Metadynamics: A Smoothly Converging and Tunable Free-Energy Method. *Physical Review Letters* **100,** 020603. ISSN: 00319007. https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.100.020603 (2 Jan. 2008).

96. Zhou, P., Jin, B., Li, H. & Huang, S. Y. HPEPDOCK: a web server for blind peptide–protein docking based on a hierarchical algorithm. *Nucleic Acids Research* **46,** W443–W450. ISSN: 0305-1048. https://dx.doi.org/10.1093/nar/gky357 (W1 July 2018).

97. Honorato, R. V. *et al.* The HADDOCK2.4 web server for integrative modeling of biomolecular complexes. *Nature Protocols* **19,** 3219–3241. ISSN: 17502799. https://www.nature.com/articles/s41596-024-01011-0 (11 Nov. 2024).

98. Baxter, R. J. Ornstein–Zernike Relation for a Disordered Fluid. *Australian Journal of Physics* **21,** 563–570. ISSN: 1446-5582. https://www.publish.csiro.au/ph/ph680563 (5 1968).

99. Blum, L. & Torruella, A. J. Invariant Expansion for Two-Body Correlations: Thermodynamic Functions, Scattering, and the Ornstein—Zernike Equation. *The Journal of Chemical Physics* **56,** 303–310. ISSN: 0021-9606. /aip/jcp/article/56/1/303/899347/Invariant-Expansion-for-Two-Body-Correlations (1 Jan. 1972).

100. Henderson, D., Abraham, F. & Barker, J. The ornstein-zernike equation for a fluid in contact with a surface. *Molecular Physics* **31,** 1291–1295. ISSN: 13623028 (4 1976).

101. Høye, J. S. & Blum, L. Solution of the Yukawa closure of the Ornstein-Zernike equation. *Journal of Statistical Physics* **16,** 399–413. ISSN: 00224715. https://link.springer.com/article/10.1007/BF01013184 (5 May 1977).

102. Kelley, C. T. & Pettitt, B. M. A fast solver for the Ornstein–Zernike equations. *Journal of Computational Physics* **197,** 491–501. ISSN: 0021-9991. https://www.sciencedirect.com/science/article/abs/pii/S002199910300648X (2 July 2004).

103. Martynov, G. A. The Ornstein-Zernike equation and critical phenomena in fluids. *Journal of Chemical Physics* **129.** ISSN: 00219606. /aip/jcp/article/129/24/244509/187781/The-Ornstein-Zernike-equation-and-critical (24 Dec. 2008).

104. Jo, S., Kim, T., Iyer, V. G. & Im, W. CHARMM-GUI: A web-based graphical user interface for CHARMM. *Journal of Computational Chemistry* **29,** 1859–1865. ISSN: 1096-987X. https://onlinelibrary.wiley.com/doi/full/10.1002/jcc.20945 (11 Aug. 2008).

105. Pettersen, E. F. *et al.* UCSF Chimera - A visualization system for exploratory research and analysis. *Journal of Computational Chemistry* **25,** 1605–1612. ISSN: 01928651. /doi/pdf/10.1002/jcc.20084%20https://onlinelibrary.wiley.com/doi/abs/10.1002/jcc.20084%20https://onlinelibrary.wiley.com/doi/10.1002/jcc.20084 (13 Oct. 2004).

106. Jia, W. *et al.* A strategy for precise and large scale identification of core fucosylated glycoproteins. *Molecular and Cellular Proteomics* **8,** 913–923. ISSN: 15359476. https://www.mcponline.org/action/showFullText?pii=S1535947620345394%20https://www.mcponline.org/action/showAbstract?pii=S1535947620345394%20https://www.mcponline.org/article/S1535-9476(20)34539-4/abstract (5 May 2009).

107. Chen, R. *et al.* Glycoproteomics Analysis of Human Liver Tissue by Combination of Multiple Enzyme Digestion and Hydrazide Chemistry. *Journal of Proteome Research* **8,** 651–661. ISSN: 15353893. https://pubs.acs.org/doi/abs/10.1021/pr8008012 (2 Feb. 2009).

108. Madeira, F. *et al.* Search and sequence analysis tools services from EMBL-EBI in 2022. *Nucleic Acids Research* **50.** https://doi.org/10.1093/nar/gkac240 (2022).

109. Webb, B. & Sali, A. Comparative Protein Structure Modeling Using MODELLER. *Current Protocols in Bioinformatics* **54,** 5.6.1–5.6.37. ISSN: 1934-340X. https://onlinelibrary.wiley.com/doi/full/10.1002/cpbi.3 (1 June 2016).

110. Abraham, M. J. *et al.* GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX* **1-2,** 19–25. ISSN: 2352-7110 (Sept. 2015).

111. Michaud-Agrawal, N., Denning, E. J., Woolf, T. B. & Beckstein, O. MDAnalysis: A toolkit for the analysis of molecular dynamics simulations. *Journal of Computational Chemistry* **32,** 2319–2327. ISSN: 1096-987X. https://onlinelibrary.wiley.com/doi/full/10.1002/jcc.21787 (10 July 2011).

112. Humphrey, W., Dalke, A. & Schulten, K. VMD: Visual molecular dynamics. *Journal of Molecular Graphics* **14,** 33–38. ISSN: 0263-7855 (1 Feb. 1996).

113. Hub, J. S., Groot, B. L. D. & Spoel, D. V. D. G-whams-a free Weighted Histogram Analysis implementation including robust error and autocorrelation estimates. *Journal of Chemical Theory and Computation* **6,** 3713–3720. ISSN: 15499618. /doi/pdf/10.1021/ct100494z (12 Dec. 2010).

114. York, W. S. *et al.* GlyGen: Computational and Informatics Resources for Glycoscience. *Glycobiology* **30,** 72–73. ISSN: 1460-2423. https://pubmed.ncbi.nlm.nih.gov/31616925/ (2 Feb. 2020).

115. Lemmin, T. & Soto, C. Glycosylator: A Python framework for the rapid modeling of glycans. *BMC Bioinformatics* **20,** 1–7. ISSN: 14712105. https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-019-3097-6%20http://www.glycosciences.de/ (1 Oct. 2019).

116. Apawu, A. K. *et al.* MRI compatible MS2 nanoparticles designed to cross the blood–brain-barrier: providing a path towards tinnitus treatment. *Nanomedicine: Nanotechnology, Biology and Medicine* **14,** 1999–2008. ISSN: 1549-9634. https://www.sciencedirect.com/science/article/abs/pii/S1549963418300753?via%3Dihub (7 Oct. 2018).

117. Huang, S. *et al.* Dual targeting effect of Angiopep-2-modified, DNA-loaded nanoparticles for glioma. *Biomaterials* **32,** 6832–6838. ISSN: 0142-9612. https://www.sciencedirect.com/science/article/abs/pii/S0142961211006351?via%3Dihub (28 Oct. 2011).

118. An, S., He, D., Wagner, E. & Jiang, C. Peptide-like Polymers Exerting Effective Glioma-Targeted siRNA Delivery and Release for Therapeutic Application. *Small* **11,** 5142–5150. ISSN: 16136829. /doi/pdf/10.1002/smll.201501167%20https://onlinelibrary.wiley.com/doi/abs/10.1002/smll.201501167%20https://onlinelibrary.wiley.com/doi/10.1002/smll.201501167 (38 Oct. 2015).

119. Régina, A. *et al.* Antitumour activity of ANG1005, a conjugate between paclitaxel and the new brain delivery vector Angiopep-2. *British Journal of Pharmacology* **155,** 185–197. ISSN: 00071188. /doi/pdf/10.1038/bjp.2008.260 (2 Sept. 2008).

120. Goto, S. *et al.* Cryo-EM structures elucidate the multiligand receptor nature of megalin. *Proceedings of the National Academy of Sciences of the United States of America* **121,** e2318859121. ISSN: 10916490. /doi/pdf/10.1073/pnas.2318859121?download=true (22 May 2024).

121. Lamiable, A. *et al.* PEP-FOLD3: faster de novo structure prediction for linear peptides in solution and in complex. *Nucleic Acids Research* **44,** W449. ISSN: 13624962. https://pmc.ncbi.nlm.nih.gov/articles/PMC4987898/ (Web Server issue July 2016).

122. Huang, S. Y. & Zou, X. An iterative knowledge-based scoring function for protein-protein recognition. *Proteins: Structure, Function and Genetics* **72,** 557–579. ISSN: 08873585. https://pubmed.ncbi.nlm.nih.gov/18247354/ (2 Aug. 2008).

123. Vangone, A. *et al.* Large-scale prediction of binding affinity in protein–small ligand complexes: the PRODIGY-LIG web server. *Bioinformatics* **35,** 1585–1587. ISSN: 1367-4803. https://dx.doi.org/10.1093/bioinformatics/bty816 (9 May 2019).

124. Duarte, J. M., Sathyapriya, R., Stehr, H., Filippis, I. & Lappe, M. Optimal contact definition for reconstruction of Contact Maps. *BMC Bioinformatics* **11,** 1–10. ISSN: 14712105. https://bmcbioinformatics.biomedcentral.com/articles/10.1186/1471-2105-11-283 (1 May 2010).

125. Acosta-Gutiérrez, S., Buckley, J. & Battaglia, G. The Role of Host Cell Glycans on Virus Infectivity: The SARS-CoV-2 Case. *Advanced Science* **10,** 2201853. ISSN: 2198-3844. /doi/pdf/10.1002/advs.202201853 (1 Jan. 2023).

126. Wolf, S. Predicting Protein–Ligand Binding and Unbinding Kinetics with Biased MD Simulations and Coarse-Graining of Dynamics: Current State and Challenges. *Journal of Chemical Information and Modeling* **63,** 2902–2910. ISSN: 1549960X. /doi/pdf/10.1021/acs.jcim.3c00151?ref=article_openPDF (10 May 2023).

127. Rijpkema, S. J., Toebes, B. J., van Vlaenderen, J., van Haren, L. & Wilson, D. A. Influence of Tacticity on the Self-Assembly of Poly(ethylene glycol)-b-poly(lactic acid) Block Copolymers. *ACS Macro Letters* **14,** 101–106. ISSN: 21611653. /doi/pdf/10.1021/acsmacrolett.4c00758?ref=article_openPDF (1 Jan. 2025).

128. Garofalo, C. *et al.* Different insight into amphiphilic PEG-PLA copolymers: Influence of macromolecular architecture on the micelle formation and cellular uptake. *Biomacromolecules* **15,** 403–415. ISSN: 15257797. https://pubs.acs.org/doi/pdf/10.1021/bm401812r?ref=article_openPDF (1 Jan. 2014).

129. Wang, N. *et al.* Design of DOX-GNRs-PNIPAM@PEG-PLA Micelle With Temperature and Light Dual-Function for Potent Melanoma Therapy. *Frontiers in Chemistry* **8,** 599740. ISSN: 22962646. www.frontiersin.org (Jan. 2021).

130. Mundel, R., Thakur, T. & Chatterjee, M. Emerging uses of PLA–PEG copolymer in cancer drug delivery. *3 Biotech* **12.** ISSN: 21905738. https://pubmed.ncbi.nlm.nih.gov/35070631/ (2 Feb. 2022).

131. Dariva, C. G. *et al.* Development of red-light cleavable PEG-PLA nanoparticles as delivery systems for cancer therapy. *Colloids and Surfaces B: Biointerfaces* **196.** ISSN: 18734367. https://pubmed.ncbi.nlm.nih.gov/32971440/ (Dec. 2020).

132. Jong, D. H. D. *et al.* Improved parameters for the martini coarse-grained protein force field. *Journal of Chemical Theory and Computation* **9,** 687–697. ISSN: 15499618. /doi/pdf/10.1021/ct300646g?ref=article_openPDF (1 Jan. 2013).

133. De Castro, R. L.-R., Ziolek, R. M., Ulmschneider, M. B. & Lorenz, C. D. Therapeutic Peptides Are Preferentially Solubilized in Specific Microenvironments within PEG-PLGA Polymer Nanoparticles. *Nano Letters* **24,** 2011–2017. ISSN: 15306992. /doi/pdf/10.1021/acs.nanolett.3c04558?ref=article_openPDF (6 Feb. 2024).

134. Michalowsky, J., Schäfer, L. V., Holm, C. & Smiatek, J. A refined polarizable water model for the coarse-grained MARTINI force field with long-range electrostatic interactions. *Journal of Chemical Physics* **146.** ISSN: 00219606. https://pubmed.ncbi.nlm.nih.gov/28178817/ (5 Feb. 2017).

135. Hunter, R. J. Zeta Potential in Colloid Science. *Zeta Potential in Colloid Science* (1981).

136. Delgado, A. V., González-Caballero, F., Hunter, R. J., Koopal, L. K. & Lyklema, J. Measurement and interpretation of electrokinetic phenomena. *Journal of Colloid*

and Interface Science **309,** 194–224. ISSN: 0021-9797. https://www.sciencedirect.com/science/article/abs/pii/S002197970700015X (2 May 2007).

137. Tandon, V., Bhagavatula, S. K., Nelson, W. C. & Kirby, B. J. Zeta potential and electroosmotic mobility in microfluidic devices fabricated from hydrophobic polymers: 1. The origins of charge. *Electrophoresis* **29,** 1092–1101. ISSN: 15222683. https://pubmed.ncbi.nlm.nih.gov/18306184/ (5 2008).

138. Dukhin, A. S., Ohshima, H., Shilov, V. N. & Goetz, P. J. Electroacoustics for concentrated dispersions. *Langmuir* **15,** 3445–3451. ISSN: 07437463. /doi/pdf/10.1021/la9813836?ref=article_openPDF (10 May 1999).

139. Knecht, V., Klasczyk, B. & Dimova, R. Macro-versus microscopic view on the electrokinetics of a water-membrane interface. *Langmuir* **29,** 7939–7948. ISSN: 07437463. /doi/pdf/10.1021/la400342m?ref=article_openPDF (25 June 2013).

140. Brkljača, Z., Namjesnik, D., Lützenkirchen, J., Předota, M. & Preočanin, T. Quartz/Aqueous Electrolyte Solution Interface: Molecular Dynamic Simulation and Interfacial Potential Measurements. *Journal of Physical Chemistry C* **122,** 24025–24036. ISSN: 19327455. /doi/pdf/10.1021/acs.jpcc.8b04035?ref=article_openPDF (42 Oct. 2018).

141. Siboulet, B., Hocine, S., Hartkamp, R. & Dufrêche, J. F. Scrutinizing Electro-Osmosis and Surface Conductivity with Molecular Dynamics. *Journal of Physical Chemistry C* **121,** 6756–6769. ISSN: 19327455. /doi/pdf/10.1021/acs.jpcc.7b00309?ref=article_openPDF (12 Mar. 2017).

142. Böckmann, R. A., Hac, A., Heimburg, T. & Grubmüller, H. Effect of sodium chloride on a lipid bilayer. *Biophysical Journal* **85,** 1647–1655. ISSN: 00063495. https://pubmed.ncbi.nlm.nih.gov/12944279/ (3 Sept. 2003).

143. Melcr, J. *et al.* Accurate Binding of Sodium and Calcium to a POPC Bilayer by Effective Inclusion of Electronic Polarization. *Journal of Physical Chemistry B* **122,** 4546–4557. ISSN: 15205207. /doi/pdf/10.1021/acs.jpcb.7b12510?ref=article_openPDF (16 Apr. 2018).

144. Sendner, C., Horinek, D., Bocquet, L. & Netz, R. R. Interfacial water at hydrophobic and hydrophilic surfaces: Slip, viscosity, and diffusion. *Langmuir* **25,** 10768–10781. ISSN: 07437463. https://pubs.acs.org/doi/pdf/10.1021/la901314b?ref=article_openPDF (18 Sept. 2009).

145. Předota, M., Machesky, M. L. & Wesolowski, D. J. Molecular Origins of the Zeta Potential. *Langmuir* **32,** 10189–10198. ISSN: 15205827. /doi/pdf/10.1021/acs.langmuir.6b02493?ref=article_openPDF (40 Oct. 2016).

146. Shokri, A., Abedin, A., Fattahi, A. & Kass, S. R. Effect of Hydrogen Bonds on pKa Values: Importance of Networking. *Journal of the American Chemical Society* **134,** 10646–10650. ISSN: 00027863. /doi/pdf/10.1021/ja3037349?ref=article_openPDF (25 June 2012).

147. Ziolek, R. M., Smith, P., Pink, D. L., Dreiss, C. A. & Lorenz, C. D. Unsupervised Learning Unravels the Structure of Four-Arm and Linear Block Copolymer Micelles. *Macromolecules* **54,** 3755–3768. ISSN: 15205835. /doi/pdf/10.1021/acs.macromol.0c02523?ref=article_openPDF (8 Apr. 2021).

Appendix A

Neural Network Prediction Benchmarking

1 Evaluation of motif prediction

AlphaFold2 and RoseTTAFold predictions were compared for three representative LRP1 motifs: CB1, B1, and the first EGF-like domain. Both methods achieved high confidence in local folding (Figure A.1.a,b). Their predictions overlapped almost perfectly (Figure A.1.c).

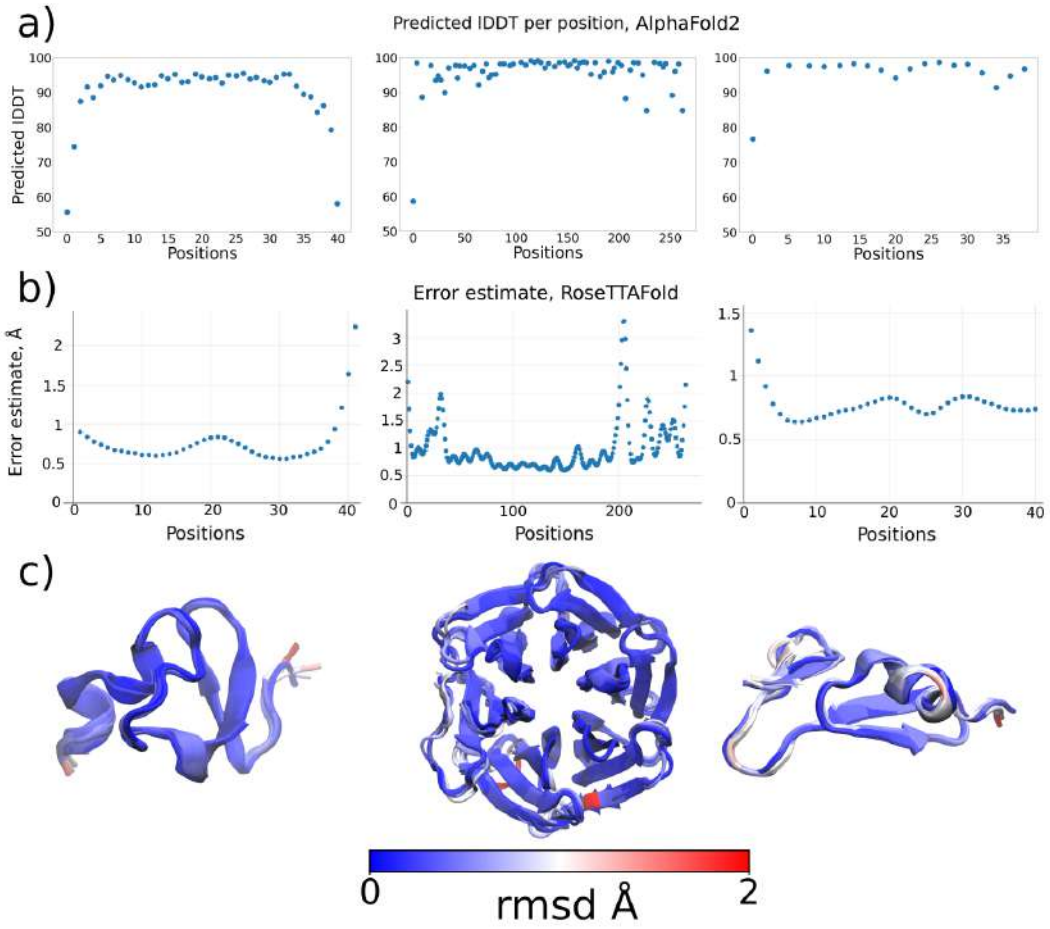


Figure A.1: Evaluation of LRP1's predicted motifs using structure predicting algorithms. a. Predicted local distance difference test lDDT per position of the AlphaFold2 prediction for each motif. b. Error estimate from RoseTTAFold for the same motifs. c. Structural alignment of the five RoseTTAFold and five AlphaFold2 predictions.

2 Comparison with LRP2

Superimposition of predicted LRP1 motifs with cryo-TEM resolved LRP2 motifs (8EM4) confirmed the predictive accuracy (Figure A.2).

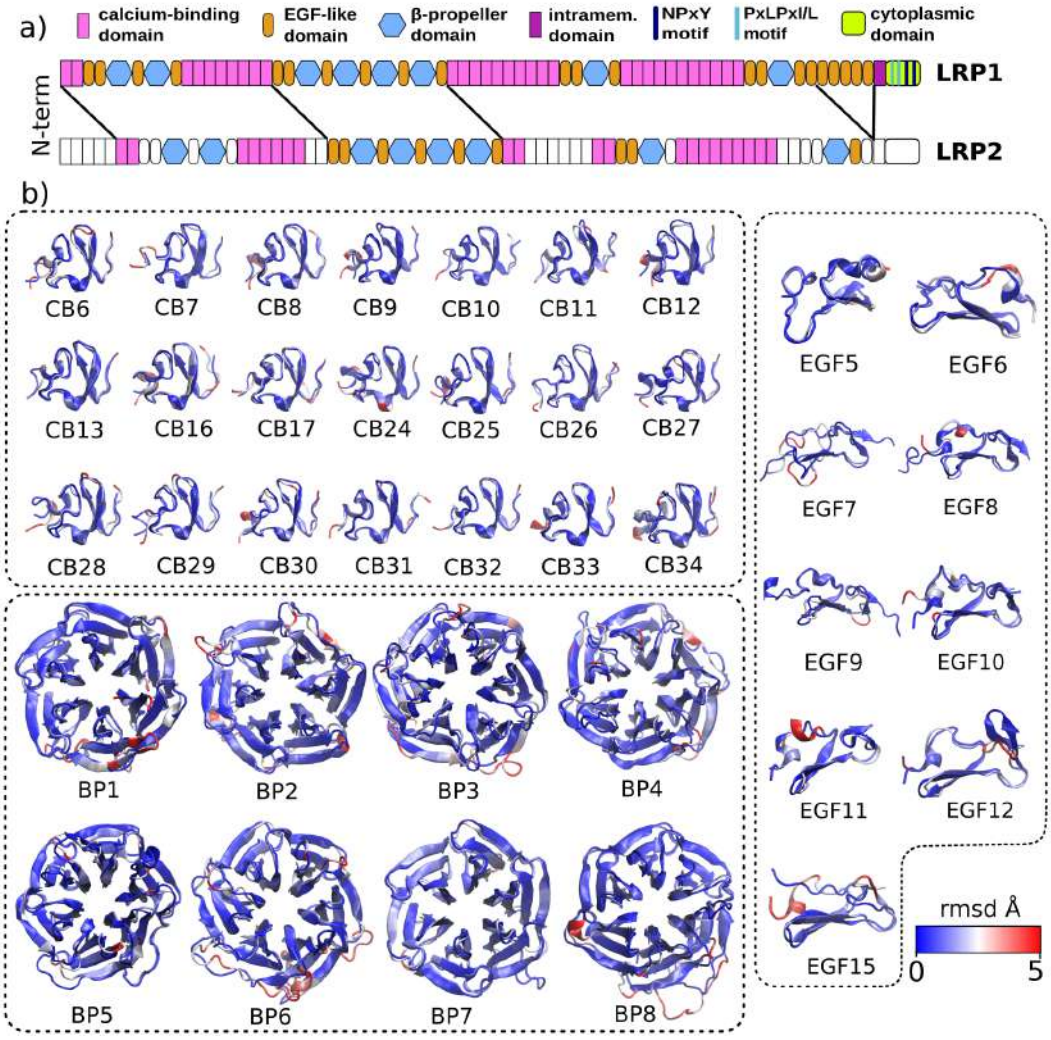


Figure A.2: a. Alignment of the LRP1 and LRP2 structural sequences. b. Structural superimposition of LRP2 cryo-TEM motifs (8EM4) with the corresponding RoseTTAFold LRP1 predictions.

3 Global prediction attempt

AlphaFold3 generated a full-length LRP1 model, but with extensive clashes and unrealistic motif overlaps, rendering it unsuitable (Figure A.3).

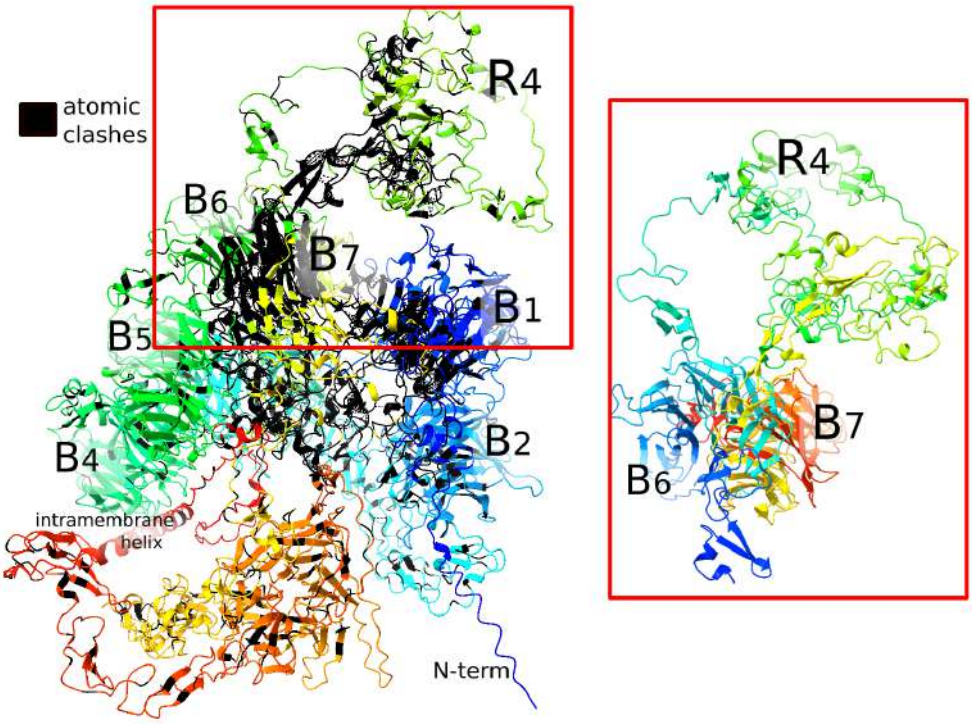


Figure A.3: Prediction of the LRP1 full protein by AlphaFold3. The model shows multiple clashes and motif overlaps, highlighting the limits of current NN tools for very large proteins.

Appendix B

Residue-specific hydration potentials

This appendix contains the complete residue-specific plots of the interaction potential $\Phi(\sigma)$ with water for all conserved amino acids studied: Aspartic acid (D, Asp), Cysteine (C, Cys), Lysine (K, Lys), Glutamic acid (E, Glu), Phenylalanine (F, Phe) and Serine (S, Ser). These figures expand on the grouped representations reported in the main Results section. Finally, we report in a Table all the residues analyzed in this work.

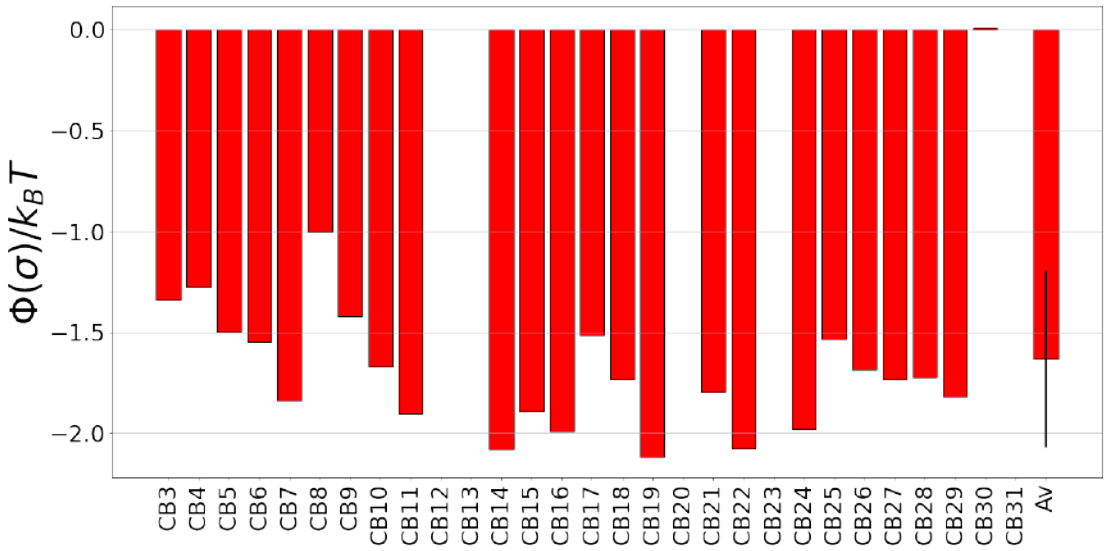


Figure B.1: Interaction potential $\Phi(\sigma)$ with wate for amino acid Asp1 in CB position 24. The histogram illustrates the interaction potential for the 31 CBs and the average (represented by the last bar). The error on the average is the standard deviation of non-null energy values.

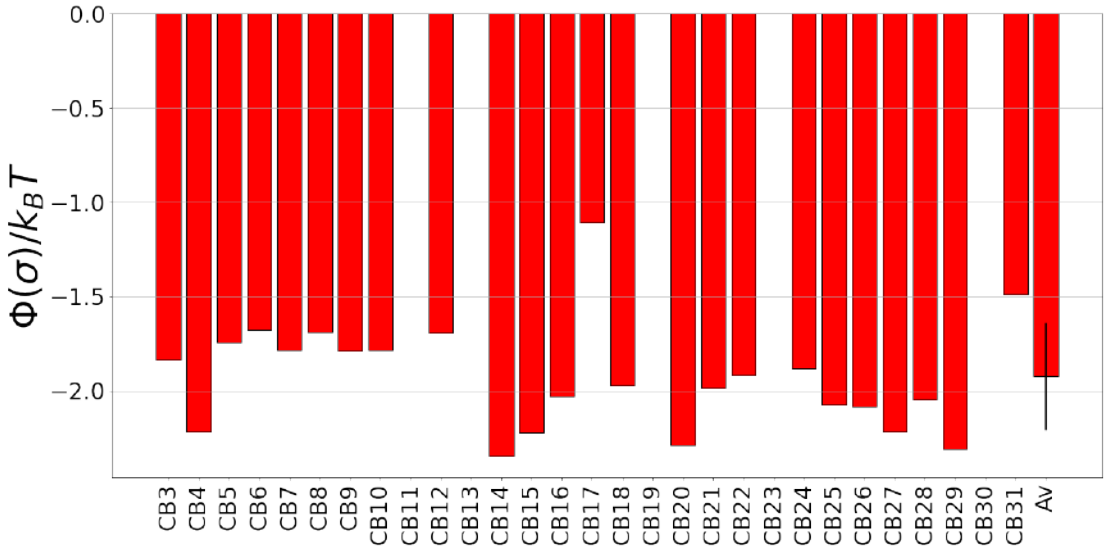


Figure B.2: Interaction potential $\Phi(\sigma)$ with wate for amino acid Asp2 in CB position 26.

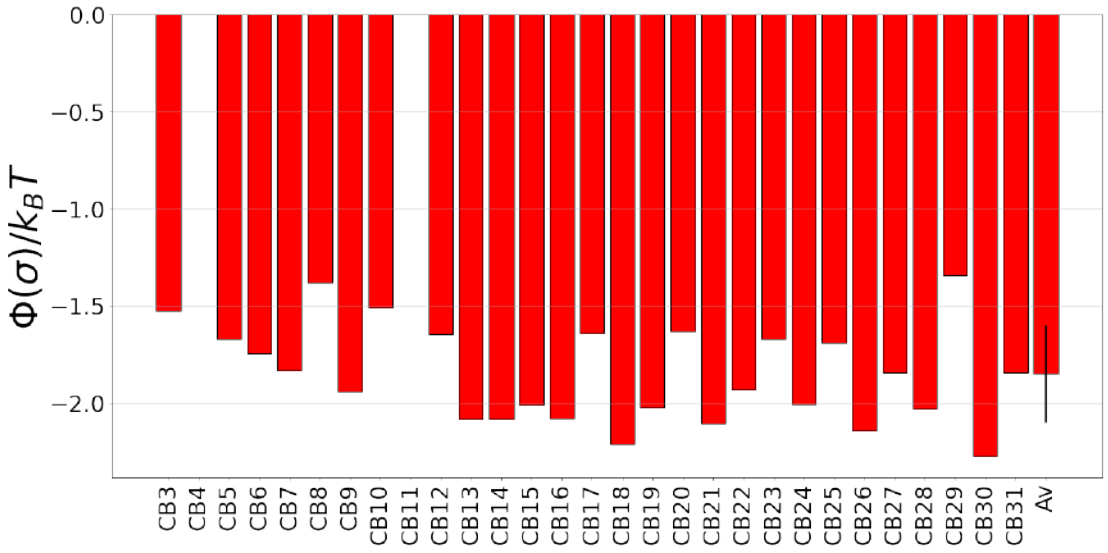


Figure B.3: Interaction potential $\Phi(\sigma)$ with wate for amino acid Asp3 in CB position 28.

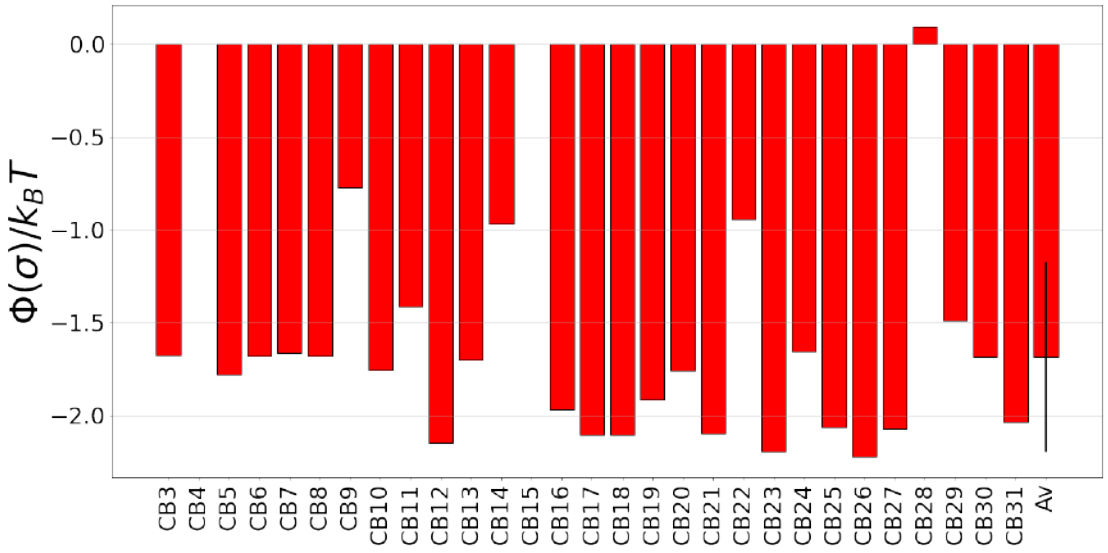


Figure B.4: Interaction potential $\Phi(\sigma)$ with wate for amino acid Asp4 in CB position 31.

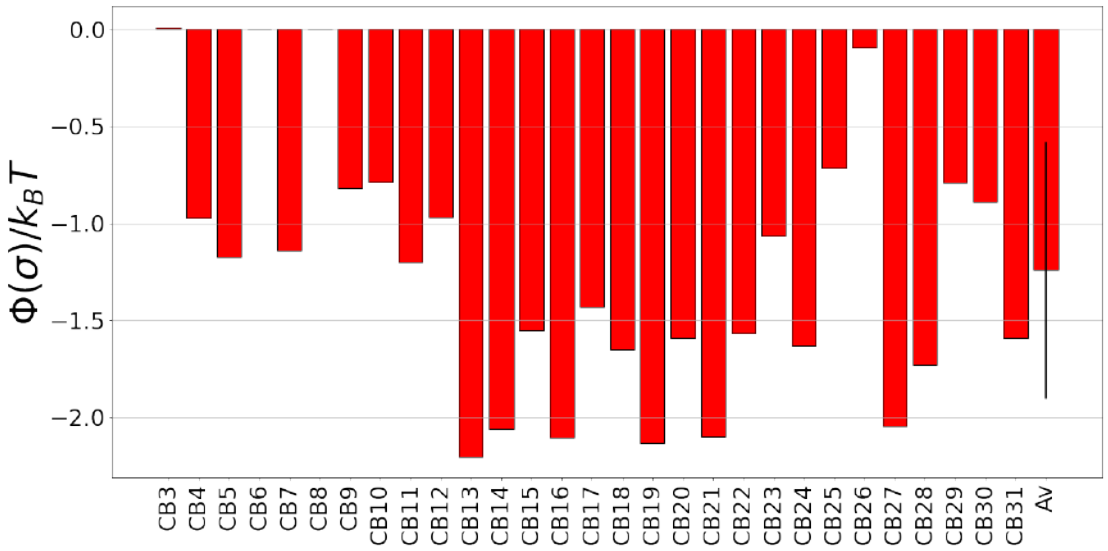


Figure B.5: Interaction potential $\Phi(\sigma)$ with wate for amino acid Asp5 in CB position 34.

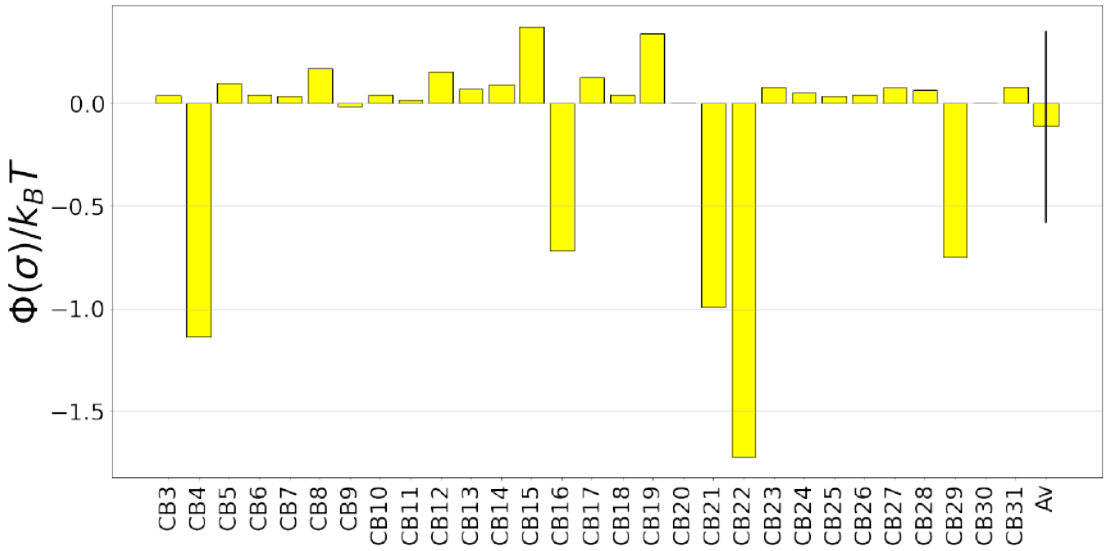


Figure B.6: Interaction potential $\Phi(\sigma)$ with wate for amino acid Cys1 in CB position 2.

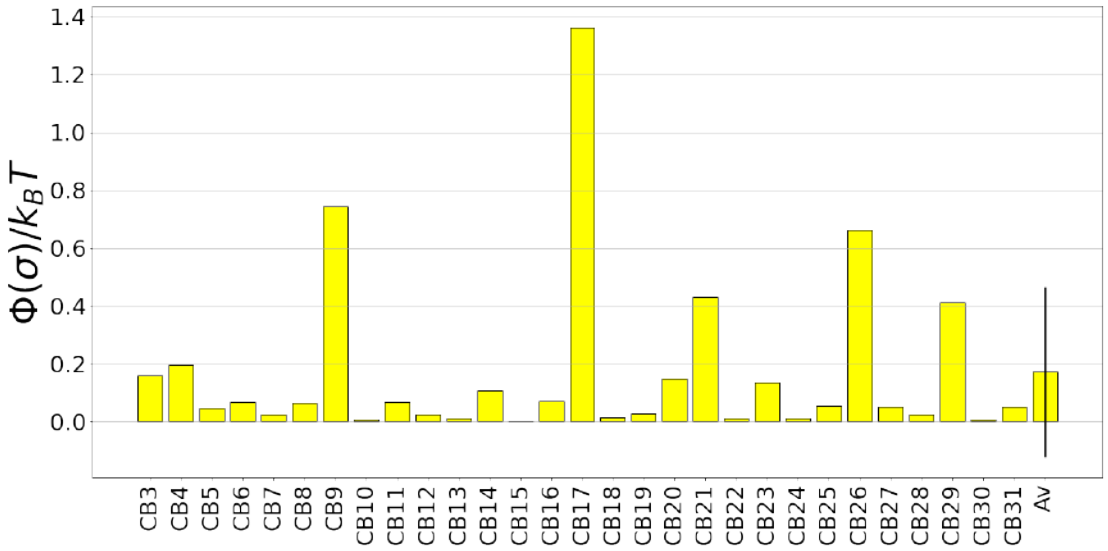


Figure B.7: Interaction potential $\Phi(\sigma)$ with wate for amino acid Cys2 in CB position 10.

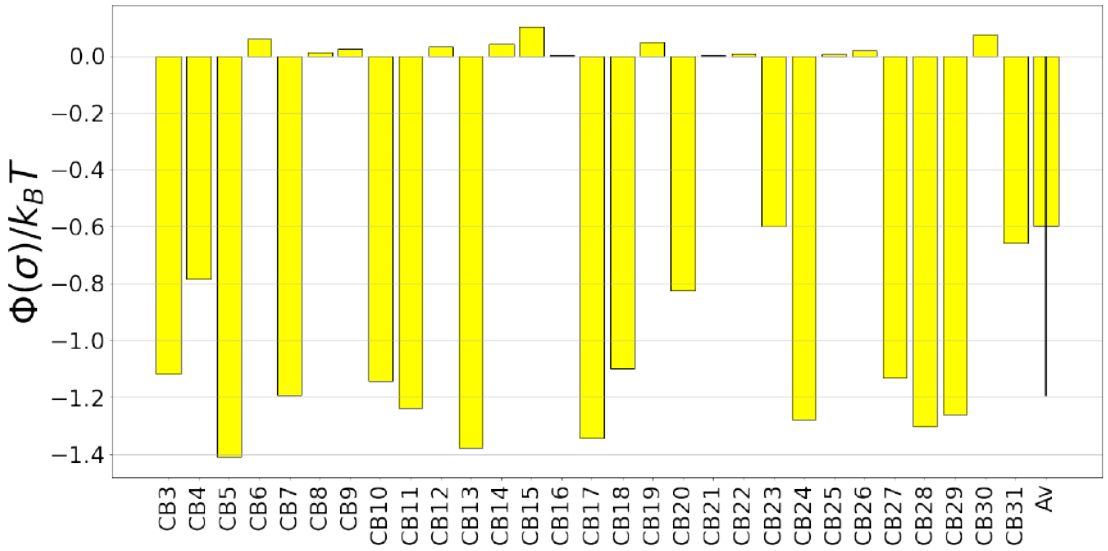


Figure B.8: Interaction potential $\Phi(\sigma)$ with wate for amino acid Cys3 in CB position 16.

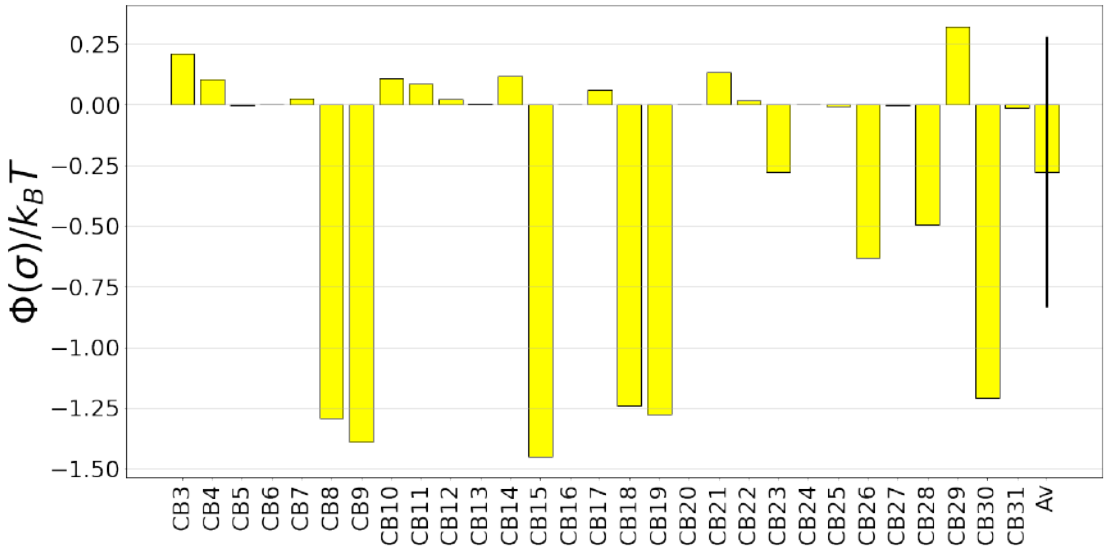


Figure B.9: Interaction potential $\Phi(\sigma)$ with wate for amino acid Cys4 in CB position 23.

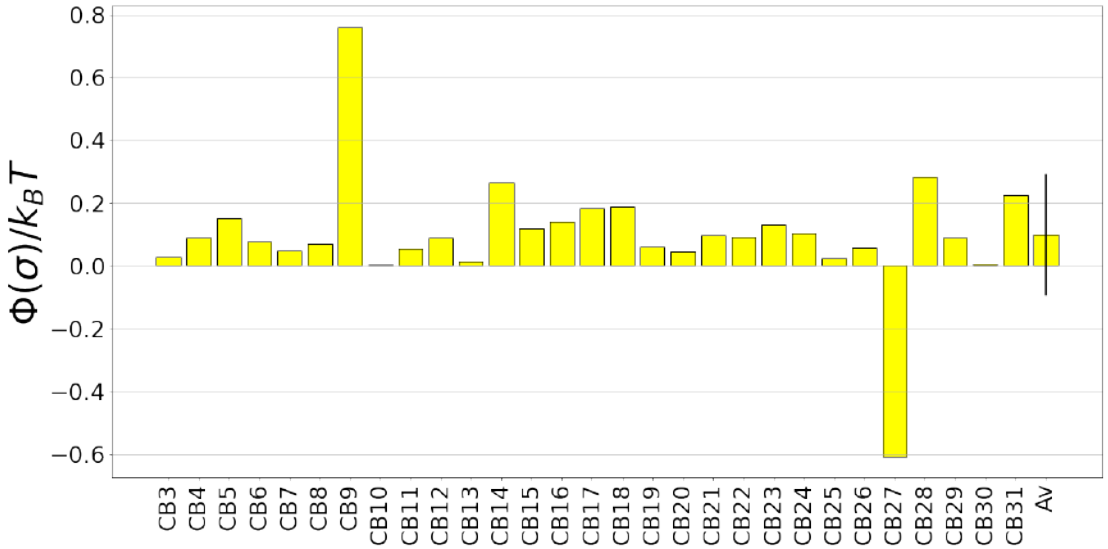


Figure B.10: Interaction potential $\Phi(\sigma)$ with wate for amino acid Cys5 in CB position 29.

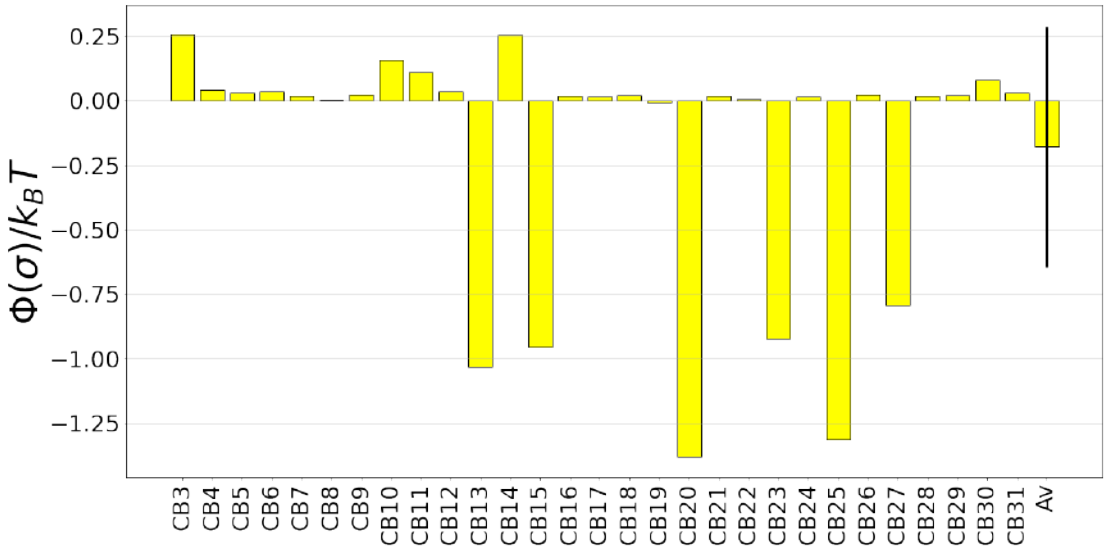


Figure B.11: Interaction potential $\Phi(\sigma)$ with wate for amino acid Cys6 in CB position 38.

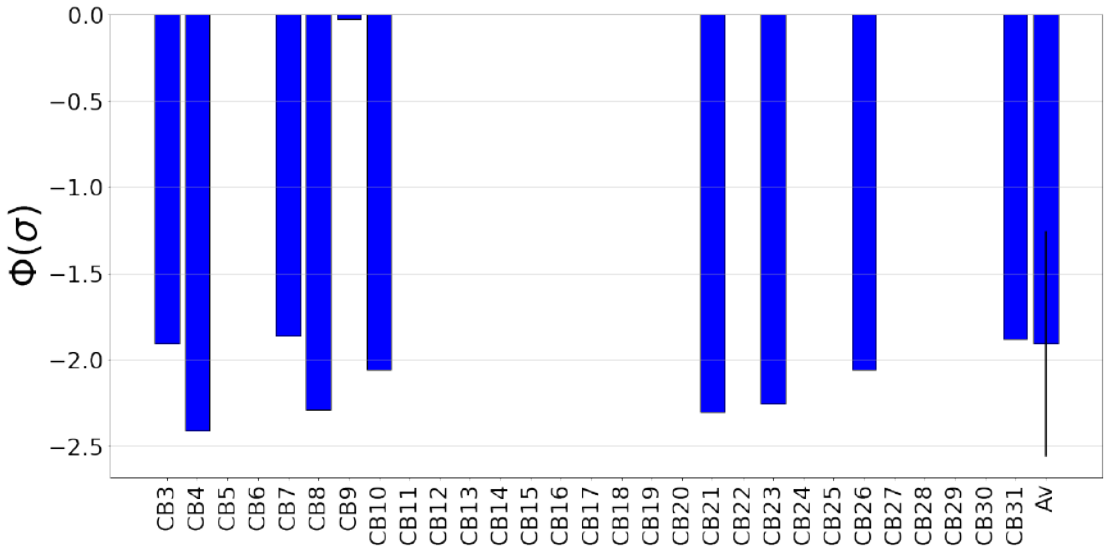


Figure B.12: Interaction potential $\Phi(\sigma)$ with wate for amino acid Lys1 in CB position 7.

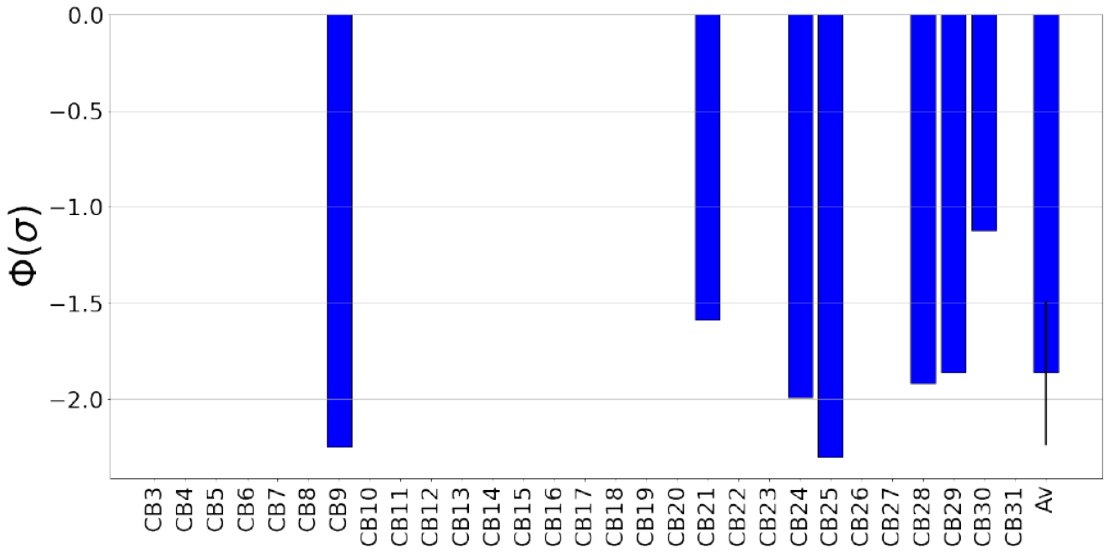


Figure B.13: Interaction potential $\Phi(\sigma)$ with wate for amino acid Lys2 in CB position 11.

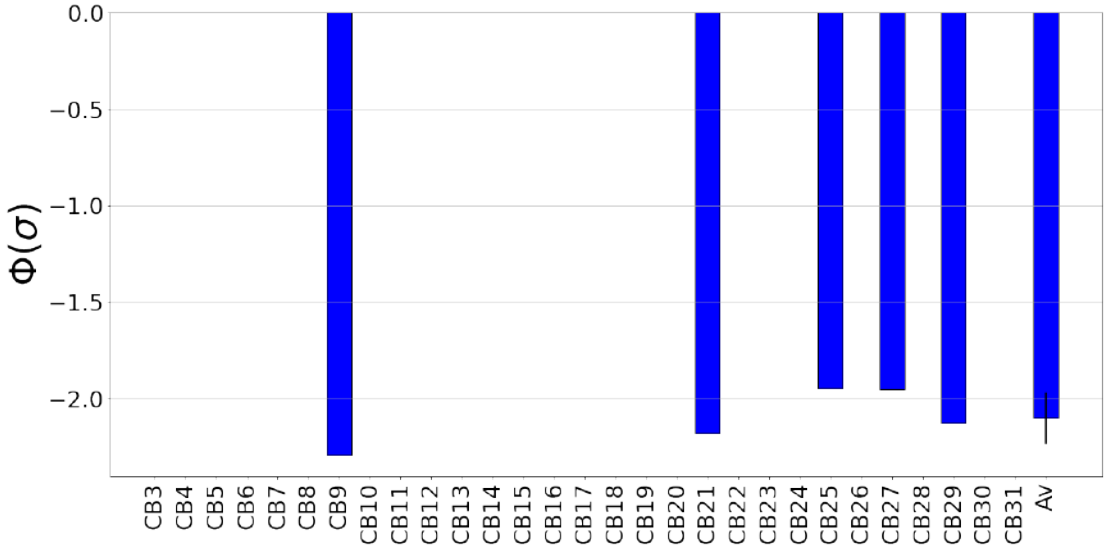


Figure B.14: Interaction potential $\Phi(\sigma)$ with wate for amino acid Lys3 in CB position 19.

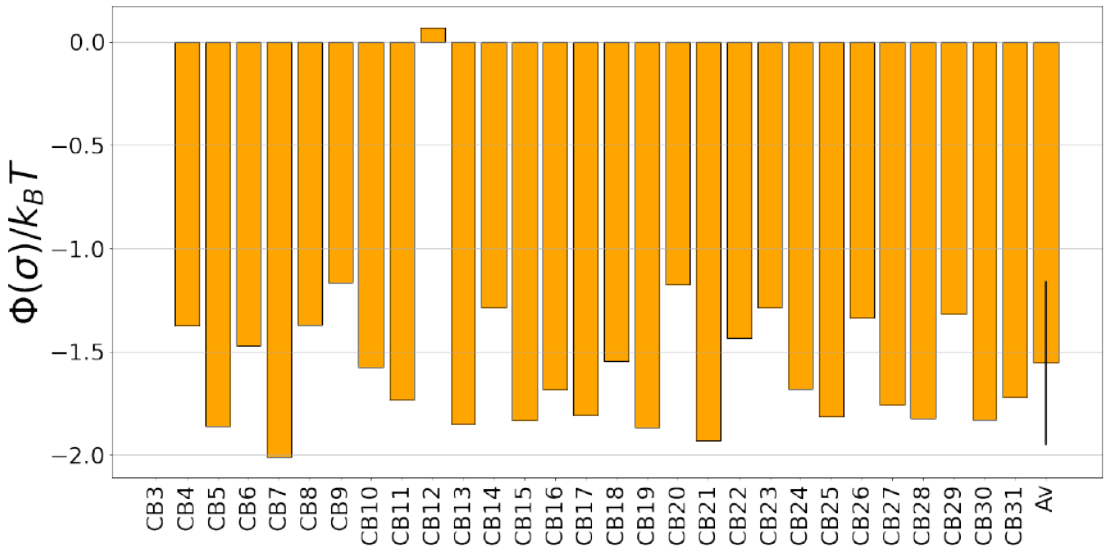


Figure B.15: Interaction potential $\Phi(\sigma)$ with wate for amino acid Glu in CB position 35.

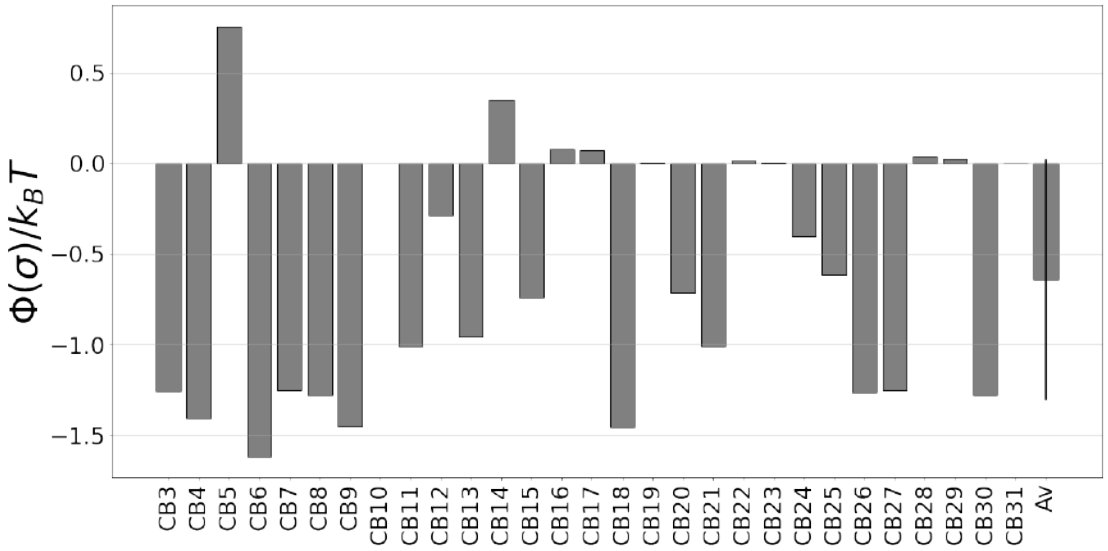


Figure B.16: Interaction potential $\Phi(\sigma)$ with wate for amino acid Phe in CB position 8.

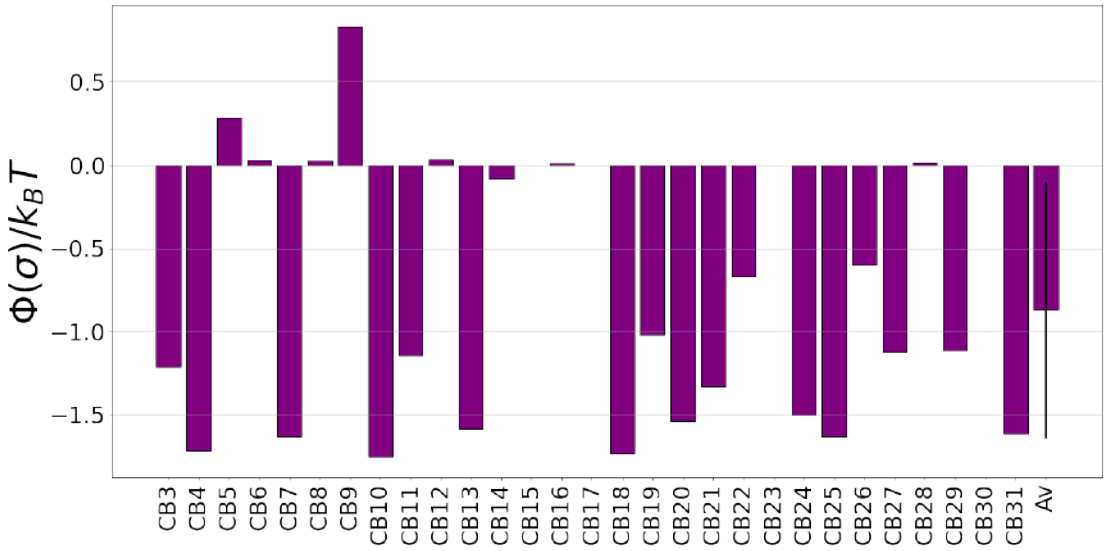


Figure B.17: Interaction potential $\Phi(\sigma)$ with wate for amino acid Ser in CB position 33.

R	R2 (CB3-10)	R3 (CB11-20)	R4 (CB21-31)
ASP1	874, 915, 956, 997, 1035, 1083, 1126, 1167	2545, -, -, 2674, 2716, 2754, 2795, 2838, 2879, -	3354, 3394, -, 3474, 3515, 3556, 3595, 3633, 3674, 3717, -
ASP2	876, 917, 958, 999, 1037, 1085, 1128, 1170	-, 2589, -, 2678, 2719, 2757, 2797, 2840, -, 2929	3357, 3396, -, 3476, 3518, 3558, 3597, 3635, 3677, -, 3765
ASP3	878, 919, 960, 1001, 1039, 1087, 1130, 1171	-, 2590, 2629, 2681, 2720, 2758, 2799, 2842, 2883, 2932	3358, 3398, 3437, 3478, 3519, 3560, 3599, 3637, 3678, 3720, 3766
ASP4	881, -, 963, 1004, 1042, 1090, 1133, 1174	2552, 2593, 2632, 2684, -, 2761, 2802, 2845, 2886, 2932	3361, 3401, 3441, 3481, 3522, 3563, 3602, 3640, 3681, 3724, 3769
ASP5	884, 925, 966, 1007, 1045, 1093, 1136, 1177	2555, 2596, 2635, 2687, 2726, 2764, 2805, 2848, 2889, 2935	3364, 3404, 3444, 3484, 3525, 3566, 3605, 3643, 3684, 3727, 3772
CYS1	854, 895, 936, 976, 1015, 1062, 1104, 1145	2524, 2566, 2605, 2644, 2696, 2734, 2774, 2818, 2858, 2904	3334, 3374, 3413, 3453, 3494, 3536, 3575, 3613, 3654, 3695, 3741
CYS2	861, 902, 943, 984, 1022, 1069, 1112, 1152	2532, 2573, 2612, 2660, 2703, 2741, 2781, 2825, 2865, 2912	3341, 3381, 3420, 3460, 3501, 3543, 3582, 3620, 3661, 3703, 3749
CYS3	866, 907, 948, 989, 1027, 1075, 1118, 1159	2537, 2578, 2617, 2666, 2708, 2746, 2787, 2830, 2870, 2917	3346, 3386, 3426, 3466, 3507, 3548, 3587, 3625, 3666, 3709, 3754
CYS4	873, 914, 955, 996, 1034, 1082, 1125, 1166	2544, 2585, 2624, 2673, 2715, 2753, 2794, 2837, 2878, 2924	3353, 3393, 3433, 3473, 3514, 3555, 3594, 3632, 3673, 3716, 3761
CYS5	879, 920, 961, 1002, 1040, 1088, 1131, 1172	2550, 2591, 2630, 2679, 2721, 2759, 2800, 2843, 2884, 2930	3359, 3399, 3439, 3479, 3520, 3561, 3600, 3638, 3679, 3722, 3767
CYS6	890, 931, 971, 1011, 1051, 1097, 1140, 1182	2561, 2600, 2639, 2688, 2730, 2769, 2812, 2853, 2897, 2939	3369, 3408, 3448, 3489, 3531, 3570, 3609, 3647, 3690, 3731, 3776
LYS1	872, 901, -, -, 1021, 1095, 1109, 1164	-, -, -, -, -, -, -, -, -, -	3342, -, 3419, -, -, 3544, -, -, -, -, 3745
LYS2	-, -, -, -, -, -, 1113, -	-, -, -, -, -, -, -, - , -, -	3345, -, -, 3464, 3502, -, -, 3621, 3667, 3704
LYS3	-, -, -, -, -, -, 1121, -	-, -, -, -, -, -, -, -, -, -	3352, -, -, -, 3513, -, 3593, -, 3672, -, -
SER	883, 923, 965, 1006, 1044, 1092, 1135, 1176	2554, 2595, 2634, 2683, -, 2763, -, 2847, 2888, 2934	3363, 3403, -, 3483, 3524, 3565, 3604, 3642, 3683, -, 3771
GLU	885, 926, 967, 1008, 1046, 1094, 1137, 1178	2556, 2597, 2636, 2685, 2727, 2765, 2806, 2849, 2890, 2936	3365, 3405, 3445, 3485, 3526, 3567, 3606, 3644, 3685, 3728, 3773
PHE	859, 900, 941, 982, 1020, 1067, 1110, -	2530, 2570, 2610, 2657, 2701, 2739, 2779, 2823, 2863, 2910	3339, 3379, 3418, 3458, 3499, 3541, 3580, 3618, 3659, 3701, 3747

Table B.1: Conserved amino acid in the primary and secondary structures along the three CB domains, called Rn.

Appendix C

Micelle charge and electrostatic profile

1 Methods

1.1 MARTINI Force Field PEG-PLA parameters

Particle	Type	Mass [u]
PEG	EO[147]	45
PLA	C3	72

Bond	**Function 1**	
	b_0 [nm]	k_b [kJ mol^{-1} nm^{-2}]
PEG–PEG	0.322	7000
PEG–PLA	0.322	7000
PLA–PLA	0.33	5000

Angle	**Function 2**		**Function 10**	
	θ_0 [deg]	[kJ mol^{-1}]	θ_0 [deg]	[kJ mol^{-1}]
PEG–PEG–PEG	135	50	135	75
PEG–PEG–PLA	135	50	/	/
PEG–PLA–PLA	135	50	/	/
PLA–PLA–PLA	120	118	/	/

Dihedral	**Function 1**		
	multiplicity	ϕ_s [deg]	k_s [kJ mol^{-1}]
PEG–PEG–PEG–PEG	4	0	0.12
PEG–PEG–PEG–PEG	3	0	0.18
PEG–PEG–PEG–PEG	2	0	0.33
PEG–PEG–PEG–PEG	1	180	1.96

Table C.1: Parameters employed in the PEG–PLA molecular dynamics simulations.

1.2 Radial charge density

To characterize the spatial distribution of charges surrounding the nanoparticle, the simulation box was partitioned into concentric spherical shells of thickness Δr. For each shell k, the total enclosed charge was calculated as

$$Q_k = \sum_{i \in \text{shell } k} q_i, \tag{C.1}$$

where q_i denotes the charge of atom i in Coulombs.

The local charge density was then obtained by normalizing to the volume of the corresponding shell:

$$\rho(r_k) = \frac{Q_k}{V_k}, \tag{C.2}$$

with the shell volume defined as

$$V_k = \frac{4}{3}\pi \left(r_{k+1}^3 - r_k^3\right), \tag{C.3}$$

where r_k and r_{k+1} are the inner and outer radii of shell k, respectively.

1.3 Electrostatic potential from Poisson's equation

The electrostatic potential was determined by solving Poisson's equation under the assumption of spherical symmetry:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{d\phi}{dr}\right) = -\frac{\rho(r)}{\varepsilon}, \tag{C.4}$$

where $\varepsilon = \varepsilon_r \varepsilon_0$ is the dielectric permittivity of the medium.

In practice, the radial electric field was first obtained from Gauss's law as

$$E(r) = \frac{1}{\varepsilon r^2} \int_0^r \rho(r')\, r'^2\, dr', \tag{C.5}$$

and the potential was then recovered by integrating the field:

$$\phi(r) = -\int_0^r E(r')\, dr'. \tag{C.6}$$

1.4 Zeta potential

The zeta potential was defined as the value of the electrostatic potential at the position of the slipping plane, r_{slip}:

$$\zeta = \phi(r_{\text{slip}}). \tag{C.7}$$

2 Radial density charge and electrostatic potential for the charged micelles

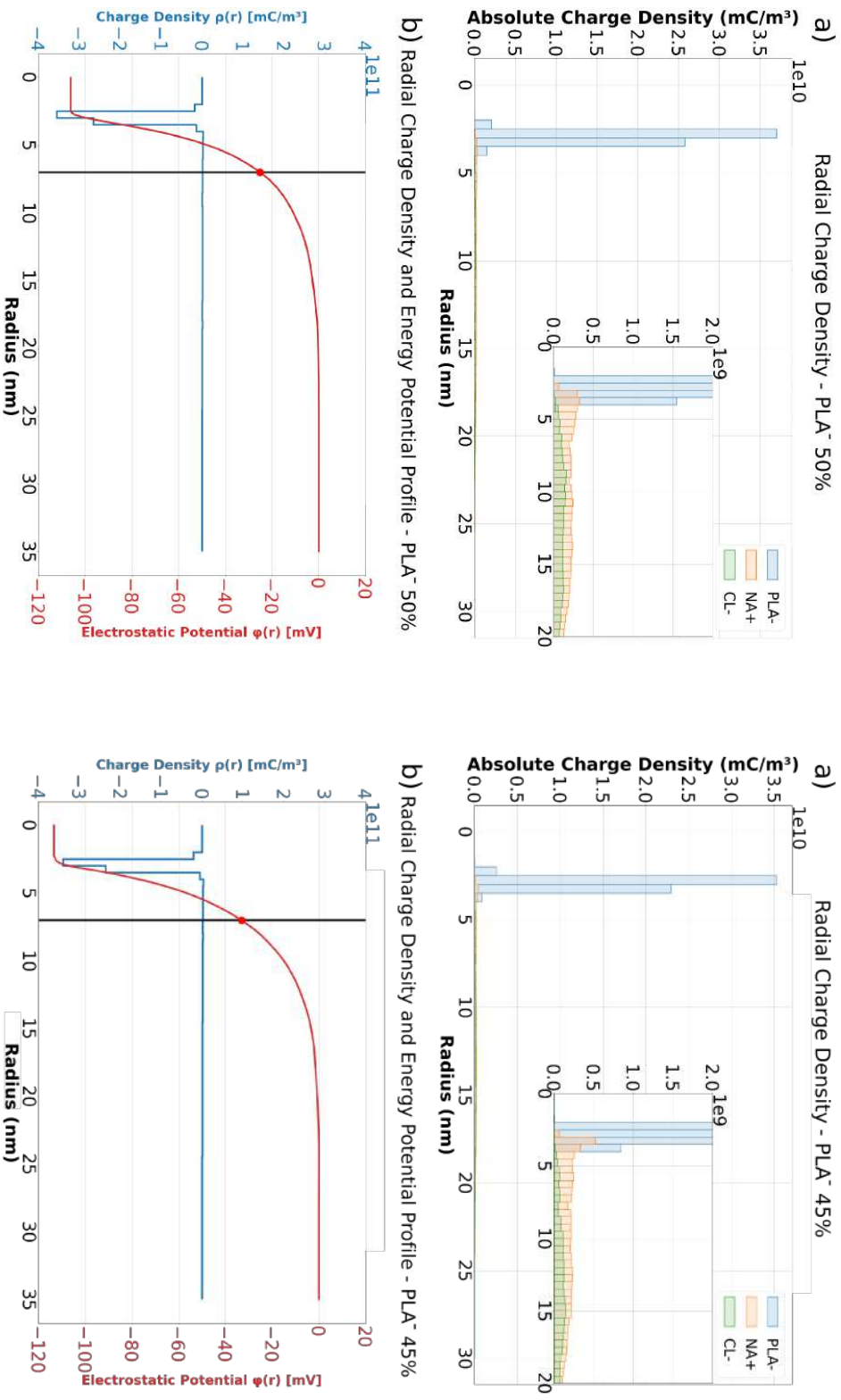


Figure C.1: a. Absolute radial charge density for charged species (50 and 45%) as a function of the distance from the micelle core. b. Radial charge density and electrostatic potential of the same charged species as a function of the distance from the micelle core. The black vertical line identifies the slip plane, and the red dot is the z-potential.

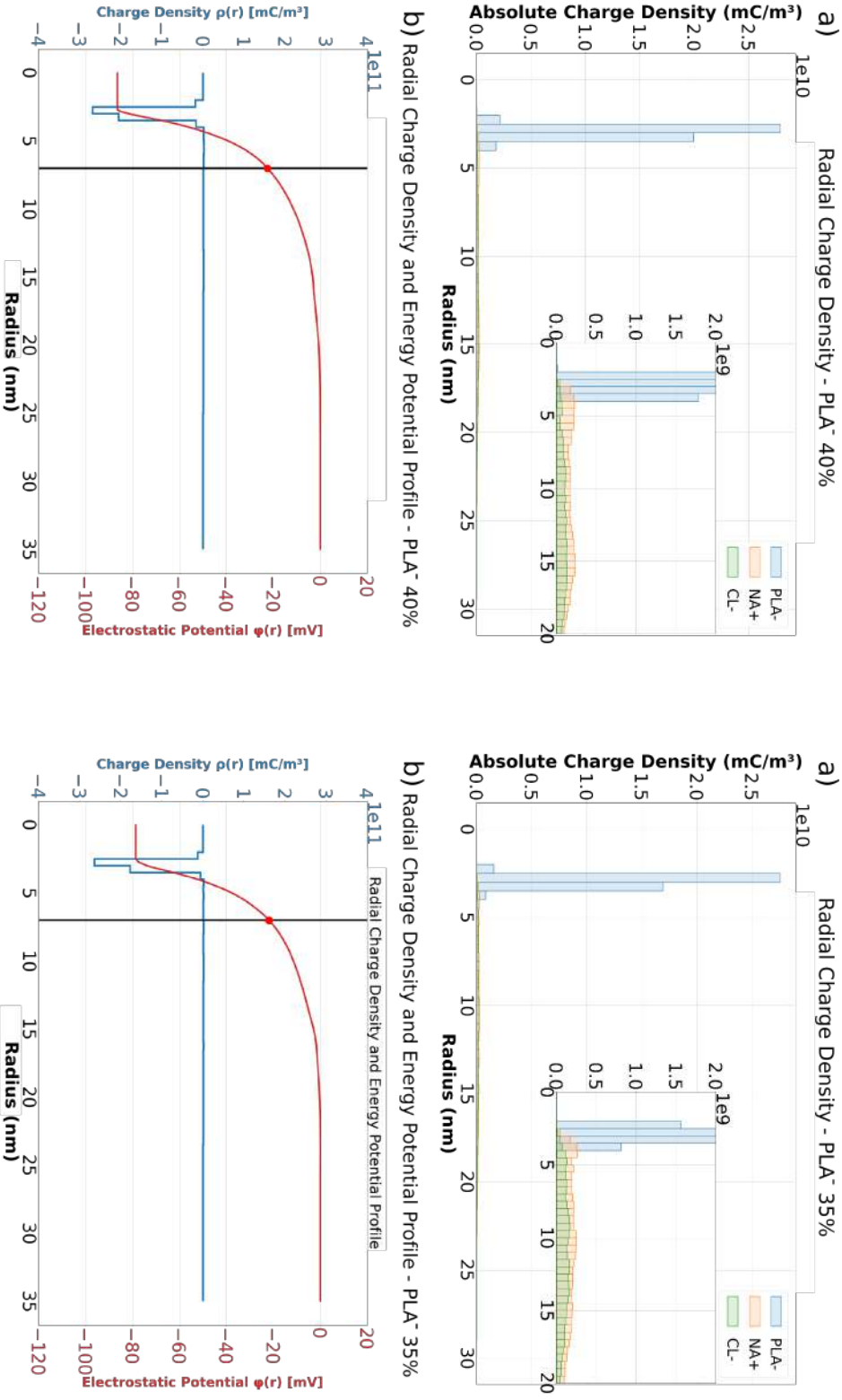


Figure C.2: a. Absolute radial charge density for charged species (40 and 35%) as a function of the distance from the micelle core. b. Radial charge density and electrostatic potential of the same charged species as a function of the distance from the micelle core. The black vertical line identifies the slip plane, and the red dot is the z-potential.

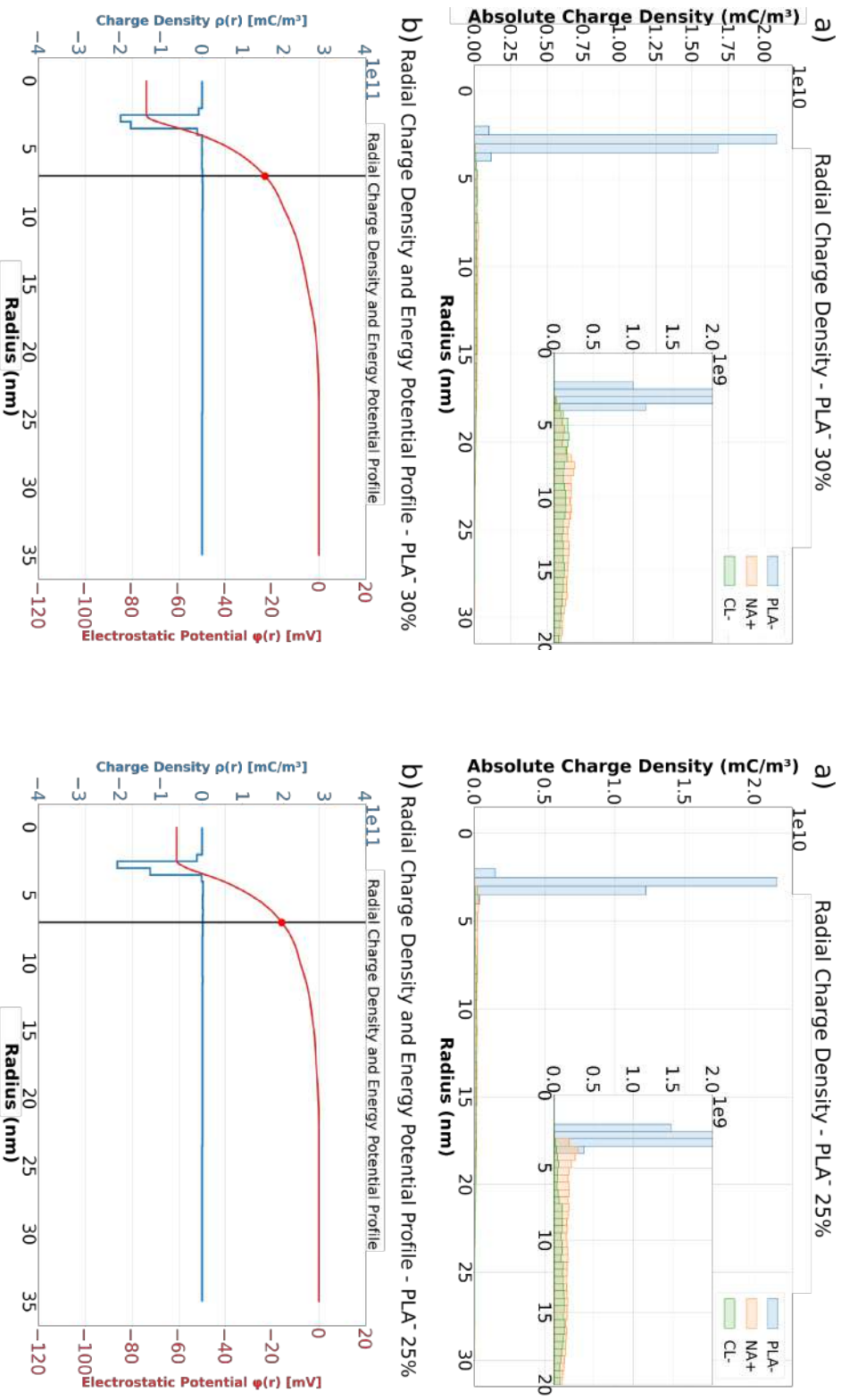


Figure C.3: a. Absolute radial charge density for charged species (30 and 25%) as a function of the distance from the micelle core. b. Radial charge density and electrostatic potential of the same charged species as a function of the distance from the micelle core. The black vertical line identifies the slip plane, and the red dot is the z-potential.

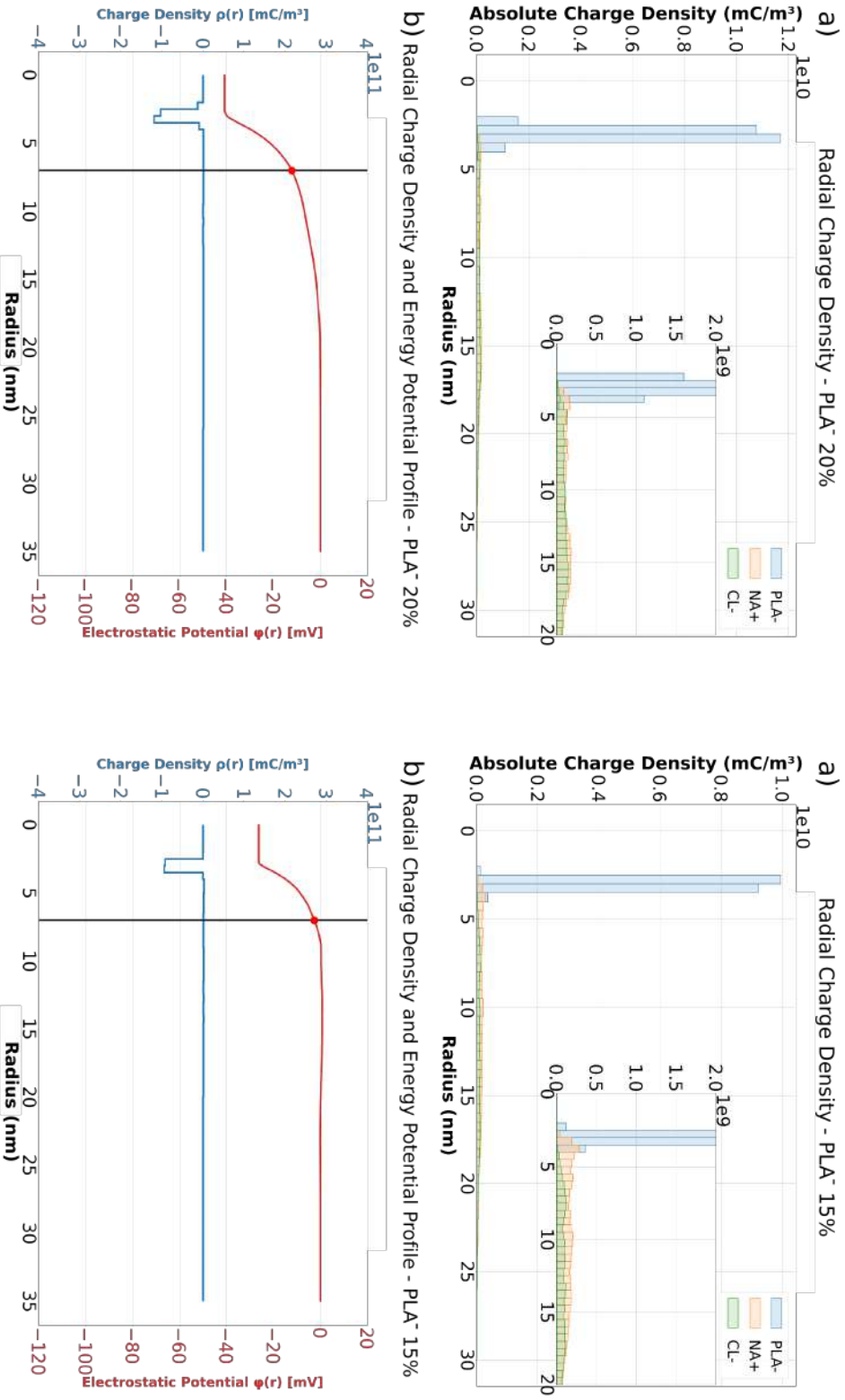


Figure C.4: a. Absolute radial charge density for charged species (20 and 15%) as a function of the distance from the micelle core. b. Radial charge density and electrostatic potential of the same charged species as a function of the distance from the micelle core. The black vertical line identifies the slip plane, and the red dot is the z-potential.

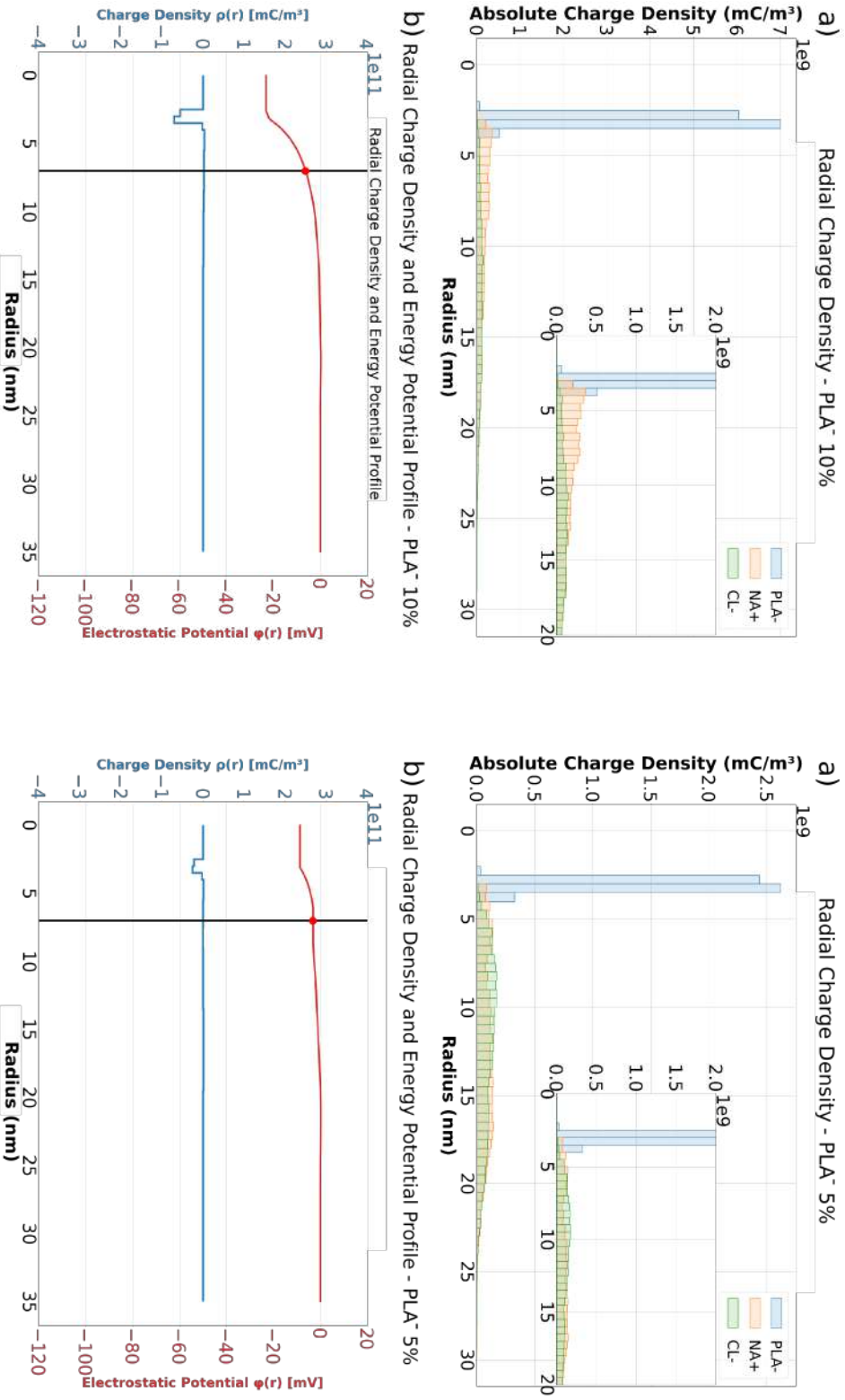


Figure C.5: a. Absolute radial charge density for charged species (10 and 5%) as a function of the distance from the micelle core. b. Radial charge density and electrostatic potential of the same charged species as a function of the distance from the micelle core. The black vertical line identifies the slip plane, and the red dot is the z-potential.

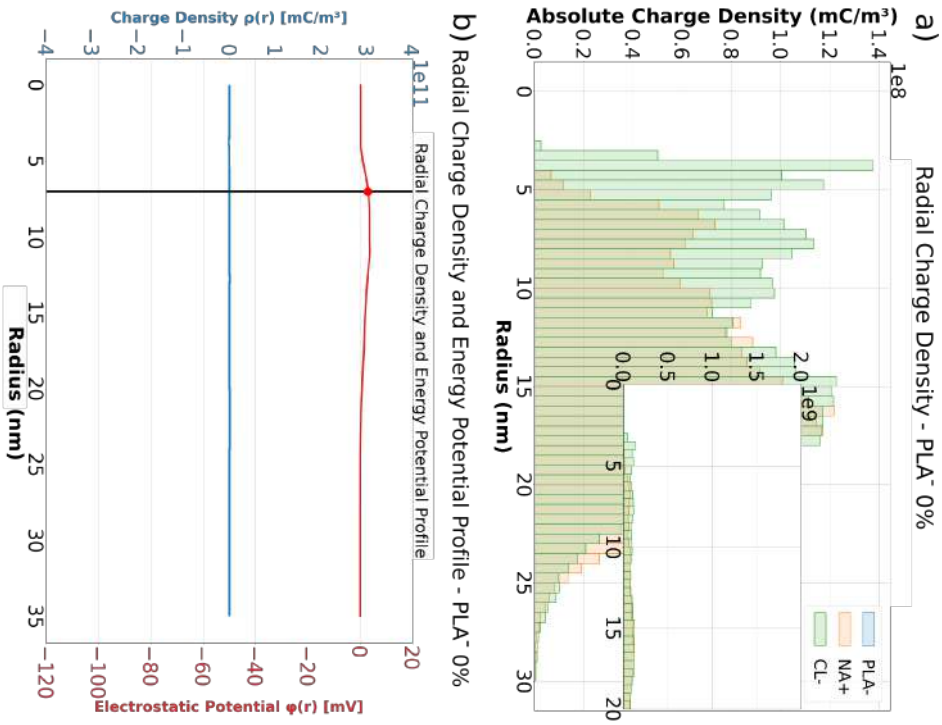


Figure C.6: a. Absolute radial charge density for a non-charged micelle as a function of the distance from the micelle core. b. Radial charge density and electrostatic potential of the same non-charged species as a function of the distance from the micelle core. The black vertical line identifies the slip plane, and the red dot is the z-potential.